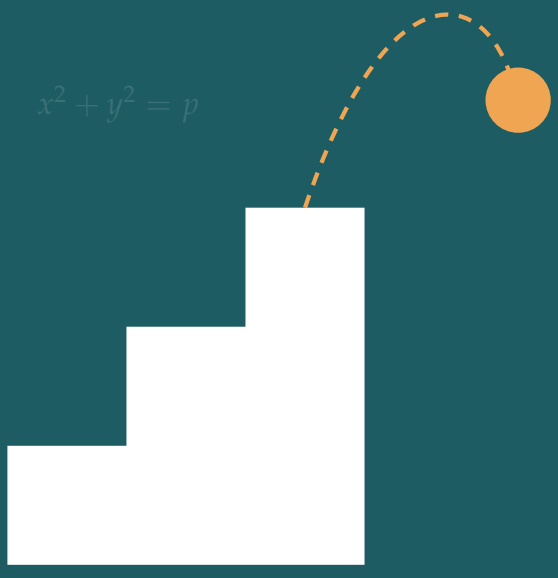
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1997 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1997 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1997 Worked Solutions

Worked solutions to all 42 questions of the 1997 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1997, Question 1

Pure • Counting Coin Combinations

STEP I 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q1

This question has a full guided treatment in *Beyond STEP*, Section 9.5, *Making Change*: https://beyondstep.keithmcnulty.org/ch09/making-change.html

The key idea

Counting "ways" only works if we have a system that lists every combination exactly once. The natural system is to fix the number of the largest coin first, then the next largest, and so on. Once only 2p and 1p coins are left, the count is immediate: the number of 2p coins decides everything. Doing it this way also means the 10p answer feeds straight into the 20p answer.

Solution

Order doesn't matter: a "way" is a choice of how many of each coin to use. So we count solutions in non-negative integers.

The basic building block: 2p and 1p coins

To make r pence from 2p and 1p coins, we choose the number b of 2p coins, which can be anything from 0 to $\lfloor r/2 \rfloor$. The 1p coins then make up the remaining $r - 2b \geq 0$ pence, in exactly one way. So the number of ways is

$$\left\lfloor \frac{r}{2} \right\rfloor + 1.$$

Making 10 pence

- *Using a 10p coin.* It makes the whole amount alone: 1 way.
- *No 10p coin.* We split by the number of 5p coins, which is 0, 1 or 2:

5p coins	left for 2p and 1p	ways $\lfloor r/2 \rfloor + 1$
2	0	1
1	5	3
0	10	6

That gives $1 + 3 + 6 = 10$ ways.

Every combination has a definite number of 10p coins and of 5p coins, so it is counted once and only once. The total is $1 + 10 = 11$ ways, as required.

Making 20 pence

Write $F(n)$ for the number of ways to make n pence from 5p, 2p and 1p coins. We have just seen that $F(10) = 10$, and clearly $F(0) = 1$ (use no coins). For $F(20)$ we split by the number of 5p coins, from 0 to 4:

5p coins	left for 2p and 1p	ways
4	0	1
3	5	3
2	10	6
1	15	8
0	20	11

so $F(20) = 1 + 3 + 6 + 8 + 11 = 29$.

Ask yourself How do the 20p and 10p coins fit in without double counting?

Split again by the large coins. Either we use the 20p coin, which must then be the only coin: 1 way. Or we don't, and we split by the number of 10p coins:

- two 10p coins: nothing left, $F(0) = 1$ way;
- one 10p coin: 10p left, $F(10) = 10$ ways;
- no 10p coin: 20p left, $F(20) = 29$ ways.

These cases are disjoint and cover everything, so the number of ways to make 20 pence is

$$1 + 1 + 10 + 29 = 41.$$

Checks

- List the 10p case by hand: 10; 5+5; 5+2+2+1, 5+2+1+1+1, 5+1 × 5; then 2 × 5, 2 × 4+1 × 2, ..., 1 × 10 (six of these). That's 11.
- A short computer count (looping over the number of each coin) gives 11 and 41. It also confirms the intermediate values $F(10) = 10$ and $F(20) = 29$, and that without the 20p coin there are 40 ways.

- Small cases are easy to check by hand: there are $F(5) = 4$ ways to make 5p from 5p, 2p and 1p coins, matching $1 + 3$.

Common slips

- Counting orders, so that 5+2+2+1 and 2+5+1+2 become different "ways". They are the same collection of coins.
- Forgetting the single-coin ways: the lone 10p coin, and the lone 20p coin.
- Giving a bare total with no visible system. The question warns that unexplained answers get no credit, so the case split itself is the point.

Beyond STEP

These are *partitions* of n into parts from a fixed set. The neat way to handle them is a generating function: the number of ways is the coefficient of x^n in

$$\frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})(1-x^{20})},$$

and our case split is just multiplying out these factors one at a time. Generating functions turn counting problems into algebra, and they lead on to Euler's and Ramanujan's work on partitions. A good place to explore this is Siklos, *Advanced Problems in Mathematics*; for the number-theory side, Davenport's *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 1997, Question 2

Pure • Inverse Trigonometric Derivatives

STEP I 1997, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q2

The key idea

In part (i) the derivative of f collapses to zero, so f is constant, but only on each interval where it is defined: the point $x = -1$ splits the real line in two, and the constant is different on each side. In part (ii) it is easier to differentiate x with respect to y and then turn the result upside down; the work is in rewriting $\sin y^2$ and $\cos y^2$ in terms of x and y, which is where the range of y matters.

Solution

Part (i)

The function is defined for every $x \neq -1$. Write $u = \dfrac{1-x}{1+x}$. By the quotient rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{-(1+x)-(1-x)}{(1+x)^2} = \frac{-2}{(1+x)^2}, \qquad 1+u^2 = \frac{(1+x)^2+(1-x)^2}{(1+x)^2} = \frac{2(1+x^2)}{(1+x)^2}.$$

Using $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan^{-1} u = \dfrac{1}{1+u^2}\dfrac{\mathrm{d}u}{\mathrm{d}x}$ (chain rule),

$$\begin{aligned} \mathrm{f}'(x) &= \frac{1}{1+x^2} + \frac{(1+x)^2}{2(1+x^2)} \cdot \frac{-2}{(1+x)^2} \\ &= \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0 \qquad (x \neq -1). \end{aligned}$$

Ask yourself Does $\mathrm{f}'(x) = 0$ mean that f is the same constant everywhere?

No. A zero derivative forces a function to be constant only on an *interval* (this comes from the mean value theorem). The domain of f is $(-\infty, -1) \cup (-1, \infty)$, which is two intervals, so f is constant on each piece, and the two constants can differ. We find each one by evaluating at a convenient point, or taking a limit.

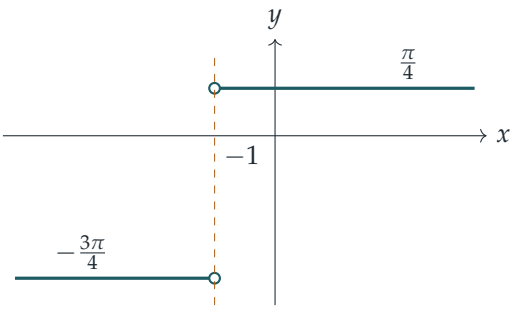
For $x > -1$: put $x = 0$. Then $f(0) = \tan^{-1} 0 + \tan^{-1} 1 = \frac{\pi}{4}$.

For $x < -1$: let $x \to -\infty$. Then $\tan^{-1} x \to -\frac{\pi}{2}$, while $\dfrac{1-x}{1+x} \to -1$, so $\tan^{-1}\left(\dfrac{1-x}{1+x}\right) \to -\frac{\pi}{4}$. Since f is constant on this interval, the constant equals the limit, $-\frac{3\pi}{4}$. (Alternatively, $x = -\sqrt{3}$ gives $\dfrac{1+\sqrt{3}}{1-\sqrt{3}} = -(2+\sqrt{3}) = -\tan\frac{5\pi}{12}$, so $f(-\sqrt{3}) = -\frac{\pi}{3} - \frac{5\pi}{12} = -\frac{3\pi}{4}$.)

Hence

$$f(x) = \begin{cases} \dfrac{\pi}{4}, & x > -1, \\ -\dfrac{3\pi}{4}, & x < -1. \end{cases}$$

The jump of π at $x = -1$ happens because $\dfrac{1-x}{1+x}$ shoots from $-\infty$ to $+\infty$ there, so its inverse tangent leaps from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$.



Part (ii)

Differentiate $x = y \sin y^2$ with respect to y, using the product rule and the chain rule:

$$\frac{dx}{dy} = \sin y^2 + 2y^2 \cos y^2.$$

Now express both trigonometric terms through x and y.

- Since $y > 0$, we may divide by y: $\sin y^2 = \dfrac{x}{y}$.
- Since $0 < y^2 < \frac{\pi}{2}$, we have $\cos y^2 > 0$, so we take the *positive* square root:

$$\cos y^2 = \sqrt{1 - \sin^2 y^2} = \sqrt{1 - \frac{x^2}{y^2}} = \frac{\sqrt{y^2 - x^2}}{y},$$

  where the last step uses $\sqrt{y^2} = y$, true because $y > 0$.

Therefore

$$\frac{dx}{dy} = \frac{x}{y} + 2y\sqrt{y^2 - x^2} = \frac{x + 2y^2\sqrt{y^2 - x^2}}{y}.$$

This is strictly positive, since $x > 0$, $y > 0$ and the square root is non-negative. So we can invert it (the inverse function rule $\frac{dy}{dx} = 1/\frac{dx}{dy}$, valid where $\frac{dx}{dy} \neq 0$):

$$\frac{dy}{dx} = \frac{y}{x + 2y^2\sqrt{y^2 - x^2}},$$

as required.

Ask yourself Why does the question insist on $0 < y < (\pi/2)^{1/2}$?

Without it, $\cos y^2$ could be negative, and the correct formula would have a minus sign in front of the square root. The restriction also makes $y \sin y^2$ strictly increasing on this range (its derivative above is positive), rising from 0 to $(\pi/2)^{1/2}$, so each x in $0 < x < (\pi/2)^{1/2}$ corresponds to exactly one y, and y really is a function of x.

Checks

- Part (i): $f(1) = \frac{\pi}{4} + \tan^{-1} 0 = \frac{\pi}{4}$ and $f(-3) = \tan^{-1}(-3) + \tan^{-1}(-2) \approx -1.2490 - 1.1071 = -2.3562 \approx -\frac{3\pi}{4}$. We checked both constants at 10,000 random points.
- Part (ii): for $x = 0.5$, solving $y \sin y^2 = 0.5$ numerically gives $y \approx 0.81355$, and the formula gives $\frac{dy}{dx} \approx 0.60284$, matching a finite-difference derivative of the numerically inverted function.
- Near $x = 0$, $y \sin y^2 \approx y^3$, so $y \approx x^{1/3}$ with $\frac{dy}{dx} \approx \frac{1}{3y^2}$; the formula gives $y/(y^3 + 2y^3) = \frac{1}{3y^2}$, which agrees.

Common slips

- Concluding $f(x) = \frac{\pi}{4}$ for all x, forgetting that $x = -1$ splits the domain.
- Writing $\cos y^2 = \pm\sqrt{\cdots}$ and not saying why the positive root is the right one.

Beyond STEP

The fact that $f' = 0$ gives a constant only on connected pieces of the domain is a consequence of the mean value theorem, one of the first big results in a university analysis course. Part (ii) is a baby case of the inverse function theorem: a function with non-zero derivative can be inverted locally, and the derivative of the inverse is the reciprocal. Abbott's *Understanding Analysis* treats both carefully.

WORKED SOLUTION

STEP I 1997, Question 3

Pure • Powers and Units Digits

STEP I 1997, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q3

The key idea

Before anything else, write a_7 as a single power of 3: cubing a power of 3 just triples the exponent. After that, each part asks a different question about one enormous number. The last digit depends only on the exponent modulo 4, because the last digits of powers of 3 repeat with period 4. The size bound needs only a crude comparison with a power of 10. The final part rests on that size bound: a huge a_7 makes $1/a_7$ negligible.

Solution

First we find a closed form. If $a_n = 3^k$ then $a_{n+1} = (3^k)^3 = 3^{3k}$, so the exponent is multiplied by 3 at each step. Starting from $a_1 = 3^1$, induction gives

$$a_n = 3^{3^{n-1}}, \qquad \text{so} \qquad a_7 = 3^{3^6} = 3^{729}.$$

Part (i)

The units digit of a product depends only on the units digits of its factors. So we can track the last digit of 3^k by multiplying the previous last digit by 3 and keeping only the last digit:

k	1	2	3	4	5	$\cdots$
units digit of 3^k	3	9	7	1	3	$\cdots$

Once we are back to 1 at $k = 4$, the pattern $3, 9, 7, 1$ repeats with period 4. (Put another way, $3^4 = 81 \equiv 1 \pmod{10}$, so $3^{4m+r} \equiv 3^r \pmod{10}$.)

? **Ask yourself** How do we find 729 modulo 4 without long division?

Since $3 \equiv -1 \pmod 4$, we have $3^6 \equiv (-1)^6 = 1 \pmod 4$. (Directly: $729 = 4 \times 182 + 1$.) Hence $3^{729} = 3^{4 \cdot 182 + 1}$ has the same units digit as 3^1, and

the units digit of a_7 is 3.

Part (ii)

Group the exponent in threes: $729 = 3 \times 243$, so

$$a_7 = 3^{729} = \left(3^3\right)^{243} = 27^{243} > 10^{243} \geq 10^{100},$$

using $27 > 10$ (and the fact that raising positive numbers to a positive power preserves order). Hence $a_7 \geq 10^{100}$, as required. The bound is far from tight: in fact a_7 has 348 digits.

Part (iii)

Split the fraction:

$$\frac{a_7 + 1}{2a_7} = \frac{1}{2} + \frac{1}{2a_7}.$$

Since $a_7 > 0$, the second term is positive. By part (ii) it is also tiny:

$$0 < \frac{1}{2a_7} \leq \frac{1}{2 \times 10^{100}} < 10^{-100}.$$

Therefore

$$0.5 < \frac{a_7 + 1}{2a_7} < 0.5 + 10^{-100} < 0.505.$$

Every number strictly between 0.5 and 0.505 rounds to 0.50 to two decimal places: the third decimal place is 0, well short of the 5 that would round up. So

$$\frac{a_7 + 1}{2a_7} = 0.50 \quad \text{to two decimal places.}$$

In fact the decimal expansion begins $0.5000\ldots$, with at least 100 zeros after the 5.

Checks

A computer handles 3^{729} exactly. Computing it by cubing 3 six times confirms $a_7 = 3^{729}$, a number with 348 digits whose last digit is 3, and that $27^{243} = 3^{729} > 10^{243}$. Using exact fractions confirms that $\frac{a_7+1}{2a_7}$ lies strictly between $\frac{1}{2}$ and $\frac{1}{2} + 10^{-100}$. On a smaller scale you can test the method of part (i): $a_2 = 3^3 = 27$ has exponent $3 \equiv 3 \pmod 4$, which predicts units digit 7, and it does.

Common slips

- Reading a_7 as $3^{3 \cdot 7}$ or 3^{18} rather than 3^{3^6}: each step *multiplies* the exponent by 3, not adds 3.
- Reducing the exponent modulo 10 instead of modulo 4. The units digits cycle with period 4, so it is the exponent modulo 4 that matters.
- In part (iii), writing "0.5" with no justification. "Correct to two places" means 0.50, and you need the bounds showing that the value cannot reach 0.505.

Beyond STEP

Part (i) is a first taste of modular arithmetic and Euler's theorem: $3^4 \equiv 1 \pmod{10}$ because $\varphi(10) = 4$. Repeated use of this reduction, tower by tower, is how number theorists find the last digits of huge power towers such as $3^{3^{3^3}}$. Davenport's *The Higher Arithmetic* develops congruences and the order of an element from scratch.

WORKED SOLUTION

STEP I 1997, Question 4

Pure • Modulus Equations

> **STEP I 1997, Question 4**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/97-s1-q4

The key idea

Each modulus term changes its formula only where its inside is zero, so the four "critical points" $x = -1, 0, 1, 2$ split the real line into five intervals. On each interval every modulus can be replaced by a plain bracket with a definite sign, the left-hand side becomes a linear function, and the equation is easy. The one thing to watch is that a root of the linear equation counts only if it lies in the interval you were working in.

Solution

Write $L(x) = |x+1| - |x| + 3|x-1| - 2|x-2|$. Recall that $|t| = t$ when $t \geq 0$ and $|t| = -t$ when $t < 0$. We work through the intervals from right to left.

$x \geq 2$. All four insides are non-negative, so

$$\begin{aligned} L(x) &= (x+1) - x + 3(x-1) - 2(x-2) \\ &= x + 1 - x + 3x - 3 - 2x + 4 = x + 2. \end{aligned}$$

The equation reads $x + 2 = x + 2$, which is true for *every* $x \geq 2$. So the whole interval $x \geq 2$ consists of solutions.

> **Ask yourself** The equation has become an identity. Have we made an algebra slip?

No. As $x \to \infty$ the coefficients of x in the moduli add up to $1 - 1 + 3 - 2 = 1$ and the constants to $1 - 0 - 3 + 4 = 2$, which exactly match the right-hand side $x + 2$. An equation involving moduli can perfectly well have an interval of solutions, and the question's "find all the solutions" is a hint that something like this might happen. A quick test: at $x = 3$, $L(3) = 4 - 3 + 6 - 2 = 5 = 3 + 2$.

$1 \le x < 2$. Now $x - 2 < 0$, so $|x-2| = -(x-2)$ and

$$L(x) = (x+1) - x + 3(x-1) + 2(x-2) = 5x - 6.$$

Then $5x - 6 = x + 2$ gives $x = 2$, which is not in $[1,2)$. (It is a solution, but we have already counted it.) No new solutions.

$0 \le x < 1$. Now $x - 1 < 0$ as well, so

$$L(x) = (x+1) - x - 3(x-1) + 2(x-2) = -x.$$

Then $-x = x + 2$ gives $x = -1$, not in $[0,1)$. No solutions.

$-1 \le x < 0$. Now $|x| = -x$ too, so

$$L(x) = (x+1) + x - 3(x-1) + 2(x-2) = x.$$

The equation $x = x + 2$ has no solutions at all.

$x < -1$. All four insides are negative, so

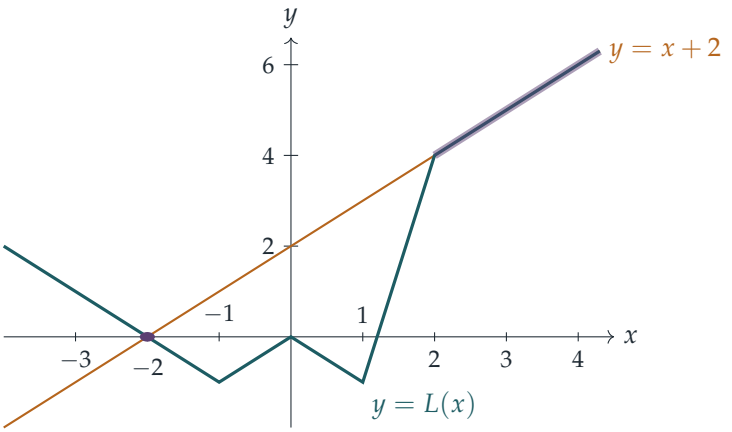
$$L(x) = -(x+1) + x - 3(x-1) + 2(x-2) = -x - 2.$$

Then $-x - 2 = x + 2$ gives $x = -2$, and $-2 < -1$, so this is a genuine solution. Check: $L(-2) = 1 - 2 + 9 - 8 = 0 = -2 + 2$.

Conclusion. The solutions are

$$x = -2 \qquad \text{and every } x \ge 2.$$

The graph shows why: $y = L(x)$ (teal) is a zigzag of line segments, which crosses the line $y = x + 2$ (amber) once at $x = -2$ and then lies along it from $x = 2$ onwards.



Checks

Substituting a few values: $L(-2) = 0 = -2 + 2$, $L(5) = 6 - 5 + 12 - 6 = 7 = 5 + 2$, while at $x = 0$ we get $L(0) = 1 + 3 - 4 = 0 \neq 2$. A computer working in exact fractions confirmed the five linear formulae for L on 100,000 random points, and found that on a grid of step 0.01 over $[-20, 20]$ the equation holds exactly at $x = -2$ and at every grid point with $x \geq 2$, and nowhere else.

Common slips

- Getting $x + 2 = x + 2$ for $x \geq 2$ and writing "no solution" or "infinitely many, so something went wrong", instead of recognising a whole interval of solutions.
- Accepting roots such as $x = -1$ (from $0 \leq x < 1$) that lie outside the interval being considered.
- Forgetting the outer region $x < -1$, which is where the isolated solution $x = -2$ lives.

Beyond STEP

A function like L is *piecewise linear*: continuous, and linear between finitely many "corners". Such functions are central to linear programming and optimisation, where minimising sums of moduli $\sum |x - a_i|$ leads to medians rather than means. They are also a first example of functions that are continuous but not differentiable at isolated points, a theme developed carefully in Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1997, Question 5

Pure • Vectors in Geometry

STEP I 1997, Question 5

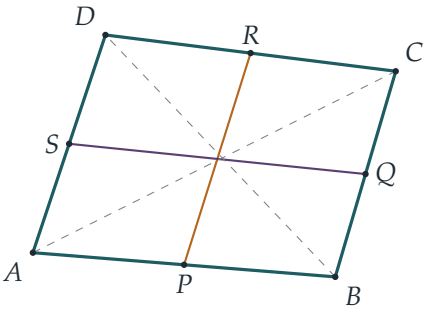
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q5

The key idea

Lengths squared are dot products, and mid-points are averages, so position vectors turn the whole identity into algebra. The useful trick is to spot that both sides secretly involve the diagonals AC and BD: once each difference is written in terms of $\overrightarrow{AC} \cdot \overrightarrow{BD}$, the identity falls out. The deduction then only needs the fact that the four rod lengths never change.

Solution



Let A, B, C, D have position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ relative to any origin, and write $|\mathbf{x}|^2 = \mathbf{x} \cdot \mathbf{x}$. The mid-points are

$$\mathbf{p} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}), \quad \mathbf{q} = \tfrac{1}{2}(\mathbf{b}+\mathbf{c}), \quad \mathbf{r} = \tfrac{1}{2}(\mathbf{c}+\mathbf{d}), \quad \mathbf{s} = \tfrac{1}{2}(\mathbf{d}+\mathbf{a}).$$

The four sides

Expanding each $|\mathbf{y}-\mathbf{x}|^2 = |\mathbf{x}|^2 + |\mathbf{y}|^2 - 2\,\mathbf{x}\cdot\mathbf{y}$, the squared terms $|\mathbf{a}|^2, \ldots, |\mathbf{d}|^2$ each appear once on each side and cancel, leaving

$$\begin{aligned} |AB|^2 + |CD|^2 - |AD|^2 - |BC|^2 &= -2\,\mathbf{a}\cdot\mathbf{b} - 2\,\mathbf{c}\cdot\mathbf{d} + 2\,\mathbf{a}\cdot\mathbf{d} + 2\,\mathbf{b}\cdot\mathbf{c} \\ &= 2(\mathbf{a}-\mathbf{c})\cdot(\mathbf{d}-\mathbf{b}). \end{aligned}$$

(Multiply out $(\mathbf{a}-\mathbf{c})\cdot(\mathbf{d}-\mathbf{b})$ to confirm the factorisation.) In words, this is $-2\,\overrightarrow{AC}\cdot\overrightarrow{BD}$.

The two mid-point joins

We have

$$\overrightarrow{PR} = \tfrac{1}{2}(\mathbf{c}+\mathbf{d}-\mathbf{a}-\mathbf{b}) = \tfrac{1}{2}\mathbf{u}, \qquad \overrightarrow{QS} = \tfrac{1}{2}(\mathbf{d}+\mathbf{a}-\mathbf{b}-\mathbf{c}) = \tfrac{1}{2}\mathbf{v},$$

say.

> **Ask yourself** How can we find $|\mathbf{u}|^2 - |\mathbf{v}|^2$ without expanding two four-term squares?

Use the difference of two squares for vectors: $|\mathbf{u}|^2 - |\mathbf{v}|^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}-\mathbf{v})$, which holds because $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$. Here

$$\mathbf{u}+\mathbf{v} = 2(\mathbf{d}-\mathbf{b}), \qquad \mathbf{u}-\mathbf{v} = 2(\mathbf{c}-\mathbf{a}),$$

so

$$|PR|^2 - |QS|^2 = \tfrac{1}{4}\left(|\mathbf{u}|^2 - |\mathbf{v}|^2\right) = (\mathbf{c}-\mathbf{a})\cdot(\mathbf{d}-\mathbf{b}) = \overrightarrow{AC}\cdot\overrightarrow{BD}.$$

Putting them together

Adding the side result to twice the mid-point result,

$$\begin{aligned} &|AB|^2 + |CD|^2 - |AD|^2 - |BC|^2 + 2\left(|PR|^2 - |QS|^2\right) \\ &\quad = 2(\mathbf{a}-\mathbf{c})\cdot(\mathbf{d}-\mathbf{b}) + 2(\mathbf{c}-\mathbf{a})\cdot(\mathbf{d}-\mathbf{b}) = 0, \end{aligned}$$

which rearranges to

$$|AB|^2 + |CD|^2 + 2|PR|^2 = |AD|^2 + |BC|^2 + 2|QS|^2,$$

as required. We never used convexity, or even that the points lie in a plane, so the identity holds for any four points, including a quadrilateral that has folded over so that two of its rods cross.

The deduction

Rearranging the identity,

$$|PR|^2 - |QS|^2 = \tfrac{1}{2}\left(|AD|^2 + |BC|^2 - |AB|^2 - |CD|^2\right).$$

The rods are rigid, so the four lengths on the right never change, however the joints flex. So $|PR|^2 - |QS|^2$ is constant throughout the motion. Equivalently, the scalar product $\overrightarrow{AC} \cdot \overrightarrow{BD}$ of the diagonals stays fixed, even though the diagonals themselves change length and direction.

Checks

- For a rectangle with sides a and b, $|AB| = |CD| = a$ and $|AD| = |BC| = b$, while PR and QS join mid-points of opposite sides, so $|PR| = b$ and $|QS| = a$. Both sides of the identity equal $2a^2 + 2b^2$.
- For a rhombus all four sides are equal, so the deduction says $|PR| = |QS|$ in every position. Indeed, the diagonals of a rhombus are perpendicular, so $\overrightarrow{AC} \cdot \overrightarrow{BD} = 0$.
- Numerically, the identity holds to within 10^{-12} on 10^5 random quadrilaterals. For a linkage with $|AB| = 3$, $|BC| = 4$, $|CD| = 5$, $|DA| = 6$ flexed through about 200 positions, $|PR|^2 - |QS|^2$ stayed at $\frac{1}{2}(36 + 16 - 9 - 25) = 9$ throughout.

Common slips

- Mixing up which sides pair with which: PR joins the mid-points of AB and CD, and it is those two sides that sit with PR in the identity.
- Using a coordinate set-up such as A at the origin and B on an axis, then forgetting that the general quadrilateral needs every other point free. Vectors avoid this bookkeeping.

Beyond STEP

The points P, Q, R, S always form a parallelogram (Varignon's theorem), and PR, QS are its diagonals. The identity here is a cousin of the parallelogram law $|\mathbf{u} + \mathbf{v}|^2 + |\mathbf{u} - \mathbf{v}|^2 = 2|\mathbf{u}|^2 + 2|\mathbf{v}|^2$, which in a first-year linear algebra course characterises exactly those norms that come from an inner product. The idea of a quantity that stays fixed while a mechanism moves, an invariant, recurs throughout mathematics. Strang's *Introduction to Linear Algebra* is a good place to see dot products and inner products developed properly.

WORKED SOLUTION

STEP I 1997, Question 6

Pure • Integral Bounds for Pi

STEP I 1997, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 4.3, *22/7 Is Bigger Than* π: https://beyondstep.keithmcnulty.org/ch04/twenty-two-sevenths.html

The key idea

The integrand is a rational function whose numerator has higher degree than its denominator, so the first job is polynomial division by x^2+1: a polynomial quotient, which integrates trivially, plus a constant remainder over $1+x^2$, which integrates to an arctangent. The arctangent is where π comes from. For the final inequalities, compare the integrand with $x^4(1-x)^4$ itself, using $1 \le 1+x^2$ on $[0,1]$.

Solution

The division

Expanding by the binomial theorem,

$$x^4(1-x)^4 = x^4(1-4x+6x^2-4x^3+x^4) = x^8-4x^7+6x^6-4x^5+x^4.$$

We divide by x^2+1, peeling off the leading term each time:

$$\begin{aligned} x^8-4x^7+6x^6-4x^5+x^4 &= x^6(x^2+1)+(-4x^7+5x^6-4x^5+x^4)\\ -4x^7+5x^6-4x^5+x^4 &= -4x^5(x^2+1)+(5x^6+x^4)\\ 5x^6+x^4 &= 5x^4(x^2+1)+(-4x^4)\\ -4x^4 &= -4x^2(x^2+1)+4x^2\\ 4x^2 &= 4(x^2+1)-4. \end{aligned}$$

Collecting the multiples of x^2+1,

$$x^4(1-x)^4 = (x^6-4x^5+5x^4-4x^2+4)(x^2+1)-4,$$

so

$$a_6 = 1, \quad a_5 = -4, \quad a_4 = 5, \quad a_3 = 0, \quad a_2 = -4, \quad a_1 = 0, \quad a_0 = 4, \quad b = -4.$$

A quick check: at $x = 0$ the right-hand side is $4 \cdot 1 - 4 = 0$, and at $x = 1$ it is $(1 - 4 + 5 - 4 + 4) \cdot 2 - 4 = 0$, matching the left-hand side both times.

The integral equal to $\frac{22}{7} - \pi$

Dividing the identity by $1 + x^2$ (which is never zero),

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Integrating term by term over $[0, 1]$, and using $\int \frac{1}{1+x^2}\,\mathrm{d}x = \arctan x$,

$$\begin{aligned}\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x &= \left[\tfrac{1}{7}x^7 - \tfrac{2}{3}x^6 + x^5 - \tfrac{4}{3}x^3 + 4x\right]_0^1 - 4\Big[\arctan x\Big]_0^1 \\ &= \tfrac{1}{7} - \tfrac{2}{3} + 1 - \tfrac{4}{3} + 4 \;-\; 4 \cdot \tfrac{\pi}{4} \\ &= \tfrac{1}{7} + 3 - \pi \\ &= \frac{22}{7} - \pi,\end{aligned}$$

since $-\frac{2}{3} - \frac{4}{3} = -2$ and $\arctan 1 = \frac{\pi}{4}$. This is the required result.

The integral of $x^4(1-x)^4$

Using the expansion from the start,

$$\int_0^1 x^4(1-x)^4\,\mathrm{d}x = \tfrac{1}{5} - \tfrac{4}{6} + \tfrac{6}{7} - \tfrac{4}{8} + \tfrac{1}{9}.$$

Over the common denominator $2520 = 5 \cdot 7 \cdot 8 \cdot 9$,

$$\frac{504 - 1680 + 2160 - 1260 + 280}{2520} = \frac{4}{2520} = \frac{1}{630}.$$

The bounds on π

Write $f(x) = x^4(1-x)^4$, so $f(x) > 0$ for $0 < x < 1$.

Upper bound. On $(0, 1)$ the integrand $f(x)/(1+x^2)$ is strictly positive and continuous, so its integral over $[0, 1]$ is strictly positive. Hence $\frac{22}{7} - \pi > 0$, that is, $\pi < \frac{22}{7}$.

Lower bound. For $0 < x < 1$ we have $1 + x^2 > 1$, and since $f(x) > 0$,

$$\frac{f(x)}{1+x^2} < f(x).$$

Ask yourself Why is the inequality between the integrals strict, not just $\leq$?

The difference $f(x) - \dfrac{f(x)}{1+x^2} = \dfrac{x^2 f(x)}{1+x^2}$ is continuous and strictly positive on the whole open interval $(0,1)$, so its integral over $[0,1]$ is strictly positive. (A continuous function that is non-negative and positive somewhere has positive integral.) Therefore

$$\frac{22}{7} - \pi = \int_0^1 \frac{f(x)}{1+x^2}\,\mathrm{d}x < \int_0^1 f(x)\,\mathrm{d}x = \frac{1}{630},$$

which rearranges to $\pi > \frac{22}{7} - \frac{1}{630}$. Combining the two bounds,

$$\frac{22}{7} > \pi > \frac{22}{7} - \frac{1}{630},$$

as required.

Checks

- Numerically, $\frac{22}{7} - \pi \approx 0.0012645$, and Simpson's rule on the first integral gives the same value to twelve decimal places.
- $\frac{1}{630} \approx 0.0015873$, comfortably bigger than 0.0012645; the bounds say $3.141270 < \pi < 3.142857$, and indeed $\pi \approx 3.141593$.
- The value $\frac{1}{630}$ also follows from the beta-integral formula $\int_0^1 x^m(1-x)^n\,\mathrm{d}x = \dfrac{m!\,n!}{(m+n+1)!}$, which gives $\dfrac{4!\,4!}{9!} = \dfrac{576}{362880} = \dfrac{1}{630}$.

Common slips

- Losing the two zero coefficients a_3 and a_1 in the long division: keep every power in place, or the remainder comes out wrong.
- Writing $\int_0^1 \frac{4}{1+x^2}\,\mathrm{d}x = 4\ln 2$ (confusing it with $\int \frac{2x}{1+x^2}\,\mathrm{d}x$); it is $4\arctan 1 = \pi$.
- Proving only $\leq$ in the final step. The question asks for strict inequalities, so say why the integrals cannot be equal.

Beyond STEP

This beautiful integral, discovered in the twentieth century, shows at a stroke that the schoolbook approximation $\frac{22}{7}$ overestimates π. The same idea, integrating $x^n(1-x)^n$ against simple functions and showing the result is small but positive, drives the classical proofs that π and e are irrational: an integer between 0 and 1 cannot exist. Integrals like $\int_0^1 x^m(1-x)^n\,\mathrm{d}x$ are values of the beta function, met in first-year analysis and probability. For the careful handling of strict inequalities between integrals, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1997, Question 7

Pure • Gaussian Quadrature

STEP I 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q7

The key idea

Both sides depend *linearly* on P, so the formula holds for every cubic as soon as it holds for the four basic cubics 1, t, t^2 and t^3. That turns the problem into four equations in four unknowns. They are nonlinear, but the odd and even powers separate neatly, and symmetry about $t = 0$ does the rest.

Solution

Write $\mathrm{P}(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$. Then

$$\int_{-1}^{1} \mathrm{P}(t)\,\mathrm{d}t = c_0 \int_{-1}^{1} 1\,\mathrm{d}t + c_1 \int_{-1}^{1} t\,\mathrm{d}t + c_2 \int_{-1}^{1} t^2\,\mathrm{d}t + c_3 \int_{-1}^{1} t^3\,\mathrm{d}t,$$

and $a_1\mathrm{P}(u_1) + a_2\mathrm{P}(u_2)$ splits in exactly the same way. So the formula holds for all cubics if and only if it holds for $\mathrm{P}(t) = t^k$, $k = 0, 1, 2, 3$. ("Only if" because each t^k is itself a cubic polynomial in the broad sense; "if" by adding up.) Since $\int_{-1}^{1} t^k\,\mathrm{d}t$ is 2, 0, $\frac{2}{3}$, 0 for $k = 0, 1, 2, 3$, we need

$$a_1 + a_2 = 2, \tag{A}$$
$$a_1 u_1 + a_2 u_2 = 0, \tag{B}$$
$$a_1 u_1^2 + a_2 u_2^2 = \tfrac{2}{3}, \tag{C}$$
$$a_1 u_1^3 + a_2 u_2^3 = 0. \tag{D}$$

Ask yourself These equations are nonlinear. Is there a combination of them that factorises?

Take (D) minus u_2^2 times (B): the a_2 terms cancel and

$$a_1 u_1^3 - a_1 u_1 u_2^2 = a_1 u_1 (u_1 - u_2)(u_1 + u_2) = 0.$$

We rule out the first three factors in turn.

- If $a_1 = 0$: then (A) gives $a_2 = 2$, (B) gives $u_2 = 0$, and (C) reads $0 = \frac{2}{3}$. Impossible.
- If $u_1 = 0$: then (B) gives $a_2 u_2 = 0$, so $a_2 u_2^2 = 0$, and again (C) reads $0 = \frac{2}{3}$. Impossible.
- If $u_1 = u_2$: the two points coincide, (B) becomes $(a_1 + a_2)u_1 = 2u_1 = 0$, so $u_1 = u_2 = 0$ and (C) fails once more.

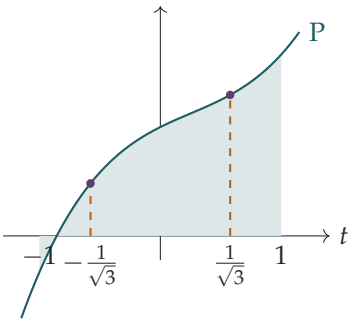
Hence $u_2 = -u_1$ with $u_1 \neq 0$. Now (B) becomes $(a_1 - a_2)u_1 = 0$, so $a_1 = a_2$, and (A) gives $a_1 = a_2 = 1$. Finally (C) gives $2u_1^2 = \frac{2}{3}$, so $u_1^2 = \frac{1}{3}$; (D) is then automatically satisfied. Therefore

$$\boxed{a_1 = a_2 = 1, \qquad u_1 = -\tfrac{1}{\sqrt{3}}, \quad u_2 = \tfrac{1}{\sqrt{3}}}$$

(or the same with u_1 and u_2 swapped), that is,

$$\int_{-1}^{1} \mathrm{P}(t)\,\mathrm{d}t = \mathrm{P}\left(-\tfrac{1}{\sqrt{3}}\right) + \mathrm{P}\left(\tfrac{1}{\sqrt{3}}\right) \quad \text{for every cubic P.}$$

The argument also shows that this is the *only* choice, apart from relabelling.



The shaded area equals the sum of the two dashed heights, whatever cubic is drawn.

Checks

- $\mathrm{P}(t) = t^2$: the integral is $\frac{2}{3}$ and the rule gives $\frac{1}{3} + \frac{1}{3}$.
- $\mathrm{P}(t) = 1 + t + t^2 + t^3$: the integral is $2 + \frac{2}{3} = \frac{8}{3}$; the rule gives $2 + 2 \cdot \frac{1}{3} = \frac{8}{3}$, since the odd terms cancel.
- The rule is *not* exact for t^4: it gives $2 \cdot \frac{1}{9} = \frac{2}{9}$, while the integral is $\frac{2}{5}$. So cubics really are the limit.
- Numerically, on 10^4 random cubics the two sides agreed to within 10^{-14}.

Common slips

- Assuming symmetry ($u_2 = -u_1$, $a_1 = a_2$) from the start. It gives the right answer quickly, but it doesn't show that no other choice works; the factorisation above does.

- Forgetting the cases $a_1 = 0$ or $u_1 = u_2$ when dividing through.
- Testing only $\mathrm{P}(t) = 1$ and t^2: the odd conditions (B) and (D) are what force the symmetry.

Beyond STEP

This is two-point *Gaussian quadrature*. With n well-chosen points and weights one can integrate every polynomial of degree up to $2n - 1$ exactly, twice what n equally spaced points manage. The best points are the zeros of the Legendre polynomials (here $\frac{1}{2}(3t^2 - 1)$, whose zeros are $\pm\frac{1}{\sqrt{3}}$), which are orthogonal on $[-1, 1]$. Orthogonal polynomials appear in first-year numerical analysis and in solving differential equations; Boyce and DiPrima, *Elementary Differential Equations*, meets Legendre's equation in its series-solutions chapter.

WORKED SOLUTION

STEP I 1997, Question 8

Pure • Roots of Exponential Equations

STEP I 1997, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q8

The key idea

A root of $x = a^x$ must be positive, since $a^x > 0$. That lets us take logarithms and turn the question into one about the zeros of $f(x) = \ln x - x \ln a$ on $x > 0$. For $a > 1$ this function rises and then falls, so it has zeros only if its maximum is at least 0. For $0 < a < 1$ it is strictly increasing, so it can cross zero at most once, and the question is whether it crosses at all.

Solution

Reducing to $f(x) = 0$

Throughout, $a > 0$, so $a^x = e^{x \ln a}$ is defined and positive for every real x. If $x \leq 0$ then $a^x > 0 \geq x$, so there are no roots with $x \leq 0$. For $x > 0$ both sides are positive, and since ln is strictly increasing,

$$x = a^x \iff \ln x = x \ln a \iff f(x) = 0, \qquad \text{where } f(x) = \ln x - x \ln a.$$

So the real roots of $x = a^x$ are exactly the zeros of f on $(0, \infty)$.

No roots when $a > e^{1/e}$

Here $a > e^{1/e} > 1$, so $\ln a > 0$. Differentiating,

$$f'(x) = \frac{1}{x} - \ln a,$$

which is positive for $0 < x < 1/\ln a$ and negative for $x > 1/\ln a$. So f increases and then decreases, and its maximum value on $(0, \infty)$ is

$$f\left(\frac{1}{\ln a}\right) = \ln\left(\frac{1}{\ln a}\right) - 1 = -\ln(\ln a) - 1.$$

Now $a > e^{1/e}$ gives $\ln a > 1/e$, and taking logarithms again (both sides positive), $\ln(\ln a) > \ln(1/e) = -1$. Hence

$$f(x) \leq -\ln(\ln a) - 1 < 0 \quad \text{for all } x > 0.$$

So f is never zero, and the equation $x = a^x$ has no real roots, as required.

Ask yourself What happens exactly at $a = e^{1/e}$, and for $1 < a < e^{1/e}$?

At $a = e^{1/e}$ the maximum is exactly 0, attained at $x = 1/\ln a = e$, so there is one (repeated) root $x = e$: the line $y = x$ is tangent to $y = a^x$ at (e, e). For $1 < a < e^{1/e}$ the maximum is positive, and $f \to -\infty$ at both ends, so there are two roots. (For $a = 1$ the only root is $x = 1$.) The question doesn't ask for these, but they show that $e^{1/e}$ is exactly the cut-off.

Exactly one root when $0 < a < 1$

Now $\ln a < 0$, so

$$f'(x) = \frac{1}{x} - \ln a = \frac{1}{x} + |\ln a| > 0 \quad \text{for all } x > 0,$$

and f is strictly increasing on $(0, \infty)$. Hence f has *at most one* zero.

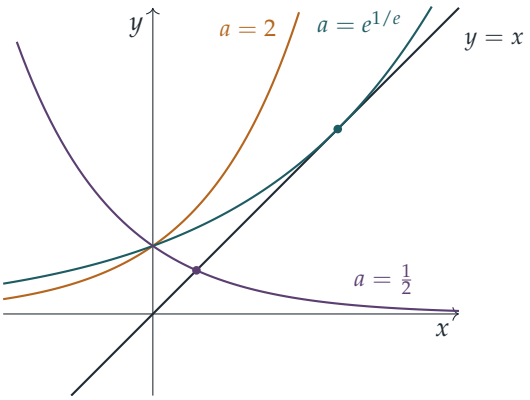
To see that it has one, note that f is continuous on $(0, \infty)$ and

$$f(1) = -\ln a > 0, \qquad f(a) = \ln a - a \ln a = (1 - a) \ln a < 0,$$

using $0 < a < 1$ in both. By the intermediate value theorem f has a zero between a and 1.

So for $0 < a < 1$ the equation $x = a^x$ has **exactly one** real root, and it lies in the interval $(a, 1)$.

The picture makes this clear: $y = a^x$ is decreasing and positive, while $y = x$ is increasing, so the two graphs cross exactly once. (Equivalently, $g(x) = a^x - x$ is strictly decreasing, with $g(0) = 1 > 0$ and $g(1) = a - 1 < 0$.)



Checks

- At the boundary, $(e^{1/e})^e = e$, so $x = e$ really is a root when $a = e^{1/e} \approx 1.4447$, matching the tangency.
- For $a = \frac{1}{2}$, bisection gives the single root $x \approx 0.6412$, which lies in $(\frac{1}{2}, 1)$ as predicted.
- Numerically, for 200 random $a > e^{1/e}$ the function $a^x - x$ stayed positive on a fine grid over $[-20, 20]$; the maximum of $\ln x - x \ln a$ matched $-1 - \ln \ln a$; for a just below $e^{1/e}$ there were two roots; and for 500 random $a \in (0, 1)$ there was exactly one sign change.

Common slips

- Taking logarithms without first noting that any root has $x > 0$: $\ln x$ is undefined otherwise, so the case $x \leq 0$ needs its own (one-line) argument.
- Showing only that f is increasing for $0 < a < 1$. That proves *at most* one root; you also need a sign change to show there is one.
- Getting the inequality the wrong way round when taking ln of $\ln a > 1/e$; it is safe because ln is increasing and both sides are positive.

Beyond STEP

The root for $0 < a < 1$ is a fixed point of the map $x \mapsto a^x$, and iterating $x_{n+1} = a^{x_n}$ leads to the "infinite power tower" $a^{a^{a^{\cdots}}}$. Euler showed that the tower converges exactly when $e^{-e} \leq a \leq e^{1/e}$, so the same cut-off $e^{1/e}$ appears again. Studying when iterations converge leads to the contraction mapping theorem of first-year analysis; Abbott, *Understanding Analysis*, covers the intermediate value theorem and the background needed.

WORKED SOLUTION

STEP I 1997, Question 9

Mechanics • Relative Velocity

STEP I 1997, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q9

The key idea

From the roadside everything is moving, but from a car everything is still: the cars become a fixed row of blocks with gaps of length c, and the chicken walks across in a straight line with its *relative* velocity. It gets through only if, while it is inside the lane of width a, it drifts sideways (relative to the cars) by no more than the gap c. That single condition gives the inequality, and the minimum speed then comes from maximising $c\sin\theta + a\cos\theta$, a standard $R\cos(\theta - \alpha)$ job.

Solution

We model the chicken as a particle and the cars as rectangles moving at speed V, each occupying the same strip of width a across the road. Take the x-axis along the road in the direction of traffic and the y-axis across it, so the chicken's velocity is $(u\cos\theta,\ u\sin\theta)$ with $0 < \theta < \pi$. Only the strip containing the cars matters: elsewhere on the road the chicken is never in danger.

The inequality

Relative to the cars, the chicken's velocity is

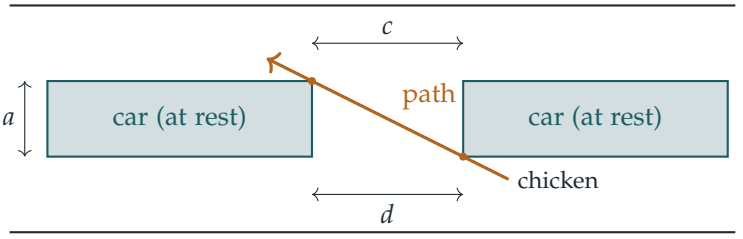
$$\mathbf{w} = (u\cos\theta - V,\ u\sin\theta).$$

In the cars' frame the cars are at rest, and the chicken moves in a straight line at constant velocity $\mathbf{w}$. It spends time

$$\tau = \frac{a}{u\sin\theta}$$

crossing the strip of width a, and during that time it slips backwards (against the traffic, relative to the cars) by

$$d = (V - u\cos\theta)\,\tau = \frac{a(V - u\cos\theta)}{u\sin\theta}.$$



The picture shows the extreme case $d = c$, where the path runs exactly along the diagonal of the gap.

Ask yourself Why must the drift d be at most c?

In the cars' frame the chicken's path through the strip covers a stretch of length $|d|$ along the road, and every point of that stretch is passed at some height inside the lane. The stationary cars leave free only gaps of length c, so the whole stretch must fit inside one gap: $|d| \le c$. (Conversely, if $|d| \le c$, the chicken can time its start so that the stretch lies inside a gap, so it crosses safely.)

In particular $d \le c$, that is,

$$a(V - u\cos\theta) \le c\,u\sin\theta, \qquad \text{so} \qquad Va \le u\,(c\sin\theta + a\cos\theta).$$

The left-hand side is positive, so $c\sin\theta + a\cos\theta > 0$ for any safe crossing, and we may divide by it:

$$u \ge \frac{Va}{c\sin\theta + a\cos\theta},$$

as required. (The other half of the condition, $d \ge -c$, stops a fast chicken from overtaking the gap and running into the car in front; it plays no part in the minimum below.)

The least speed and the crossing time

Write $c\sin\theta + a\cos\theta = \sqrt{a^2+c^2}\,\cos(\theta - \alpha)$, where α is the acute angle with

$$\cos\alpha = \frac{a}{\sqrt{a^2+c^2}}, \qquad \sin\alpha = \frac{c}{\sqrt{a^2+c^2}}, \qquad \tan\alpha = \frac{c}{a}.$$

So $c\sin\theta + a\cos\theta \le \sqrt{a^2+c^2}$, with equality only when $\theta = \alpha$. Hence for every direction

$$u \ge \frac{Va}{c\sin\theta + a\cos\theta} \ge \frac{Va}{\sqrt{a^2+c^2}},$$

and both inequalities are equalities when $\theta = \alpha$ and $u = Va/\sqrt{a^2+c^2}$. That choice really is safe: it gives $d = c$ exactly (the path of the figure, corner to corner), and $-c \le d \le c$ holds. So the least possible uniform speed is $Va/\sqrt{a^2+c^2}$, at angle $\alpha = \arctan(c/a)$.

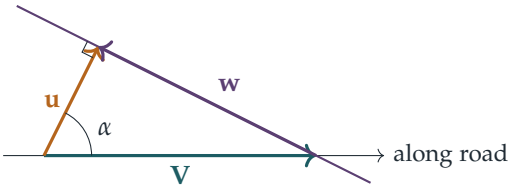
The chicken crosses the full breadth b with velocity component $u\sin\alpha$ across the road, so the time taken is

$$\frac{b}{u\sin\alpha} = b\cdot\frac{\sqrt{a^2+c^2}}{Va}\cdot\frac{\sqrt{a^2+c^2}}{c} = \frac{b\,(a^2+c^2)}{Vac} = \frac{b}{V}\left(\frac{c}{a}+\frac{a}{c}\right),$$

which is the required result.

Ask yourself Why does the least speed come with $\tan\theta = c/a$?

Look at the velocities. The chicken's velocity is $\mathbf{u} = \mathbf{V} + \mathbf{w}$, and at best the relative velocity $\mathbf{w}$ points along the diagonal of the gap, direction $(-c, a)$. So the tip of $\mathbf{u}$ lies on the line through the tip of $\mathbf{V}$ in that direction, and the shortest $\mathbf{u}$ is the perpendicular from the origin to that line. Its direction (a, c) is at angle $\arctan(c/a)$ to the road, and its length is $V\sin(\text{angle of the diagonal}) = Va/\sqrt{a^2+c^2}$.



Checks

- Units: $Va/\sqrt{a^2+c^2}$ is a speed, and $\frac{b}{V}\left(\frac{c}{a}+\frac{a}{c}\right)$ is a length over a speed, a time.
- Limits: if $\theta = \frac{\pi}{2}$ (straight across), the bound is $u \geq Va/c$: the chicken must cross the width a in the time c/V that a gap takes to pass. That is clearly right.
- Numbers: with $a = 1$, $c = 2$, $V = 2$, $b = 5$ the least speed is $2/\sqrt{5} \approx 0.894$ at $\tan\theta = 2$, and the crossing time is $\frac{5}{2}\left(2+\frac{1}{2}\right) = 6.25$. A direct simulation of cars and chicken in the road frame agrees: it finds a safe start time at 1.01 times this speed and none at 0.99 times it.

Common slips

- Using the crossing time for the whole road, $b/(u\sin\theta)$, in the safety condition. Only the time spent in the lane of width a matters.
- Getting the sign of the relative velocity wrong: relative to the cars the chicken moves at $u\cos\theta - V$ along the road, not $u\cos\theta + V$.

Beyond STEP

Changing to the frame in which the obstacles are at rest is a Galilean transformation, and choosing the frame in which a problem is static is one of the most useful habits in

mechanics. The minimisation is also a small optimisation problem: the shortest vector from a point to a line is the perpendicular, which you will meet again in linear algebra as orthogonal projection. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems on relative motion in this spirit.

WORKED SOLUTION

STEP I 1997, Question 10

Mechanics • Circular Motion

STEP I 1997, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q10

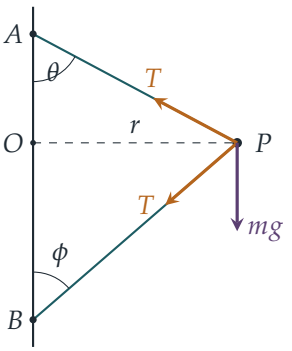
The key idea

Because the ring is smooth, the string runs freely through it, so the tension is the *same* on both sides of P. Resolve vertically (no vertical acceleration) and horizontally (acceleration $r\omega^2$ towards the axis), then divide one equation by the other so that T and m cancel. The sum-to-product formulae then turn the ratio into a single tangent of a half-angle.

Solution

The model and the forces

We treat P as a particle on a light inextensible string; the ring is smooth, so the tension T is the same in PA and PB. Let O be the centre of the circle, on the line AB with $OP = r$. The forces on P are its weight mg, a tension T along PA and a tension T along PB.



The string PA makes angle θ with the vertical (the angle at A between AB and AP), so its pull on P has components $T\cos\theta$ upwards and $T\sin\theta$ towards the axis. Similarly PB makes angle ϕ with the vertical at B, and its pull on P has components $T\cos\phi$

downwards and $T\sin\phi$ towards the axis.

The equations of motion

P moves in a horizontal circle at constant speed, so it has no vertical acceleration and an acceleration $r\omega^2$ towards O. Resolving,

$$\begin{aligned} \text{vertically:} \quad & T\cos\theta - T\cos\phi = mg, \\ \text{horizontally:} \quad & T\sin\theta + T\sin\phi = mr\omega^2. \end{aligned}$$

These hold whatever the shape of the triangle (if P were above A, say, $\cos\theta$ would simply be negative). Since $mg > 0$, the first equation gives $\cos\theta > \cos\phi$, so $\phi > \theta$: the lower string is the more steeply inclined to the axis, as the diagram suggests.

Showing the result

Dividing the vertical equation by the horizontal one,

$$\frac{\cos\theta - \cos\phi}{\sin\theta + \sin\phi} = \frac{g}{r\omega^2}.$$

By the sum-to-product formulae,

$$\begin{aligned} \cos\theta - \cos\phi &= 2\sin\tfrac{1}{2}(\theta+\phi)\,\sin\tfrac{1}{2}(\phi-\theta), \\ \sin\theta + \sin\phi &= 2\sin\tfrac{1}{2}(\theta+\phi)\,\cos\tfrac{1}{2}(\phi-\theta). \end{aligned}$$

Since $0 < \theta < \pi$ and $0 < \phi < \pi$ (they are angles of a genuine triangle APB), we have $0 < \frac{1}{2}(\theta+\phi) < \pi$, so $\sin\frac{1}{2}(\theta+\phi) \neq 0$ and it cancels:

$$\tan\frac{\phi-\theta}{2} = \frac{g}{r\omega^2},$$

as required.

The tension

Ask yourself The answer may contain $\sin\frac{1}{2}(\theta+\phi)$ but not $\phi-\theta$. How do we get rid of $\cos\frac{1}{2}(\phi-\theta)$?

From the horizontal equation and the second sum-to-product formula,

$$2T\sin\tfrac{1}{2}(\theta+\phi)\cos\tfrac{1}{2}(\phi-\theta) = mr\omega^2.$$

We know the *tangent* of $\alpha = \frac{1}{2}(\phi-\theta)$, so we can find its cosine from a right-angled triangle with opposite side g and adjacent side $r\omega^2$. We have $0 < \phi-\theta < \pi$, so $0 < \alpha < \frac{\pi}{2}$ and $\cos\alpha > 0$:

$$\cos\tfrac{1}{2}(\phi-\theta) = \frac{r\omega^2}{\sqrt{r^2\omega^4 + g^2}}.$$

Substituting and cancelling $r\omega^2$,

$$T = \frac{m\sqrt{r^2\omega^4 + g^2}}{2\sin\frac{1}{2}(\theta+\phi)}.$$

There is a nice way to see this: $\sqrt{(mg)^2 + (mr\omega^2)^2}$ is the size of the resultant of the two tensions, and two equal forces T at angle $\pi - (\theta + \phi)$ to each other (the angle APB) have a resultant of size $2T\sin\frac{1}{2}(\theta+\phi)$.

Checks

- As $\omega \to \infty$ the right-hand side tends to 0, so $\phi \to \theta$. The ring is flung out until it sits level with the midpoint of AB, as we'd expect when gravity is negligible.
- As $\omega \to 0$, $\tan\frac{1}{2}(\phi - \theta) \to \infty$, so $\phi - \theta \to \pi$: the string hangs straight down, which is the only possibility with no rotation.
- Numerically, we placed A, B and P at random, found T and ω^2 directly from the vector equation of motion, measured θ and ϕ from the coordinates, and compared with both results: the worst relative error over about 8000 cases was 3×10^{-12}.

Common slips

- Using two different tensions T_1, T_2. With a smooth ring there is one string and one tension; with two different tensions the problem is not determined.
- Getting the sign of the vertical component of the tension in PB wrong: that string pulls P *down* towards B.
- Taking the negative square root for $\cos\frac{1}{2}(\phi - \theta)$, which would give a negative tension.

Beyond STEP

For a given string length, the ring lies on an ellipse with foci A and B, and the string makes equal angles with the tangent at P: the reflection property of the ellipse. So the resultant of the tensions is along the normal to the ellipse, just like the reaction of a smooth wire bent into that shape. Thinking of constraints as forces normal to a surface is the starting point of Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems on rotating systems of this kind.

WORKED SOLUTION

STEP I 1997, Question 11

Mechanics • Motion with Resistance

STEP I 1997, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

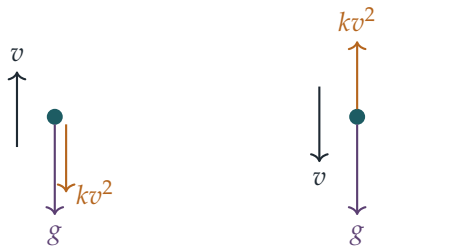
step.maths.org/questions/97-s1-q11

The key idea

The resistance depends on velocity, but we want velocity as a function of *distance*, not time. So we write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}s} = \frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{2}v^2\right)$. With $w = v^2$ as the unknown, the equation of motion becomes a *linear* first-order equation, which an integrating factor solves at once. The last part turns on one physical fact: resistance always opposes the motion, so it changes direction when the particle turns round.

Solution

We model the particle as a point of unit mass moving in a vertical line, acted on by its weight g (downwards) and a resistance of magnitude kv^2 directed against its velocity. Measure s upwards from the point of projection.



The upward motion

While the particle rises its velocity v is upwards, so the resistance kv^2 acts downwards, together with the weight. Newton's second law (mass 1) gives

$$v\frac{\mathrm{d}v}{\mathrm{d}s} = -g - kv^2,$$

using $\frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds}$. Put $w = v^2$, so that $\frac{dw}{ds} = 2v\frac{dv}{ds}$:

$$\frac{dw}{ds} + 2kw = -2g.$$

This is linear in w. Multiplying by the integrating factor e^{2ks},

$$\frac{d}{ds}\left(w\,e^{2ks}\right) = -2g\,e^{2ks} \quad \Longrightarrow \quad w\,e^{2ks} = -\frac{g}{k}\,e^{2ks} + C.$$

At $s = 0$, $w = u^2$, so $C = u^2 + \frac{g}{k}$. Hence

$$v^2 = \left(u^2 + \frac{g}{k}\right)e^{-2ks} - \frac{g}{k} = u^2e^{-2ks} + \frac{g}{k}\left(e^{-2ks} - 1\right),$$

which is $(*)$, as required. (Separating variables in $v\,dv/(g + kv^2) = -ds$ and getting a logarithm works just as well.)

Maximum height

The particle stops rising when $v = 0$; since the right-hand side of $(*)$ strictly decreases in s, this happens exactly once, at the greatest height H. Setting $v = 0$ in $(*)$:

$$\left(u^2 + \frac{g}{k}\right)e^{-2kH} = \frac{g}{k} \quad \Longrightarrow \quad e^{2kH} = 1 + \frac{ku^2}{g},$$

so

$$H = \frac{1}{2k}\ln\left(1 + \frac{ku^2}{g}\right).$$

The downward path

Ask yourself The derivation of $(*)$ never mentioned the direction of motion. Why shouldn't it carry on holding?

It did use the direction, in the sign of the resistance. On the way down the velocity is downwards, so the resistance kv^2 now acts *upwards*, against the weight. With s still measured upwards and v the upward velocity, the equation becomes

$$v\frac{dv}{ds} = -g + kv^2, \qquad \text{so} \qquad \frac{dw}{ds} - 2kw = -2g,$$

a different differential equation. So $(*)$ does not hold on the downward path.

To see this concretely, solve the new equation with $w = 0$ at $s = H$. The integrating factor e^{-2ks} gives $w = \frac{g}{k} + De^{2ks}$, and $w(H) = 0$ forces $D = -\frac{g}{k}e^{-2kH}$:

$$v^2 = \frac{g}{k}\left(1 - e^{-2k(H-s)}\right) \qquad (0 \le s \le H).$$

Using $u^2 + \frac{g}{k} = \frac{g}{k}e^{2kH}$, equation $(*)$ can be written as $v^2 = \frac{g}{k}\left(e^{2k(H-s)} - 1\right)$. Putting $x = 2k(H - s) \ge 0$, the two expressions are $\frac{g}{k}(1 - e^{-x})$ and $\frac{g}{k}(e^x - 1)$, and

$$(e^x - 1) - (1 - e^{-x}) = e^x + e^{-x} - 2 = \left(e^{x/2} - e^{-x/2}\right)^2 > 0 \quad \text{for } x > 0.$$

So $(*)$ gives the wrong speed at every point below the top (they agree only at $s = H$, where both are zero); it overestimates the speed on the way down.

There is also a quick physical argument. Equation $(*)$ gives v^2 as a function of height alone, so if it held on the way down the particle would pass each height at the same speed as on the way up, and return to the start with speed u. But the resistance does negative work throughout the motion, so the particle has lost energy and must return more slowly. Indeed, at $s = 0$ the correct formula gives

$$v^2 = \frac{g}{k}\left(1 - \frac{1}{1 + ku^2/g}\right) = \frac{u^2}{1 + ku^2/g} < u^2.$$

Checks

- As $k \to 0$, $\ln(1 + ku^2/g) \approx ku^2/g$, so $H \to u^2/(2g)$, the familiar no-resistance height; and $(*)$ tends to $v^2 = u^2 - 2gs$ (expand $e^{-2ks} \approx 1 - 2ks$).
- Numerically, with $u = 10$, $k = 0.05$, $g = 9.8$, stepping the equation of motion gives $H \approx 4.1224$, matching $\frac{1}{0.1}\ln(1 + \frac{5}{9.8}) = 4.12245$; the particle returns with $v^2 \approx 66.22$, matching $u^2/(1 + ku^2/g)$, not $u^2 = 100$. Halfway down, $v^2 \approx 36.51$ while $(*)$ would give 44.87.

Common slips

- Writing the resistance as $-kv^2$ in a single equation for the whole motion: since $v^2 \geq 0$ this term always points the same way, which is exactly why one formula can't cover both legs. (A single equation would need $-kv|v|$.)
- Using $\frac{dv}{dt}$ and solving for v in terms of t, then struggling to eliminate t; $v\frac{dv}{ds}$ goes straight to the answer.

Beyond STEP

Quadratic drag is a good model for a ball moving fast through air, and the downward equation shows a *terminal velocity*: as the fall lengthens, $v^2 \to g/k$. Writing the drag as $-kv|v|$ gives one nonlinear equation for the whole motion, and a lot of first-year mechanics is about such equations, where the force depends on velocity and energy is not conserved. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats velocity-dependent forces and terminal velocity carefully.

WORKED SOLUTION

STEP I 1997, Question 12

Probability and Statistics • Central Limit Theorem

STEP I 1997, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q12

The key idea

Solve the quadratic explicitly, so that X becomes a simple function of T. Then every question about X turns into a question about T, whose distribution we know completely. A neat simplification is that $1 - T$ is also uniform on $[0, 1]$. The last part is about the sum of many independent copies, which is a signal to use the central limit theorem.

Solution

The distribution of X

By the quadratic formula the roots of $x^2 + 2x + T = 0$ are $x = -1 \pm \sqrt{1 - T}$. These are real because $0 \leq T \leq 1$, and the larger root takes the plus sign:

$$X = -1 + \sqrt{1 - T}.$$

As T runs over $[0, 1]$, X runs over $[-1, 0]$.

Let $U = 1 - T$. Since T is uniform on $[0, 1]$, so is U: reflecting the interval about its midpoint doesn't change the uniform density. So $X = \sqrt{U} - 1$ with $U \sim \mathrm{U}[0, 1]$.

The probability that $X > -\frac{1}{3}$

Since $X + 1 = \sqrt{U}$ and both sides are non-negative,

$$X > -\tfrac{1}{3} \iff \sqrt{U} > \tfrac{2}{3} \iff U > \tfrac{4}{9}.$$

Squaring keeps the inequality's direction because both sides are non-negative. Hence

$$\mathrm{P}\left(X > -\tfrac{1}{3}\right) = \mathrm{P}\left(U > \tfrac{4}{9}\right) = 1 - \tfrac{4}{9} = \frac{5}{9}.$$

(Equivalently, $T < \frac{5}{9}$.)

Mean and variance

For $U \sim \mathrm{U}[0,1]$ the density is 1 on $[0,1]$, so

$$\mathrm{E}\big(\sqrt{U}\big) = \int_0^1 u^{1/2}\,\mathrm{d}u = \frac{2}{3}, \qquad \mathrm{E}\big((\sqrt{U})^2\big) = \mathrm{E}(U) = \frac{1}{2}.$$

Therefore

$$\mathrm{E}(X) = \mathrm{E}\big(\sqrt{U}\big) - 1 = \frac{2}{3} - 1 = -\frac{1}{3}.$$

Subtracting a constant doesn't change a variance, so

$$\mathrm{Var}(X) = \mathrm{Var}\big(\sqrt{U}\big) = \mathrm{E}(U) - \big(\mathrm{E}\sqrt{U}\big)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18},$$

as required.

Ask yourself Do we need the probability density function of X to find these moments?

No. For any function g, $\mathrm{E}\big(g(U)\big) = \int_0^1 g(u)\,\mathrm{d}u$, so we can work with U directly. If you'd like the density anyway: $\mathrm{P}(X \le x) = \mathrm{P}\big(U \le (x+1)^2\big) = (x+1)^2$ for $-1 \le x \le 0$, so X has density $2(x+1)$ on $[-1,0]$. Integrating $x \cdot 2(x+1)$ over that interval also gives $-\frac{1}{3}$.

The sum of 800 roots

The X_i are independent, because the experiments are, and each has mean $-\frac{1}{3}$ and variance $\frac{1}{18}$. So

$$\mathrm{E}(Y) = 800 \cdot \left(-\tfrac{1}{3}\right) = -\frac{800}{3}, \qquad \mathrm{Var}(Y) = 800 \cdot \frac{1}{18} = \frac{400}{9},$$

where adding the variances uses independence. The standard deviation of Y is $\frac{20}{3}$.

Since 800 is large, the central limit theorem says that Y is approximately $\mathrm{N}\left(-\frac{800}{3}, \frac{400}{9}\right)$. We want

$$\mathrm{P}(Y \le K) \approx \Phi\left(\frac{K + \frac{800}{3}}{20/3}\right) = 0.08.$$

From tables, $\Phi(1.405) \approx 0.92$, so by symmetry $\Phi(-1.405) \approx 0.08$. Hence

$$\frac{K + \frac{800}{3}}{20/3} \approx -1.405,$$

$$K \approx -\frac{800}{3} - 1.405 \times \frac{20}{3} \approx -266.67 - 9.37 \approx -276.$$

Ask yourself Shouldn't there be a continuity correction?

No. Each X_i is a continuous random variable, so Y is too, and no continuity correction is needed. The correction is only for approximating a discrete distribution, such as a binomial, by a normal one.

Checks

- The root is right: when $T = 0$ the equation is $x(x+2) = 0$, whose larger root is $0 = -1 + \sqrt{1}$.
- $\mathrm{E}(X) = -\frac{1}{3}$ lies in $[-1, 0]$, and the density $2(x+1)$ leans towards 0, so a mean above the midpoint $-\frac{1}{2}$ makes sense.
- A simulation with 400 000 values of T gave $\mathrm{P}(X > -\frac{1}{3}) \approx 0.555$, mean ≈ -0.3337 and variance ≈ 0.0556. Simulating Y directly 4000 times gave $\mathrm{P}(Y \leq -276.03) \approx 0.083$, which is consistent with 0.08.

Common slips

- Taking the minus sign in the quadratic formula. That gives the smaller root.
- Writing $\mathrm{Var}(Y) = 800^2 \mathrm{Var}(X)$. That is the variance of $800X_1$, not of a sum of 800 independent copies.
- Using $+1.405$ instead of -1.405. A probability of 0.08 is in the lower tail, so K must lie below the mean.

Beyond STEP

The technique of writing $X = g(U)$ for a uniform U runs backwards too. If F is any continuous distribution function, then $F^{-1}(U)$ has distribution function F. This *inverse transform method* is how computers simulate random variables from uniform random numbers. The central limit theorem, used here as a black box, is proved in a first probability course using moment generating or characteristic functions. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

WORKED SOLUTION

STEP I 1997, Question 13

Probability and Statistics • Conditional Probability

STEP I 1997, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q13

The key idea

Whether Mr Blond should switch on the light depends only on what kind of trap is waiting, so the whole question comes down to one conditional probability: the chance that the trap is an anaconda rather than a bomb, given that there is no canary. The identity of the assassin matters only because it affects which weapon is used. So we combine the two layers (assassin, then weapon) with the law of total probability, condition on "no canary", and then compare the two risks of death.

Solution

Setting up

Let M_i be the event that Mr i set the trap, for $i = 1, 2, 3, 4$, and let A, B and C be the events that the trap is an anaconda, a bomb or a canary. We are given

$$P(M_i) = \frac{i}{10}, \qquad P(A \mid M_i) = \frac{i+1}{10}, \quad P(B \mid M_i) = \frac{i}{10}, \quad P(C \mid M_i) = \frac{9-2i}{10}.$$

These are consistent: $\frac{1+2+3+4}{10} = 1$, and for each i the three weapon probabilities are non-negative and add to $\frac{(i+1)+i+(9-2i)}{10} = 1$. Exactly one assassin and exactly one weapon are involved, so the events $M_i \cap A$, $M_i \cap B$ and $M_i \cap C$ partition all the possibilities.

What the silence tells him

By the law of total probability,

$$P(A) = \sum_{i=1}^{4} P(M_i)P(A \mid M_i) = \frac{1\cdot 2 + 2\cdot 3 + 3\cdot 4 + 4\cdot 5}{100} = \frac{40}{100},$$

$$P(B) = \sum_{i=1}^{4} P(M_i)P(B \mid M_i) = \frac{1+4+9+16}{100} = \frac{30}{100}.$$

So $P(C') = P(A) + P(B) = \frac{70}{100}$. (As a check, $P(C) = \frac{7+10+9+4}{100} = \frac{30}{100}$.)

Ask yourself Do we need to work out which assassin is now most likely?

No. Hearing no singing does change the odds on each assassin, but the risk of death depends only on the weapon. Since $A \subseteq C'$ and $B \subseteq C'$, conditioning on C' simply rescales the two probabilities:

$$P(A \mid C') = \frac{P(A)}{P(C')} = \frac{40}{70} = \frac{4}{7}, \qquad P(B \mid C') = \frac{P(B)}{P(C')} = \frac{30}{70} = \frac{3}{7}.$$

The information about the assassins is already built into these two numbers.

Comparing the two choices

Let K be the event that Mr Blond is killed. Once we condition on C', the trap is an anaconda or a bomb, and his fate depends only on the trap and his choice.

Light on. A bomb kills him with probability $\frac{1}{2}$, and an anaconda kills him with probability $1 - \frac{2}{3} = \frac{1}{3}$. So

$$P(K \mid C', \text{ on}) = \frac{3}{7} \cdot \frac{1}{2} + \frac{4}{7} \cdot \frac{1}{3} = \frac{9}{42} + \frac{8}{42} = \frac{17}{42}.$$

Light off. A bomb never kills him, and an anaconda kills him with probability $\frac{1}{2}$. So

$$P(K \mid C', \text{ off}) = \frac{3}{7} \cdot 0 + \frac{4}{7} \cdot \frac{1}{2} = \frac{2}{7} = \frac{12}{42}.$$

Since $\frac{12}{42} < \frac{17}{42}$, **Mr Blond should enter the flat without switching on the light**. His chance of survival is then $\frac{5}{7}$ (about 71%), compared with $\frac{25}{42}$ (about 60%) if he switches it on.

The reason is that the anaconda is the more likely trap, but it is the bomb that the light really makes worse. Switching on the light lowers the anaconda risk from $\frac{1}{2}$ to $\frac{1}{3}$ but raises the bomb risk from 0 to $\frac{1}{2}$. With probability $\frac{4}{7}$ of an anaconda, the light saves $\frac{4}{7} \cdot \frac{1}{6} = \frac{4}{42}$; with probability $\frac{3}{7}$ of a bomb, it costs $\frac{3}{7} \cdot \frac{1}{2} = \frac{9}{42}$. The cost is larger.

Checks

- Before he listens, the risks would be $\frac{30}{100} \cdot \frac{1}{2} + \frac{40}{100} \cdot \frac{1}{3}$ for the light on and $\frac{40}{100} \cdot \frac{1}{2}$ for the light off. These are the answers above multiplied by $\frac{7}{10}$, as they should be, since the canary contributes nothing to either comparison.
- The light-off strategy is better whenever $p = P(A \mid C')$ satisfies $\frac{1}{2}p < \frac{1}{2}(1-p) + \frac{1}{3}p$, that is, whenever $p < \frac{3}{4}$. Here $p = \frac{4}{7}$, comfortably below that.
- A simulation of 10^6 evenings gives a death rate of about 0.404 with the light on and 0.286 with it off, matching $\frac{17}{42} \approx 0.405$ and $\frac{2}{7} \approx 0.286$.

Common slips

- Forgetting to condition on "no canary" and working with $P(A) = 0.4$ and $P(B) = 0.3$ as if they were the whole story. The advice comes out the same, but the probabilities of death are wrong.
- Mixing up survival and death: the anaconda figure with the light on is a $\frac{2}{3}$ chance of *survival*, so a $\frac{1}{3}$ chance of death.
- Treating the assassins as equally likely, or ignoring the assassin layer and averaging the weapon probabilities.

Beyond STEP

This is a small problem in *decision theory*: we choose the action that minimises expected loss under the posterior distribution, and here the "loss" is simply the probability of death. The observation that only $P(A \mid C')$ matters, and the threshold $p < \frac{3}{4}$ at which the best action switches, are typical of how such decisions are analysed: the data change the posterior, and the posterior decides the action. For more on conditional probability and Bayesian updating, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 1997, Question 14

Probability and Statistics • Minimising Expected Cost

STEP I 1997, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q14

The key idea

The cost T is a function of the random variable X, so its expectation is an integral of that function against the density. Splitting the integral at $x = y$ gives $\mathrm{E}(T)$ as an ordinary function of y, and then minimising it is a first-year calculus exercise: differentiate, set to zero, and check that the stationary point really is a minimum.

Solution

We take $y \geq 0$, since a negative amount of preparation makes no sense. Write $T = g(X)$, where

$$g(x) = \begin{cases} y & x \leq y, \\ y + r + s(x - y) & x > y. \end{cases}$$

Then

$$\begin{aligned} \mathrm{E}(T) &= \int_0^\infty g(x)\, \mathrm{e}^{-x}\, \mathrm{d}x \\ &= \int_0^\infty y\, \mathrm{e}^{-x}\, \mathrm{d}x + \int_y^\infty \big(r + s(x - y)\big) \mathrm{e}^{-x}\, \mathrm{d}x, \end{aligned}$$

where we have pulled out the cost y, which is paid whatever happens. The first integral is y, since the density integrates to 1.

For the second, substitute $u = x - y$, so $\mathrm{e}^{-x} = \mathrm{e}^{-y}\mathrm{e}^{-u}$ and u runs from 0 to ∞:

$$\begin{aligned} \int_y^\infty \big(r + s(x - y)\big) \mathrm{e}^{-x}\, \mathrm{d}x &= \mathrm{e}^{-y} \int_0^\infty (r + su)\, \mathrm{e}^{-u}\, \mathrm{d}u \\ &= \mathrm{e}^{-y} \Big(r \int_0^\infty \mathrm{e}^{-u}\, \mathrm{d}u + s \int_0^\infty u\, \mathrm{e}^{-u}\, \mathrm{d}u \Big) \\ &= \mathrm{e}^{-y}(r + s), \end{aligned}$$

using $\int_0^\infty u\, \mathrm{e}^{-u}\, \mathrm{d}u = \big[-u\mathrm{e}^{-u}\big]_0^\infty + \int_0^\infty \mathrm{e}^{-u}\, \mathrm{d}u = 1$ (integration by parts). Hence

$$C(y) = \mathrm{E}(T) = y + (r + s)\, \mathrm{e}^{-y}.$$

Ask yourself Is there a way to see the factor $e^{-y}(r+s)$ without integrating?

Yes. Flood damage happens with probability $P(X > y) = e^{-y}$. Given that it happens, the excess $X - y$ is again exponential with mean 1 (the exponential distribution is *memoryless*), so the expected damage is $r + s \cdot 1$. Multiplying gives $(r+s)e^{-y}$.

Now minimise. Differentiating,

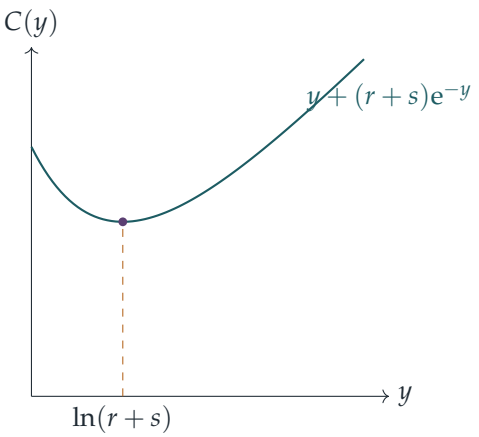
$$C'(y) = 1 - (r+s)e^{-y}, \qquad C''(y) = (r+s)e^{-y} > 0.$$

So $C'(y) = 0$ exactly when $e^y = r+s$, that is $y = \ln(r+s)$. Since $C'' > 0$ for all y, C is convex, C' is increasing, and so C decreases for $y < \ln(r+s)$ and increases for $y > \ln(r+s)$. The stationary point is therefore the global minimum, and the expected total cost is minimised by taking

$$y = \ln(r+s),$$

as required. The minimum expected cost is $C\big(\ln(r+s)\big) = \ln(r+s) + 1$.

This needs $\ln(r+s) \geq 0$, i.e. $r+s \geq 1$, which is implicitly assumed. If $r+s < 1$ then $C'(y) > 0$ for every $y \geq 0$, so the best choice is $y = 0$: the damage is so cheap that it isn't worth preparing at all.



(Sketch drawn with $r+s = 3$.)

Checks

- At $y = 0$ we always pay damage, and $E(T) = r + s\,E(X) = r+s$, which matches $C(0)$.
- With $r = 2$, $s = 3$ the formula gives $y = \ln 5 \approx 1.609$ and minimum cost $1 + \ln 5 \approx 2.609$; a fine grid search of $C(y)$ finds the same minimiser, and a simulation of 4×10^5 years at $y = 1$ gave a mean cost of 2.844 against the formula's $1 + 5e^{-1} \approx 2.839$.

- As $r+s$ grows, y increases: dearer floods justify more preparation, as common sense says.

Common slips

- Forgetting that the preparation cost y is paid in *every* year, not only when $X \leq y$.
- Integrating $sx\,\mathrm{e}^{-x}$ but losing the $-sy\,\mathrm{e}^{-x}$ term, which gives $\mathrm{e}^{-y}(r+s+sy)$ instead of $\mathrm{e}^{-y}(r+s)$.
- Finding the stationary point without showing it is a minimum.

Beyond STEP

This is a small example of decision-making under uncertainty: choose an action to minimise expected loss. The same structure, balancing a sure cost against an expected penalty, underlies the "newsvendor problem" in operations research and the idea of Bayes decisions in statistics. The memoryless property used above characterises the exponential distribution among continuous distributions; Grimmett and Welsh, *Probability: An Introduction*, covers it along with expectations of functions of random variables.

WORKED SOLUTION

STEP II 1997, Question 1

Pure • Digit Sums and Counting

STEP II 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q1

The key idea

Don't list the numbers: count them, then add up the contribution of each place value separately. Every allowed digit turns up equally often in each of the last three places, so the sums practically do themselves. Better still, there is a symmetry: replacing each digit d by $7-d$ swaps $0 \leftrightarrow 7$ and $2 \leftrightarrow 5$, so the numbers pair off with a common sum, and the factorisation drops straight out.

Solution

Which numbers qualify?

A qualifying number between 1000 and 6000 has four digits, and its thousands digit must be one of $0, 2, 5, 7$. It can't be 0 (that gives fewer than four digits) and it can't be 7 (that gives a number over 7000). So the thousands digit is 2 or 5, and every number from 2000 to 5777 built this way lies in range. The endpoints 1000 and 6000 themselves contain the digits 1 and 6, so it doesn't matter whether "between" includes them.

The other three digits can each be any of the four allowed digits, independently. So there are

$$2 \times 4 \times 4 \times 4 = 128$$

numbers.

Adding by place value

Thousands. Half of the 128 numbers, that is 64, start with 2 and the other 64 start with 5. The thousands digits contribute

$$64 \times (2+5) \times 1000 = 448\,000.$$

Hundreds, tens and units. Fix one of these places. Each of the four digits appears there in exactly $128/4 = 32$ of the numbers, because the remaining digits can be chosen in the

same number of ways whichever digit sits in that place. The digit total in that place is therefore $32 \times (0+2+5+7) = 32 \times 14 = 448$. The three places carry weights 100, 10 and 1, so together they contribute

$$448 \times (100+10+1) = 448 \times 111 = 49\,728.$$

Adding, the required sum is

$$S = 448 \times 1000 + 448 \times 111 = 448 \times 1111 = 497\,728.$$

As a product of primes

We have $448 = 64 \times 7 = 2^6 \times 7$, and $1111 = 11 \times 101$, where 101 is prime (it isn't divisible by 2, 3, 5 or 7, and $11^2 > 101$). Hence

$$S = 2^6 \times 7 \times 11 \times 101.$$

Ask yourself Is there a way to see the answer without any place-value bookkeeping?

Yes. If n qualifies, so does $7777 - n$: subtracting from 7777 digit by digit involves no borrowing, and it sends each digit d to $7 - d$, which maps $\{0, 2, 5, 7\}$ to itself and swaps the leading digits 2 and 5. So the 128 numbers split into 64 pairs, each adding to 7777 (no number is paired with itself, since 7777 is odd). Therefore

$$S = 64 \times 7777 = 2^6 \times 7 \times 1111 = 2^6 \times 7 \times 11 \times 101,$$

in agreement with the work above.

Checks

A short computer search over 1000 to 6000 finds exactly 128 qualifying numbers with sum 497 728, and $2^6 \times 7 \times 11 \times 101 = 64 \times 7777 = 497\,728$. As a sanity check on size, the average qualifying number is $7777/2 = 3888.5$, comfortably in the middle of the range 2000 to 5777.

Common slips

- Allowing 7 as the leading digit (giving numbers over 6000) or forgetting that 0 cannot lead.
- Saying each digit appears $128/4$ times in the thousands place too: only 2 and 5 appear there, 64 times each.
- Stopping at 497 728, or at 448×1111, without breaking 1111 into 11×101.

Beyond STEP

Counting how often each digit appears in each place is the simplest instance of a *double-counting* argument: summing the same quantity in two orders, here by number and by place. The $7777 - n$ pairing is a symmetry argument of the kind that also gives Gauss's formula $1 + 2 + \cdots + n = \frac{1}{2}n(n+1)$. Numbers whose digits come from a fixed set, and their sums, are a staple of elementary number theory and combinatorics; Davenport's *The Higher Arithmetic* is a lovely next step.

WORKED SOLUTION

STEP II 1997, Question 2

Pure • Recurrence Relations

STEP II 1997, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q2

The key idea

The chain of equalities is really a recurrence in disguise: each link says that $x_n + 2/x_{n+1} = 3$, so each term determines the next. Compute the first few terms as fractions and look hard at the numerators and denominators. Powers of 2 appear quickly, and once we have a guess, the recurrence gives exactly the step an induction needs.

Solution

Finding the pattern

Reading the chain one equality at a time gives

$$\frac{2}{x_1} = 3 \qquad \text{and} \qquad x_n + \frac{2}{x_{n+1}} = 3 \quad (n \geq 1),$$

so that

$$x_1 = \frac{2}{3}, \qquad x_{n+1} = \frac{2}{3 - x_n}.$$

(The recurrence only makes sense if $x_n \neq 3$; we shall see that every x_n lies between 0 and 1.) Working out the first few terms:

$$x_1 = \frac{2}{3}, \qquad x_2 = \frac{2}{3 - \frac{2}{3}} = \frac{6}{7}, \qquad x_3 = \frac{2}{3 - \frac{6}{7}} = \frac{14}{15}, \qquad x_4 = \frac{2}{3 - \frac{14}{15}} = \frac{30}{31}.$$

The denominators $3, 7, 15, 31$ are each one less than a power of 2, namely $2^{n+1} - 1$, and each numerator is one less than its denominator. So we guess

$$x_n = \frac{2^{n+1} - 2}{2^{n+1} - 1} = 1 - \frac{1}{2^{n+1} - 1}.$$

Proof by induction

Let $P(n)$ be the statement $x_n = \dfrac{2^{n+1} - 2}{2^{n+1} - 1}$.

Base case. For $n = 1$ the formula gives $\frac{4-2}{4-1} = \frac{2}{3} = x_1$, so $P(1)$ is true.

Inductive step. Suppose $P(k)$ is true for some $k \geq 1$. To keep the algebra tidy, write $N = 2^{k+1}$, so that $x_k = \dfrac{N-2}{N-1}$. Then

$$3 - x_k = \frac{3(N-1) - (N-2)}{N-1} = \frac{2N-1}{N-1},$$

which is not zero (indeed it is greater than 2, since $x_k < 1$), so we may divide by it:

$$x_{k+1} = \frac{2}{3 - x_k} = \frac{2(N-1)}{2N-1} = \frac{2N-2}{2N-1}.$$

? **Ask yourself** Why is this the formula with $n = k+1$?

Because $2N = 2^{k+2} = 2^{(k+1)+1}$. So

$$x_{k+1} = \frac{2^{(k+1)+1} - 2}{2^{(k+1)+1} - 1},$$

which is exactly $P(k+1)$.

Since $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for every $k \geq 1$, $P(n)$ is true for all $n \geq 1$, by induction. This also confirms that $0 < x_n < 1$ for every n, so no denominator $3 - x_n$ is ever zero.

An "otherwise" route: linearising

The guess can also be *derived*. A recurrence of the form $x_{n+1} = 2/(3 - x_n)$ becomes linear if we write $x_n = 2a_n/a_{n+1}$ for a new sequence (a_n) with non-zero terms. Substituting,

$$\frac{2a_{n+1}}{a_{n+2}} = \frac{2}{3 - \frac{2a_n}{a_{n+1}}} = \frac{2a_{n+1}}{3a_{n+1} - 2a_n}, \qquad \text{so} \qquad a_{n+2} = 3a_{n+1} - 2a_n.$$

The auxiliary equation $t^2 - 3t + 2 = 0$ has roots 1 and 2, so $a_n = A + B \cdot 2^n$. Taking $a_1 = 1$ (we are free to scale a) and $a_2 = 3$ (so that $x_1 = 2a_1/a_2 = \frac{2}{3}$) gives $a_n = 2^n - 1$, and then $x_n = \dfrac{2(2^n - 1)}{2^{n+1} - 1}$, the same formula.

Checks

- The formula with $n = 0$ gives $x_0 = 0$, and indeed the first equation $3 = 2/x_1$ is the general link $x_0 + 2/x_1 = 3$ with $x_0 = 0$. That's a nice consistency check.
- As $n \to \infty$, $x_n \to 1$. A limit L of the recurrence must satisfy $L = 2/(3 - L)$, that is $L^2 - 3L + 2 = 0$, so $L = 1$ or $L = 2$, and 1 is the one our sequence approaches from below.
- Iterating $x_{n+1} = 2/(3 - x_n)$ in exact fractions from $x_1 = \frac{2}{3}$ matched the formula for every n up to 40; for example $x_5 = \frac{62}{63}$.

Common slips

- Misreading the chain: the equation is $x_n + 2/x_{n+1} = 3$, not $x_{n+1} + 2/x_n = 3$. Getting this wrong gives a different sequence.
- Stopping after "the pattern continues". The question asks for a proof, and a few terms are only evidence.
- In the inductive step, forgetting to rewrite $2 \cdot 2^{k+1}$ as $2^{(k+1)+1}$, so the conclusion is never visibly the statement $P(k+1)$.

Beyond STEP

Recurrences of the form $x_{n+1} = (ax_n + b)/(cx_n + d)$ are iterations of a Möbius transformation, and they can always be linearised as in the "otherwise" route. Equivalently, we can track powers of the 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, whose eigenvalues (here 1 and 2) control the long-term behaviour. The fixed points 1 and 2 are attracting and repelling respectively, which is why the sequence settles at 1. For the matrix viewpoint, see Strang, *Introduction to Linear Algebra*; for the convergence of sequences in general, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1997, Question 3

Pure • Partial Fractions Integration

STEP II 1997, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q3

The key idea

The two quadratics in the denominators multiply to give $x^4 + 4$. That's the classic "difference of two squares in disguise": $x^4 + 4 = (x^2 + 2)^2 - (2x)^2$. Once we have the partial fractions, each piece is a standard integral. We split each numerator into a multiple of the derivative of its denominator, which integrates to a logarithm, plus a constant, which gives an arctangent after completing the square.

Solution

The partial fractions

First note that

$$(x^2 + 2x + 2)(x^2 - 2x + 2) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4.$$

So, multiplying the required identity through by $x^4 + 4$, we need

$$(ax + b)(x^2 - 2x + 2) + (cx + d)(x^2 + 2x + 2) = 1 \quad \text{for all } x.$$

Expanding the left-hand side gives

$$(a + c)x^3 + (b - 2a + d + 2c)x^2 + (2a - 2b + 2c + 2d)x + (2b + 2d).$$

Comparing coefficients with the constant polynomial 1:

$$\begin{aligned} x^3: &\quad a + c = 0, \\ x^2: &\quad b + d - 2a + 2c = 0, \\ x^1: &\quad 2(a + c) + 2(d - b) = 0, \\ x^0: &\quad 2(b + d) = 1. \end{aligned}$$

The first and third equations give $c = -a$ and $d = b$. The fourth then gives $b = d = \frac{1}{4}$. Finally, the second becomes $\frac{1}{2} - 4a = 0$, so $a = \frac{1}{8}$. Therefore

$$a = \tfrac{1}{8}, \qquad b = \tfrac{1}{4}, \qquad c = -\tfrac{1}{8}, \qquad d = \tfrac{1}{4},$$

that is,

$$\frac{1}{x^4+4} = \frac{1}{8}\left(\frac{x+2}{x^2+2x+2} + \frac{2-x}{x^2-2x+2}\right).$$

(We could have predicted $c = -a$ and $d = b$ without any algebra. Replacing x by $-x$ leaves $\frac{1}{x^4+4}$ unchanged and swaps the two denominators.)

The integral

Ask yourself How do we integrate $\dfrac{x+2}{x^2+2x+2}$?

The derivative of the denominator is $2x+2$. So we write the numerator as a multiple of $2x+2$ plus whatever is left over:

$$x+2 = \tfrac{1}{2}(2x+2)+1, \qquad 2-x = -\tfrac{1}{2}(2x-2)+1.$$

Completing the square, $x^2 \pm 2x + 2 = (x \pm 1)^2 + 1$. Hence

$$\frac{1}{x^4+4} = \frac{1}{16}\cdot\frac{2x+2}{x^2+2x+2} - \frac{1}{16}\cdot\frac{2x-2}{x^2-2x+2} + \frac{1}{8}\cdot\frac{1}{(x+1)^2+1} + \frac{1}{8}\cdot\frac{1}{(x-1)^2+1}.$$

Both quadratics are positive for all real x, so every term is continuous on $[0,1]$. The first two terms have the form f'/f with $f > 0$ and integrate to $\ln f$. The last two integrate to arctangents. So

$$\begin{aligned}
\int_0^1 \frac{\mathrm{d}x}{x^4+4} &= \frac{1}{16}\Big[\ln(x^2+2x+2) - \ln(x^2-2x+2)\Big]_0^1 + \frac{1}{8}\Big[\tan^{-1}(x+1) + \tan^{-1}(x-1)\Big]_0^1 \\
&= \frac{1}{16}\big[(\ln 5 - \ln 1) - (\ln 2 - \ln 2)\big] \\
&\quad + \frac{1}{8}\big[(\tan^{-1} 2 + \tan^{-1} 0) - (\tan^{-1} 1 + \tan^{-1}(-1))\big] \\
&= \frac{1}{16}\ln 5 + \frac{1}{8}\big[\tan^{-1} 2 - \big(\tfrac{\pi}{4} - \tfrac{\pi}{4}\big)\big] \\
&= \frac{1}{16}\ln 5 + \frac{1}{8}\tan^{-1} 2,
\end{aligned}$$

as required. The $\ln 2$ terms and the $\frac{\pi}{4}$ terms cancel in pairs, which is a pleasing consequence of the symmetry we noticed above.

Checks

- At $x = 0$ the partial fractions give $\frac{1}{8}\left(\frac{2}{2} + \frac{2}{2}\right) = \frac{1}{4} = \frac{1}{0+4}$. At $x = 1$ they give $\frac{1}{8}\left(\frac{3}{5} + \frac{1}{1}\right) = \frac{1}{5}$, which is also correct.
- The integrand decreases from $\frac{1}{4}$ to $\frac{1}{5}$ on $[0,1]$, so the integral must lie between 0.2 and 0.25. Numerically, $\frac{1}{16}\ln 5 + \frac{1}{8}\tan^{-1} 2 \approx 0.100589 + 0.138394 = 0.238983$. Simpson's rule on the integral gives 0.2389834593, in agreement to about 16 significant figures.

Common slips

- Trying to factorise $x^4 + 4$ over the reals by looking for real roots. It has none, so the factors must be quadratics. Adding and subtracting $4x^2$ to complete the square is the trick.
- Forgetting the lower limit of the arctangent terms. $\tan^{-1}(0+1) = \frac{\pi}{4}$ and $\tan^{-1}(0-1) = -\frac{\pi}{4}$ only cancel when both are included.

Beyond STEP

The factorisation $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$ is Sophie Germain's identity, $a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2)$. It shows, for example, that $n^4 + 4$ is prime only when $n = 1$. More generally, every real polynomial factorises into real linear and quadratic factors, because complex roots come in conjugate pairs. So partial fractions, and hence logs and arctangents, are enough to integrate any rational function. With contour integration you can evaluate $\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^4+4} = \frac{\pi}{4}$ almost without computation; Needham's *Visual Complex Analysis* explains why.

WORKED SOLUTION

STEP II 1997, Question 4

Pure • Remainder Theorem

STEP II 1997, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q4

The key idea

Division by a polynomial of degree d leaves a remainder of degree less than d, and the identity dividend $=$ divisor $\times$ quotient $+$ remainder holds for *every* x. So we substitute the values of x that make the divisor vanish: the quotient disappears and only the remainder is left. Everything in this question is that one trick, used three times.

Solution

The remainder theorem

Dividing $\mathrm{p}(x)$ by the linear polynomial $x-a$ gives a quotient $\mathrm{s}(x)$ and a remainder R whose degree is less than 1, so R is a constant:

$$\mathrm{p}(x) = (x-a)\,\mathrm{s}(x) + R \qquad \text{for all } x.$$

Since this is an identity in x, we may put $x = a$. The first term on the right becomes $0 \cdot \mathrm{s}(a) = 0$, leaving $\mathrm{p}(a) = R$. So the remainder is $\mathrm{p}(a)$, as required.

Part (i)

By the remainder theorem, the given remainders say

$$\mathrm{p}(1) = 3, \qquad \mathrm{p}(2) = 1, \qquad \mathrm{p}(3) = 5.$$

Putting $x = 1, 2, 3$ into $\mathrm{p}(x) = (x-1)(x-2)(x-3)\mathrm{q}(x) + \mathrm{r}(x)$ kills the first term each time, so

$$\mathrm{r}(1) = 3, \qquad \mathrm{r}(2) = 1, \qquad \mathrm{r}(3) = 5.$$

Ask yourself We know nothing about $\mathrm{q}(x)$. Is that a problem?

No: at $x = 1, 2, 3$ the factor $(x-1)(x-2)(x-3)$ is zero, so whatever q is, it contributes nothing. Three values pin down r because it has at most three coefficients.

A convenient way to write a polynomial of degree at most two is

$$\mathrm{r}(x) = A(x-1)(x-2) + B(x-1) + C,$$

because each substitution then removes terms one at a time:

$$\begin{aligned} x = 1: &\quad C = 3, \\ x = 2: &\quad B + C = 1 \;\Rightarrow\; B = -2, \\ x = 3: &\quad 2A + 2B + C = 5 \;\Rightarrow\; 2A - 4 + 3 = 5 \;\Rightarrow\; A = 3. \end{aligned}$$

Hence

$$\mathrm{r}(x) = 3(x-1)(x-2) - 2(x-1) + 3 = 3x^2 - 11x + 11.$$

Checking: $\mathrm{r}(1) = 3 - 11 + 11 = 3$, $\mathrm{r}(2) = 12 - 22 + 11 = 1$, $\mathrm{r}(3) = 27 - 33 + 11 = 5$.

Part (ii)

(The question writes $\mathrm{P}_n(x)$ in one place; it means the same polynomial $\mathrm{P}(x)$.) By the remainder theorem we need

$$\mathrm{P}(a) = a \qquad \text{for } a = 0, 1, \ldots, n.$$

The polynomial x already does this, but it has the wrong degree. So look at $\mathrm{D}(x) = \mathrm{P}(x) - x$ instead: we need $\mathrm{D}(a) = 0$ for $a = 0, 1, \ldots, n$, so by the factor theorem (the remainder theorem with remainder 0) each of $x, x-1, \ldots, x-n$ divides $\mathrm{D}(x)$. These $n+1$ linear factors are distinct, so their product divides $\mathrm{D}(x)$, and the product has degree $n+1$.

Ask yourself How do we make $\mathrm{P}(x)$ have degree exactly $n+1$?

Take $\mathrm{D}(x)$ to be that product itself (times a non-zero constant if we like). Adding x cannot cancel its leading term x^{n+1}, because $n+1 \geq 2$. So

$$\mathrm{P}(x) = x + x(x-1)(x-2)\cdots(x-n)$$

has degree $n+1$ and $\mathrm{P}(a) = a + 0 = a$ for each integer a with $0 \leq a \leq n$. This is a polynomial of the kind required.

In fact this argument finds them all: if P has degree $n+1$ and the stated remainders, then $\mathrm{D}(x) = \mathrm{P}(x) - x$ has degree at most $n+1$ and is divisible by the product of degree $n+1$, so $\mathrm{D}(x) = k\,x(x-1)\cdots(x-n)$ for a constant k, and $k \neq 0$ for the degree to be $n+1$. The general answer is

$$\mathrm{P}(x) = x + k\,x(x-1)(x-2)\cdots(x-n), \qquad k \neq 0.$$

Checks

- For $n = 1$, $\mathrm{P}(x) = x + x(x-1) = x^2$: dividing by x leaves 0 and dividing by $x-1$ leaves 1, as needed.

- In (i), take any q, say $\mathrm{q}(x) = x$, form p and divide by $x-1$, $x-2$, $x-3$ by synthetic division: the remainders come out as $3, 1, 5$.
- We checked by computer the remainder theorem on 1000 random polynomials, part (i) with 100 random choices of q, and part (ii) for $n = 1, \ldots, 11$ with $k = 1, -3, 7$; every remainder agreed exactly.

Common slips

- Forgetting that in part (ii) the polynomial $\mathrm{P}(x) = x$ itself satisfies the conditions but has degree 1, not $n+1$; the product term is what fixes the degree.
- Solving three simultaneous equations in the coefficients of $ax^2 + bx + c$ is fine but slower; the nested form $A(x-1)(x-2) + B(x-1) + C$ avoids the elimination.

Beyond STEP

Part (i) is an instance of polynomial interpolation: through any m points with distinct x-values passes exactly one polynomial of degree less than m, and the nested form we used is Newton's form of it. The same reasoning, in the language of rings, is the Chinese remainder theorem for polynomials: knowing a polynomial modulo $x-1$, $x-2$ and $x-3$ is the same as knowing it modulo their product. Allenby's *Rings, Fields and Groups* develops polynomial rings and this theorem.

WORKED SOLUTION

STEP II 1997, Question 5

Pure • Complex Exponential Mapping

STEP II 1997, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q5

This question has a full guided treatment in *Beyond STEP*, Section 5.7, *The Complex Exponential as a Map*: https://beyondstep.keithmcnulty.org/ch05/complex-exponential.html

The key idea

The relation is just $z = \mathrm{e}^{u}\mathrm{e}^{\mathrm{i}v} = \mathrm{e}^{w}$, written in modulus–argument form: u controls the modulus of z and v controls its argument. Once you see that, every part is about polar coordinates. The one subtlety is that the argument is only defined up to multiples of 2π, so the map is many-to-one, and to get a one-to-one correspondence we must cut the u–v plane down to a horizontal strip of the right height, taking care over which edges to include.

Solution

Finding w when $z = \mathrm{i}\,\mathrm{e}$

Since $\mathrm{e}^{u} > 0$ and $|\cos v + \mathrm{i}\sin v| = 1$, the relation says that

$$|z| = \mathrm{e}^{u}, \qquad \arg z = v \pmod{2\pi}.$$

For $z = \mathrm{i}\,\mathrm{e}$ we have $|z| = \mathrm{e}$ and $\arg z = \frac{\pi}{2}$. So $\mathrm{e}^{u} = \mathrm{e}$, giving $u = 1$ (the real exponential is one-to-one), and $\cos v + \mathrm{i}\sin v = \mathrm{i}$, so $\cos v = 0$ and $\sin v = 1$, giving $v = \frac{\pi}{2} + 2k\pi$. Hence

$$w = 1 + \mathrm{i}\left(\tfrac{\pi}{2} + 2k\pi\right), \qquad k \in \mathbb{Z}.$$

There are infinitely many such w, spaced 2π apart on the vertical line $u = 1$.

The images of the lines $u =$ constant

On the line $u = c$, the point z has modulus e^c and argument v. As v runs over all real numbers, $\cos v + \mathrm{i}\sin v$ runs round the unit circle (infinitely often), so z traces the

$$\text{circle } x^2 + y^2 = \mathrm{e}^{2c}, \quad \text{centre } O, \text{ radius } \mathrm{e}^c.$$

Larger c gives a larger circle; $c = 0$ gives the unit circle; as $c \to -\infty$ the circles shrink towards O but never reach it, since $\mathrm{e}^c > 0$.

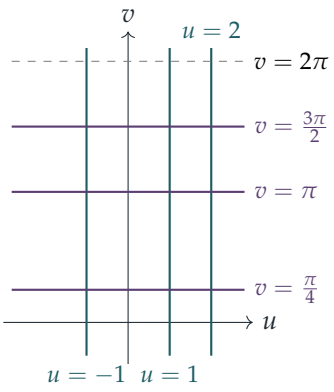
The images of the lines $v =$ constant

On the line $v = c$, we have $z = \mathrm{e}^u(\cos c + \mathrm{i}\sin c)$, a positive multiple of the fixed unit vector $(\cos c, \sin c)$. As u runs over $\mathbb{R}$, the multiple e^u takes every value in $(0, \infty)$ exactly once. So the image is the

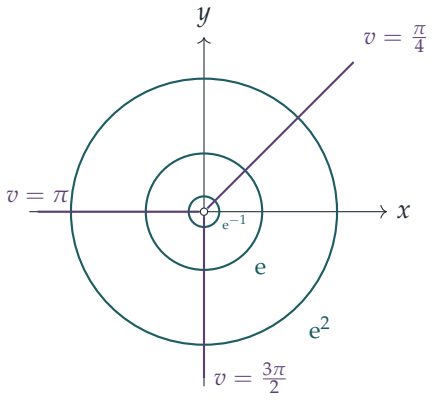
$$\text{half-line from } O \text{ at angle } c \text{ to the positive } x\text{-axis, with } O \text{ itself excluded.}$$

Lines $v = c$ and $v = c + 2k\pi$ give the same half-line. In Cartesian form, for example, $v = \frac{\pi}{4}$ gives $y = x$ with $x > 0$, and $v = \pi$ gives the negative x-axis.

The sketch shows the grid of lines in the u–v plane and its image: vertical lines become circles and horizontal lines become rays.



u–v plane



x–y plane

(The circles are not drawn to scale: the radii e^{-1}, e, e^2 grow very fast. The open circle marks O, which is never reached.)

Subsets in one-to-one correspondence with the first quadrant

A point z is in the open first quadrant exactly when $z \neq 0$ and it has an argument strictly between 0 and $\frac{\pi}{2}$. Take

$$W_1 = \left\{(u, v) : u \in \mathbb{R},\ 0 < v < \tfrac{\pi}{2}\right\}, \qquad W_2 = \left\{(u, v) : u \in \mathbb{R},\ 2\pi < v < \tfrac{5\pi}{2}\right\}.$$

Every point of W_1 maps into the first quadrant, since $e^u \cos v > 0$ and $e^u \sin v > 0$ there. Conversely, a point z of the first quadrant has $z = e^u(\cos v + i \sin v)$ exactly when $u = \ln|z|$ and $v = \theta + 2k\pi$, where $\theta \in (0, \frac{\pi}{2})$ is its principal argument. Precisely one of these values, $k = 0$, lies in $(0, \frac{\pi}{2})$, so z has exactly one preimage in W_1. Hence W_1 is in one-to-one correspondence with the quadrant. The same argument with $k = 1$ works for W_2, which is W_1 shifted up by 2π.

(Any strip $\{2k\pi < v < 2k\pi + \frac{\pi}{2}\}$ would do equally well.)

Subsets in one-to-one correspondence with $0 < |z| < 1$

Now we need $|z| = e^u < 1$, that is $u < 0$, and every argument must be hit exactly once, so v must run over an interval of length exactly 2π.

> **Ask yourself** Should the strip include its top edge, its bottom edge, both or neither?

Exactly one. If we included both, say $0 \le v \le 2\pi$, then $(u, 0)$ and $(u, 2\pi)$ would both map to the same point e^u on the positive real axis, so the correspondence would fail to be one-to-one. If we included neither, the positive real axis would not be reached at all. Including just one edge makes every argument appear exactly once. So take

$$W_3 = \{(u, v) : u < 0,\ 0 \le v < 2\pi\}, \qquad W_4 = \{(u, v) : u < 0,\ -\pi < v \le \pi\}.$$

For W_3: if $0 < |z| < 1$ then $u = \ln|z| < 0$ is forced, and among the arguments $\theta + 2k\pi$ of z exactly one lies in $[0, 2\pi)$, because that interval has length 2π and contains only one of its endpoints. So each such z has exactly one preimage in W_3, and each point of W_3 maps to a z with $0 < e^u < 1$. For W_4 the same argument works with the interval $(-\pi, \pi]$, which corresponds to the principal argument. Both are half-planes $u < 0$ cut down to half-open strips of height 2π.

Checks

- Put $u = 1$, $v = \frac{\pi}{2} + 2\pi$: $z = e(\cos\frac{5\pi}{2} + i\sin\frac{5\pi}{2}) = ie$, as required.
- Numerically, for thousands of random (u, v) we found $|z| = e^u$ and $\arg z \equiv v$ to within 10^{-9}. For random points in each target set, we found exactly one preimage in each of $W_1, \ldots, W_4$. And the points $(-1, 0)$ and $(-1, 2\pi)$ both give $z = e^{-1} \approx 0.3679$, which confirms that a closed strip would double-count.

Common slips

- Giving only $w = 1 + \frac{\pi}{2}i$ and forgetting the $+2k\pi$.
- Including the origin in the images: e^u is never 0, so the circles never shrink to a point and the rays start at an excluded point.
- Using a closed strip $0 \le v \le 2\pi$ (or an open one $0 < v < 2\pi$) for W_3: the first

double-counts the positive real axis and the second misses it.

Beyond STEP

The map here is the complex exponential $w \mapsto e^w$. It sends Cartesian grid lines to polar ones, and it is periodic with period $2\pi i$. Choosing a strip on which it is one-to-one is exactly choosing a *branch of the complex logarithm*, its inverse. The edge you leave out is the *branch cut*, which explains why $\log z$ cannot be defined continuously on all of $\mathbb{C} \setminus \{0\}$. Tristan Needham's *Visual Complex Analysis* has beautiful pictures of this mapping and of conformal maps in general.

WORKED SOLUTION

STEP II 1997, Question 6

Pure • Trigonometric Equations

STEP II 1997, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q6

The key idea

Write everything in terms of $t = \tan\theta$. The triple-angle formula turns the equation into a cubic in t, and once one easy root is spotted the remaining quadratic is exactly the one in the first part, so its roots are handled by a double-angle condition rather than by evaluating $\tan^{-1}(1 \pm \sqrt{2})$. The cotangent equation is the same equation in disguise: the substitution $\theta = \frac{\pi}{2} - \psi$ swaps cot for tan.

Solution

The opening result

Let $t = \tan\phi$ with $t^2 = 2t + 1$. Then $1 - t^2 = -2t$. Also $t \neq 0$, because $t = 0$ does not satisfy $t^2 = 2t + 1$, and $t^2 \neq 1$ (otherwise $2t = 0$). So $\tan 2\phi$ is defined and

$$\tan 2\phi = \frac{2t}{1 - t^2} = \frac{2t}{-2t} = -1,$$

as required.

The converse holds too, and we shall need it. If $\tan 2\phi = -1$ then $2t = -(1 - t^2)$, that is $t^2 = 2t + 1$. So

$$\tan^2\phi = 2\tan\phi + 1 \iff \tan 2\phi = -1.$$

The equation $\tan\theta = 2 + \tan 3\theta$

For a solution, both $\tan\theta$ and $\tan 3\theta$ must be defined. Put $t = \tan\theta$. Then

$$\tan 3\theta = \frac{3t - t^3}{1 - 3t^2},$$

and $1 - 3t^2 \neq 0$ because $\tan 3\theta$ is defined. Hence

$$\tan\theta - \tan 3\theta = \frac{t(1 - 3t^2) - (3t - t^3)}{1 - 3t^2} = \frac{-2t - 2t^3}{1 - 3t^2}.$$

Setting this equal to 2 and multiplying through by $1-3t^2$ (which is non-zero) gives

$$-t-t^3 = 1-3t^2 \quad\Longleftrightarrow\quad t^3-3t^2+t+1=0.$$

The coefficients add up to zero, so $t=1$ is a root, and

$$t^3-3t^2+t+1=(t-1)(t^2-2t-1).$$

Neither $t=1$ nor $t=1\pm\sqrt{2}$ satisfies $3t^2=1$, so every root gives a genuine solution.

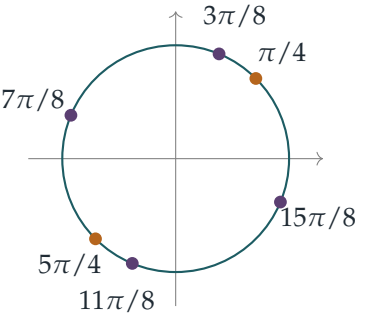
Ask yourself How do we turn $\tan\theta = 1\pm\sqrt{2}$ into rational multiples of π?

We use the first part. The roots of $t^2-2t-1=0$ are exactly the values of $\tan\theta$ with $t^2=2t+1$, and by the equivalence above these are exactly the θ with $\tan 2\theta=-1$.

- $\tan\theta=1$: $\theta=\frac{\pi}{4}+n\pi$. In $(0,2\pi)$ this gives $\theta=\frac{\pi}{4},\ \frac{5\pi}{4}$.
- $\tan 2\theta=-1$: $2\theta=\frac{3\pi}{4}+n\pi$, so $\theta=\frac{3\pi}{8}+\frac{n\pi}{2}$. Here 2θ ranges over $(0,4\pi)$, so $n=0,1,2,3$ and $\theta=\frac{3\pi}{8},\ \frac{7\pi}{8},\ \frac{11\pi}{8},\ \frac{15\pi}{8}$.

(For instance $\tan\frac{3\pi}{8}=1+\sqrt{2}$ and $\tan\frac{7\pi}{8}=-\tan\frac{\pi}{8}=1-\sqrt{2}$.) So the solutions in $0<\theta<2\pi$ are

$$\theta=\frac{\pi}{4},\ \frac{3\pi}{8},\ \frac{7\pi}{8},\ \frac{5\pi}{4},\ \frac{11\pi}{8},\ \frac{15\pi}{8}.$$



The amber points come from $\tan\theta=1$; the plum points, a quarter-turn apart, come from $\tan 2\theta=-1$.

The equation $\cot\theta = 2+\cot 3\theta$

Ask yourself Do we have to redo the whole calculation with cot?

No. Put $\theta=\frac{\pi}{2}-\psi$. Then $\cot\theta=\cot\left(\frac{\pi}{2}-\psi\right)=\tan\psi$. Also

$$\cot 3\theta=\cot\left(\frac{3\pi}{2}-3\psi\right)=\cot\left(\frac{\pi}{2}-3\psi\right)=\tan 3\psi,$$

using the fact that cot has period π. Each side is defined exactly when the corresponding tan is. So the equation becomes

$$\tan\psi = 2 + \tan 3\psi,$$

which is the previous equation. The range $-\frac{3\pi}{2} < \theta < \frac{\pi}{2}$ becomes $0 < \psi < 2\pi$, which is exactly the previous range. So ψ takes the six values found above, and $\theta = \frac{\pi}{2} - \psi$ gives

$$\theta = \frac{\pi}{4},\ \frac{\pi}{8},\ -\frac{3\pi}{8},\ -\frac{3\pi}{4},\ -\frac{7\pi}{8},\ -\frac{11\pi}{8}.$$

Checks

- At $\theta = \frac{\pi}{4}$: $\tan 3\theta = \tan\frac{3\pi}{4} = -1$, and $2 + (-1) = 1 = \tan\frac{\pi}{4}$. At $\theta = \frac{3\pi}{8}$: $\tan\frac{9\pi}{8} = \tan\frac{\pi}{8} = \sqrt{2} - 1$, and $2 + \sqrt{2} - 1 = 1 + \sqrt{2} = \tan\frac{3\pi}{8}$.
- A cubic in t has at most three roots, and tan has period π, so we expect at most $3 \times 2 = 6$ solutions in an interval of length 2π. We found six.
- Numerically, scanning $\tan x - \tan 3x - 2$ on a fine grid over $(0, 2\pi)$ finds sign changes (away from the poles) only at $x = \frac{2\pi}{8}, \frac{3\pi}{8}, \frac{7\pi}{8}, \frac{10\pi}{8}, \frac{11\pi}{8}, \frac{15\pi}{8}$. The same scan for the cotangent equation finds exactly the six values listed.

Common slips

- Stopping at $\tan 2\theta = -1$ with only $n = 0, 1$. Since θ runs over $(0, 2\pi)$, 2θ runs over $(0, 4\pi)$, and that gives four solutions, not two.
- Losing the root $t = 1$ by dividing by something without checking it, or forgetting $\frac{5\pi}{4}$.
- In the last part, solving from scratch and getting the range wrong. The substitution maps the given interval exactly onto $(0, 2\pi)$, and that is no accident.

Beyond STEP

The identity $\tan 3\theta = (3t - t^3)/(1 - 3t^2)$ is the real and imaginary parts of $(1 + \mathrm{i}t)^3$, and in general $\tan n\theta$ is a ratio of polynomials in $\tan\theta$ read off from $(1 + \mathrm{i}t)^n$. The roots of these polynomials are tangents of rational multiples of π, which is why their factorisations, like $t^2 - 2t - 1$ here, are so tidy. Needham's *Visual Complex Analysis* shows how multiple-angle formulae drop out of the geometry of complex multiplication.

WORKED SOLUTION

STEP II 1997, Question 7

Pure • Curve Sketching

STEP II 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q7

The key idea

The right-hand side depends on x only through x^2, so think of it as a quadratic in $u = x^2$ and complete the square: that gives the largest possible value of y^2 in one line, with no calculus. For the sketch, the equation is unchanged when $x \to -x$ or $y \to -y$, so we only need to understand the first quadrant and then reflect. The places worth pinning down are where the curve meets the axes, the slope at the origin, and the slope where the curve meets the x-axis away from the origin.

Solution

Maximum and minimum values

If $a = 0$ the equation says $y^2 = -x^4$, which forces $x = y = 0$: the "curve" is the single point $(0, 0)$, and the only value of y is 0. From now on take $a \neq 0$. Only a^2 appears, so we may as well think of $a > 0$ and write $|a|$ at the end.

Since $y^2 \geq 0$, we need $x^2(a^2 - x^2) \geq 0$, that is $x = 0$ or $x^2 \leq a^2$. So the curve lives in the strip $-|a| \leq x \leq |a|$. Completing the square in x^2,

$$y^2 = a^2x^2 - x^4 = \frac{a^4}{4} - \left(x^2 - \frac{a^2}{2}\right)^2 \leq \frac{a^4}{4},$$

with equality exactly when $x^2 = \frac{1}{2}a^2$. Hence $|y| \leq \frac{1}{2}a^2$ everywhere on the curve, and both $y = \frac{1}{2}a^2$ and $y = -\frac{1}{2}a^2$ are attained, at $x = \pm|a|/\sqrt{2}$. Therefore

$$\boxed{y_{\max} = \frac{a^2}{2}, \qquad y_{\min} = -\frac{a^2}{2}, \qquad \text{both at } x = \pm\frac{|a|}{\sqrt{2}}.}$$

? **Ask yourself** How does this square with calculus? Where is $\mathrm{d}y/\mathrm{d}x = 0$?

Differentiating implicitly, $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2a^2x - 4x^3 = 2x(a^2 - 2x^2)$. Where $y \neq 0$, the slope is zero exactly when $x^2 = \frac{1}{2}a^2$ (the root $x = 0$ has $y = 0$), and these are the four points found above. The points with $y = 0$ need separate care because we can't divide by y there; we deal with them next. Both methods agree, but completing the square proves at once that the stationary values are the overall maximum and minimum.

Shape of the curve

Take $a > 0$. Solving for y gives two branches,

$$y = x\sqrt{a^2 - x^2} \qquad \text{and} \qquad y = -x\sqrt{a^2 - x^2}, \qquad -a \leq x \leq a,$$

which are reflections of each other in the x-axis. On the first branch,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \sqrt{a^2 - x^2} - \frac{x^2}{\sqrt{a^2 - x^2}} = \frac{a^2 - 2x^2}{\sqrt{a^2 - x^2}}.$$

- **At the origin** the slope is a, and on the other branch it is $-a$. So the curve passes through the origin twice and crosses itself there, with tangents $y = \pm ax$.
- **At** $x = \pm a$ we have $y = 0$, and as $x \to a^-$ the slope tends to $-\infty$. So the curve meets the x-axis at $(\pm a, 0)$ with a *vertical* tangent; it's smooth and rounded there, not pointed.
- **In between**, the first branch rises from $(0, 0)$ to its maximum $\frac{1}{2}a^2$ at $x = a/\sqrt{2}$ and falls back to $(a, 0)$. The other quadrants follow by symmetry in both axes.

The result is a figure of eight lying on its side, with loops in $x \geq 0$ and $x \leq 0$, width $2|a|$ and total height a^2.

A tidy way to see the whole curve at once is to put $x = a\sin t$. Then $a^2 - x^2 = a^2\cos^2 t$, so $y^2 = a^4 \sin^2 t \cos^2 t$, and the curve is traced by

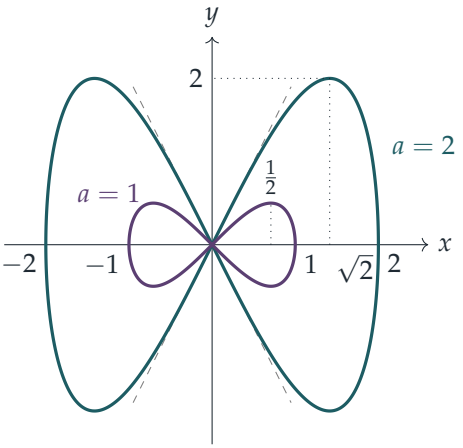
$$x = a\sin t, \qquad y = \tfrac{1}{2}a^2 \sin 2t, \qquad 0 \leq t < 2\pi,$$

which shows again that $|y| \leq \frac{1}{2}a^2$.

The cases $a = 1$ and $a = 2$

	meets x-axis at	tangents at O	extreme points
$a = 1$	$x = 0,\ \pm 1$	$y = \pm x$	$\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{2}\right)$
$a = 2$	$x = 0,\ \pm 2$	$y = \pm 2x$	$\left(\pm\sqrt{2}, \pm 2\right)$

To place the two curves relative to each other: for $0 < |x| \leq 1$ we have $x^2(1 - x^2) < x^2(4 - x^2)$, so the $a = 1$ curve lies strictly inside the $a = 2$ curve, and the only common point is the origin. For example, at $x = 1$ the $a = 1$ curve is at $y = 0$ while the $a = 2$ curve is at $y = \pm\sqrt{3}$.



The dashed lines are the tangents $y = \pm 2x$ for $a = 2$; for $a = 1$ the tangents at the origin are $y = \pm x$, at 45°.

Checks

- A grid search over $|x| \leq |a|$ for 50 random values of a found the largest $|y|$ to be $\frac{1}{2}a^2$ in every case, at $|x| = |a|/\sqrt{2} \approx 0.7071|a|$. For $a = 2$ it gives $y_{\max} = 2.000000$ at $|x| \approx 1.41421$.
- The maximum scales as a^2, as it must: replacing (x, y) by (ax, a^2y) turns the $a = 1$ curve into the general one. So the $a = 2$ curve is the $a = 1$ curve stretched by 2 horizontally and by 4 vertically, and that's why it's so much taller.

Common slips

- Drawing only $y = x\sqrt{a^2 - x^2}$. The equation defines y^2, so both signs of y belong to the graph, giving two loops rather than an S-shape.
- Drawing sharp corners at $(\pm a, 0)$. The slope there is infinite, so the curve crosses the x-axis vertically and smoothly.
- Forgetting that the curve crosses itself at the origin with slopes $\pm a$. These slopes are what tell the two curves apart near O.

Beyond STEP

The origin here is a *node*, a point where an algebraic curve crosses itself: near O the equation looks like $y^2 \approx a^2x^2$, the pair of lines $y = \pm ax$. Classifying singular points by their lowest-order terms like this is the first step in studying algebraic curves. The trick $x = a \sin t$ shows that this curve can be parametrised by trigonometric functions; curves of this kind are called Lissajous figures (here with frequencies in the ratio 1 : 2).

For more practice at sketching curves like this one, and at the STEP way of thinking generally, read Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1997, Question 8

Pure • Inequalities by Integration

STEP II 1997, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q8

The key idea

Each expression of the form $(x^p - 1)/p$ is an integral in disguise: it is $\int_1^x t^{p-1}\,\mathrm{d}t$. So comparing two such expressions comes down to comparing the integrands t^{p-1} and t^{q-1}, which is easy once we notice that the comparison flips at $t = 1$. The final inequality is then one more integration, this time of the inequality we have just proved.

Solution

Integrating an inequality

If $\mathrm{f}(t) \geq \mathrm{g}(t)$ on $[a, b]$, then $\mathrm{f} - \mathrm{g} \geq 0$ there, so the region between the graph of $\mathrm{f} - \mathrm{g}$ and the t-axis lies on or above the axis and its signed area is non-negative:

$$\int_a^b \mathrm{f}(t)\,\mathrm{d}t - \int_a^b \mathrm{g}(t)\,\mathrm{d}t = \int_a^b \big(\mathrm{f}(t) - \mathrm{g}(t)\big)\,\mathrm{d}t \geq 0.$$

Note that this needs $a \leq b$: if the limits are the wrong way round, the inequality between the integrals reverses.

The case $x \geq 1$

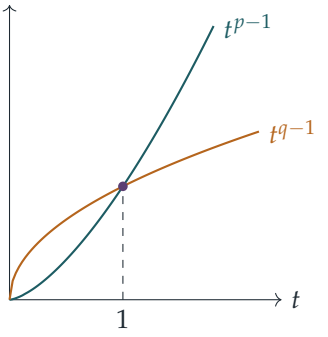
For $t \geq 1$ and $p > q$ we have $t^{p-1} = t^{q-1}\,t^{p-q} \geq t^{q-1}$, because $t^{p-q} \geq 1$ when $t \geq 1$ and $p - q > 0$. Since $x \geq 1$, the interval $[1, x]$ has its limits in the right order, so by the first part

$$\int_1^x t^{p-1}\,\mathrm{d}t \;\geq\; \int_1^x t^{q-1}\,\mathrm{d}t.$$

As $p, q > 0$, both integrals can be evaluated directly:

$$\int_1^x t^{p-1}\,\mathrm{d}t = \Big[\frac{t^p}{p}\Big]_1^x = \frac{x^p - 1}{p}, \qquad \int_1^x t^{q-1}\,\mathrm{d}t = \frac{x^q - 1}{q}.$$

Hence $\dfrac{x^p - 1}{p} \geq \dfrac{x^q - 1}{q}$ for $x \geq 1$, as required.



The sketch (drawn with $p = 2.5$, $q = 1.5$) shows the point: the larger power wins for $t > 1$ and loses for $0 < t < 1$.

The case $0 \le x \le 1$

Ask yourself For $0 < t < 1$ the inequality between t^{p-1} and t^{q-1} goes the other way. Why does the conclusion not reverse too?

Two reversals cancel. For $0 < t \le 1$ we have $t^{p-q} \le 1$, so $t^{p-1} \le t^{q-1}$. But now $x \le 1$, so the interval in the right order is $[x, 1]$, not $[1, x]$.

First take $0 < x \le 1$. Integrating $t^{q-1} \ge t^{p-1}$ over $[x, 1]$ (both are continuous there, as $x > 0$) gives

$$\int_x^1 t^{q-1}\,\mathrm{d}t \;\ge\; \int_x^1 t^{p-1}\,\mathrm{d}t, \qquad \text{that is,} \qquad \frac{1-x^q}{q} \;\ge\; \frac{1-x^p}{p}.$$

Multiplying by -1 reverses the inequality, giving $\dfrac{x^p-1}{p} \ge \dfrac{x^q-1}{q}$.

For $x = 0$ we check directly: the inequality reads $-\frac{1}{p} \ge -\frac{1}{q}$, which is true because $p > q > 0$ gives $\frac{1}{p} < \frac{1}{q}$. (We treat $x = 0$ separately because if $q < 1$ then t^{q-1} is unbounded near 0.) So the inequality holds for all $0 \le x \le 1$, and together with the previous case, for every $x \ge 0$.

The final inequality

Write $\phi_p(t) = \dfrac{t^p - 1}{p}$. We have now shown that $\phi_p(t) \ge \phi_q(t)$ for every $t \ge 0$ whenever $p > q > 0$. Both functions are continuous on $[0, \infty)$, so for $x > 0$ we may integrate this inequality over $[0, x]$:

$$\int_0^x \frac{t^p-1}{p}\,\mathrm{d}t \;\ge\; \int_0^x \frac{t^q-1}{q}\,\mathrm{d}t.$$

Now

$$\int_0^x \frac{t^p-1}{p}\,\mathrm{d}t = \frac{1}{p}\Big(\frac{x^{p+1}}{p+1} - x\Big) = \frac{x}{p}\Big(\frac{x^p}{p+1} - 1\Big),$$

and similarly for q. So

$$\frac{x}{p}\Big(\frac{x^p}{p+1}-1\Big) \;\geq\; \frac{x}{q}\Big(\frac{x^q}{q+1}-1\Big),$$

and dividing by $x > 0$ (which keeps the direction of the inequality) gives

$$\frac{1}{p}\Big(\frac{x^p}{p+1}-1\Big) \;\geq\; \frac{1}{q}\Big(\frac{x^q}{q+1}-1\Big).$$

At $x = 0$ this reads $-\frac{1}{p} \geq -\frac{1}{q}$, which we have already seen is true. This completes the proof for all $x \geq 0$.

Checks

- At $x = 1$ the middle inequality is $0 \geq 0$: the two curves cross there, consistent with equality.
- Take $p = 2$, $q = 1$, $x = 3$ in the final inequality: the left side is $\frac{1}{2}(3-1) = 1$ and the right side is $\frac{3}{2} - 1 = \frac{1}{2}$, so $1 \geq \frac{1}{2}$.
- A random test of 200,000 triples (p, q, x) with $0 < q < p$ and $x \geq 0$, including $x = 0$, $x = 1$ and tiny x, found no counterexample to either inequality.

Common slips

- Integrating over $[1, x]$ when $x < 1$ and keeping the inequality the same way round: the limits must be in increasing order before you integrate an inequality.
- Forgetting the end point $x = 0$, where t^{q-1} may not even be defined.
- Trying to prove the last part from scratch by calculus, rather than integrating the inequality you have just established.

Beyond STEP

As $q \to 0$, $(x^q - 1)/q \to \ln x$, so the result shows that $(x^p - 1)/p$ decreases to $\ln x$ as p decreases to 0: a neat family of bounds on the logarithm, which is also the basis of the Box–Cox transformation in statistics. The principle that integration preserves inequalities (monotonicity of the integral) is one of the first theorems proved about the Riemann integral in an analysis course. See Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1997, Question 9

Mechanics • Friction and Rigid-Body Equilibrium

STEP II 1997, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

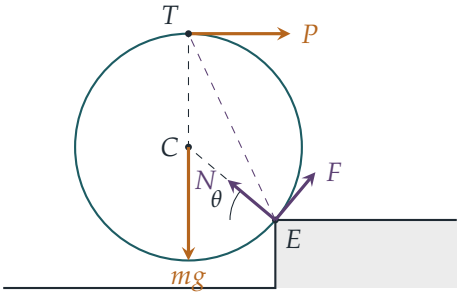
step.maths.org/questions/97-s2-q9

The key idea

Because the sphere is drawn *slowly*, it is in equilibrium at every instant, so in each position we have a statics problem. Once the sphere has left the ground only three forces act on it: the weight, the pull at the top and the reaction at the edge of the step. Taking moments about the edge gets rid of the unknown reaction and gives the pull. The three-force rule (the three lines of action must meet at a point) then tells us exactly which way the reaction points, and that gives the friction condition almost for free.

Solution

Let the radius be $r = \frac{1}{2}d$, so the step has height $\frac{1}{4}d = \frac{1}{2}r$. We model the sphere as a rigid uniform body touching the step only at the edge E (a horizontal line, which in the vertical cross-section through the centre is a point). Let C be the centre and T the highest point, where the horizontal force P acts towards the step. Let θ be the angle that EC makes with the horizontal.



The geometry

With E as origin, C is at height $r\sin\theta$ above E and a horizontal distance $r\cos\theta$ from it, and T is a further r above C. At the start, when the sphere rests on the ground, C is at

height r above the ground, so $\frac{1}{2}r$ above E; hence $\sin\theta = \frac{1}{2}$ and $\theta = \frac{\pi}{6}$. The sphere is fully on the step when C is vertically above E, i.e. $\theta = \frac{\pi}{2}$. So the movement corresponds to θ increasing from $\frac{\pi}{6}$ to $\frac{\pi}{2}$.

The maximum force

While the sphere still touches the ground, the ground carries part of the weight, and it can only push upwards. The sphere starts to rise when P is just large enough for the ground reaction to fall to zero. From then on the forces are P at T, the weight mg at C, and the reaction of the edge at E, with normal component N along EC and friction F perpendicular to EC.

Taking moments about E (the reaction has no moment there), the arm of P is the height of T above E, which is $r(1+\sin\theta)$, and the arm of the weight is $r\cos\theta$. Equilibrium gives

$$P\,r(1+\sin\theta) = mg\,r\cos\theta \quad\Longrightarrow\quad P = \frac{mg\cos\theta}{1+\sin\theta}.$$

On $\frac{\pi}{6} \le \theta \le \frac{\pi}{2}$ the numerator decreases and the denominator increases, so P is decreasing in θ. (Before lift-off, P only builds up to this value.) The maximum is therefore at $\theta = \frac{\pi}{6}$, the moment of leaving the ground:

$$P_{\max} = \frac{mg\cdot\frac{\sqrt{3}}{2}}{1+\frac{1}{2}} = \frac{mg}{\sqrt{3}}.$$

The friction condition

Ask yourself Which way does the reaction at the edge point?

Three non-parallel forces in equilibrium have concurrent lines of action. The line of P is the horizontal through T and the line of the weight is the vertical through C, which also passes through T. So they meet at T, and the total reaction at E must act along ET.

Now $CE = CT = r$, so triangle CET is isosceles. The angle at C between CT (vertically up) and CE (pointing down at angle θ below the horizontal) is $\frac{\pi}{2}+\theta$, so each base angle is

$$\angle CET = \tfrac{1}{2}\left(\pi - \tfrac{\pi}{2} - \theta\right) = \tfrac{\pi}{4} - \tfrac{\theta}{2}.$$

This is the angle between the total reaction and the normal EC, so

$$\frac{F}{N} = \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right).$$

We can confirm this by resolving. The reaction is $(-P,\ mg)$, since it balances P and the weight. Its component along the unit normal $(-\cos\theta, \sin\theta)$ and along the unit tangent

$(\sin\theta, \cos\theta)$ are

$$N = P\cos\theta + mg\sin\theta = mg\,\frac{\cos^2\theta + \sin\theta + \sin^2\theta}{1+\sin\theta} = mg,$$
$$F = mg\cos\theta - P\sin\theta = mg\cos\theta\left(1 - \frac{\sin\theta}{1+\sin\theta}\right) = \frac{mg\cos\theta}{1+\sin\theta},$$

so $F/N = \cos\theta/(1+\sin\theta)$, which equals $\tan(\frac{\pi}{4} - \frac{\theta}{2})$ by the half-angle formulae. Both F and N are positive throughout, so the edge really does push on the sphere and friction acts up the tangent, as drawn.

There is no slipping at E only if $F \le \mu N$ in every position, i.e.

$$\mu \ge \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \quad \text{for all } \tfrac{\pi}{6} \le \theta \le \tfrac{\pi}{2}.$$

The right-hand side decreases as θ increases, so the most demanding position is again $\theta = \frac{\pi}{6}$, where it equals $\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$. At that instant friction must be at least $1/\sqrt{3}$ of the normal reaction; with $\mu = 1/\sqrt{3}$ exactly, friction would be limiting and the sphere on the point of slipping, so to draw it up safely without slipping we need $\mu > 1/\sqrt{3}$, as required.

Checks

- At $\theta = \frac{\pi}{2}$ the centre is directly above the edge: $P = 0$ and $F = 0$, as they should be.
- $N = mg$ throughout, which makes sense: the reaction is $(-P, mg)$, and its component along the normal works out to exactly mg.
- Numerically, stepping θ from $\frac{\pi}{6}$ to $\frac{\pi}{2}$ with moments and resolution done separately, the largest value of P/mg and of F/N is $0.57735\ldots = 1/\sqrt{3}$, both at the start.

Common slips

- Taking the moment arm of P about E as $r\sin\theta$ (the height of C) instead of $r(1+\sin\theta)$ (the height of T).
- Assuming the reaction at the edge is purely normal. It cannot be: the normal passes through C, and the three forces would then fail to meet at a point.

Beyond STEP

The three-force argument is a special case of a general fact: a rigid body is in equilibrium exactly when the total force and the total moment about any one point are both zero, and for three forces this forces concurrency. The angle between the reaction and the normal, here $\frac{\pi}{4} - \frac{\theta}{2}$, is the "angle of friction" idea, which turns friction problems into geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body statics and dynamics from these principles.

WORKED SOLUTION

STEP II 1997, Question 10

Mechanics • Explosions and Elastic Impacts

STEP II 1997, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q10

The key idea

Split every velocity into a part along the axis and a part across it. The cylinder is smooth, so it can only push radially: it never changes the axial velocities, and an elastic bounce simply reverses the transverse velocity. Momentum conservation in the explosion then forces the two fragments to fly apart along a single diameter, and the whole problem becomes one-dimensional motion between two walls, with the axial motion going along for the ride.

Solution

Setting up

Model the fragments as particles (they are small), the cylinder as smooth and fixed, and ignore gravity as instructed. Take the z-axis along the cylinder's axis, and let the explosion happen at the origin at time $t = 0$.

Just after the explosion, write the velocity of the fragment of mass qM as an axial part w_1 plus a transverse part $\mathbf{u}_1$ (perpendicular to the axis), and similarly w_2, $\mathbf{u}_2$ for the fragment of mass $(1-q)M$. No external force acts during the explosion, so momentum is conserved:

$$qMw_1 + (1-q)Mw_2 = Mv, \qquad qM\mathbf{u}_1 + (1-q)M\mathbf{u}_2 = \mathbf{0}.$$

The second equation says the transverse velocities point in opposite directions, so both fragments move out from the axis along the same diameter, in opposite senses. Call that diameter the x-direction, with fragment 1 (mass qM) moving towards $x = +L$ at speed u_1 and fragment 2 towards $x = -L$ at speed u_2, where

$$qu_1 = (1-q)u_2. \tag{1}$$

Since $q \leq \frac{1}{2}$, we have $u_1 \geq u_2$: the lighter fragment is the faster.

Each fragment hits the wall head-on (it is moving along a radius), so the smooth wall's impulse is along that radius. That reverses the transverse velocity (the bounce is

perfectly elastic) and leaves the axial velocity alone. So each fragment keeps moving along the same diameter, and its axial velocity is constant throughout.

Showing that $q = \frac{3}{8}$

Axial motion. The axial positions are $z_1 = w_1 t$ and $z_2 = w_2 t$. The fragments meet at some time $t > 0$, so $w_1 = w_2$, and then the momentum equation gives $w_1 = w_2 = v$. Both fragments keep pace with where the original body would have been.

Transverse motion. Fragment 1 reaches $x = L$ at $t = L/u_1$ and then comes back, so after its bounce

$$x_1 = L - u_1\left(t - \frac{L}{u_1}\right) = 2L - u_1 t.$$

In the same way, after its bounce fragment 2 is at $x_2 = -2L + u_2 t$. They collide after both have bounced, so $x_1 = x_2$ gives

$$t = \frac{4L}{u_1 + u_2}, \qquad x = 2L - \frac{4Lu_1}{u_1 + u_2} = 2L\,\frac{u_2 - u_1}{u_1 + u_2}.$$

From (1), $u_2 = \frac{q}{1-q}u_1$, so $u_1 + u_2 = \frac{u_1}{1-q}$ and $u_2 - u_1 = \frac{(2q-1)u_1}{1-q}$. Hence the collision point is

$$x = 2L(2q - 1).$$

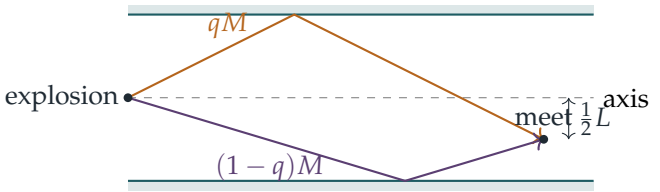
We need $|x| = \frac{1}{2}L$, so $2q - 1 = \pm\frac{1}{4}$, giving $q = \frac{5}{8}$ or $q = \frac{3}{8}$. Since $q \le \frac{1}{2}$,

$$q = \tfrac{3}{8},$$

as required. The collision is at $x = -\frac{1}{2}L$, on the heavier fragment's side of the axis.

Ask yourself Is this really the first meeting, after exactly one bounce each?

With $q = \frac{3}{8}$, (1) gives $u_2 = \frac{3}{5}u_1$. Fragment 1 bounces at $t = L/u_1$ and fragment 2 at $t = \frac{5}{3}L/u_1$, while the collision time is $4L/(u_1 + u_2) = \frac{5}{2}L/u_1$, which is later than both. Could they meet earlier? Before fragment 1 bounces they are moving apart, and on its way back fragment 1 would only reach fragment 2 at $t = 2L/(u_1 - u_2) = 5L/u_1$, long after fragment 2 has turned round. So the meeting we found is the first one, and each fragment has bounced exactly once.



The figure shows the plane containing the axis and the common diameter (axial direction across the page); it is drawn to scale for $q = \frac{3}{8}$ and the speeds found below.

The energy imparted

The collision time is

$$t = \frac{4L}{u_1 + u_2} = \frac{4L(1-q)}{u_1} = \frac{5L}{2u_1}.$$

Setting this equal to $5L/v$ gives $u_1 = \frac{1}{2}v$, and then $u_2 = \frac{3}{5}u_1 = \frac{3}{10}v$.

The axial velocities are unchanged at v, so the kinetic energy gained in the explosion is purely the transverse kinetic energy:

$$\begin{aligned}\Delta E &= \tfrac{1}{2}\cdot\tfrac{3}{8}M\left(\tfrac{1}{2}v\right)^2 + \tfrac{1}{2}\cdot\tfrac{5}{8}M\left(\tfrac{3}{10}v\right)^2 \\ &= \tfrac{1}{2}Mv^2\left(\tfrac{3}{32} + \tfrac{9}{160}\right) = \tfrac{1}{2}Mv^2\cdot\tfrac{3}{20} = \frac{3Mv^2}{40}.\end{aligned}$$

The velocity after coalescence

Just before the collision, each fragment's transverse velocity has been reversed by its bounce: fragment 1 has transverse momentum $-qMu_1$ and fragment 2 has $+(1-q)Mu_2$. By (1) these add to zero. The axial momentum is $qMv + (1-q)Mv = Mv$. Momentum is conserved in the coalescence, so the combined body of mass M has velocity

$$v \text{ along the direction of the axis,}$$

the same as the original body, but travelling on a line parallel to the axis at distance $\frac{1}{2}L$ from it. All $\frac{3}{40}Mv^2$ of energy from the explosion is lost in the coalescence.

Checks

- $q = \frac{1}{2}$ gives $x = 0$: equal fragments are mirror images and must meet on the axis, as they should.
- With $L = v = 1$, $u_1 = 0.5$ and $u_2 = 0.3$: fragment 1 is at $2 - 0.5\times 5 = -0.5$ and fragment 2 at $-2 + 0.3\times 5 = -0.5$ when $t = 5$. A time-stepped simulation with reflections at $x = \pm 1$ reproduced the meeting at $t = 5.000$, $x = -0.500$, after one bounce each.
- The final transverse momentum $-\frac{3}{8}\cdot\frac{1}{2} + \frac{5}{8}\cdot\frac{3}{10} = 0$, so the final velocity is purely axial.

Common slips

- Letting the wall change the axial velocity. A smooth wall exerts only a normal (radial) force.
- Forgetting that the bounce reverses the transverse momentum. The total transverse momentum only comes back to zero after *both* fragments have bounced.
- Keeping $q = \frac{5}{8}$: that is the same configuration with the fragments' labels swapped, ruled out by $q \le \frac{1}{2}$.

Beyond STEP

Separating motion into components along and across a symmetry direction is the beginning of a big idea: a symmetry of the system gives a conserved quantity. Here the cylinder looks the same all along its length, so axial momentum is conserved even with the walls present (Noether's theorem makes this precise). The question also shows that an explosion followed by a perfectly inelastic collision can give the energy back as heat while the axial motion carries on regardless. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops momentum and centre-of-mass motion in this style.

WORKED SOLUTION

STEP II 1997, Question 11

Mechanics • Projectile Motion

STEP II 1997, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

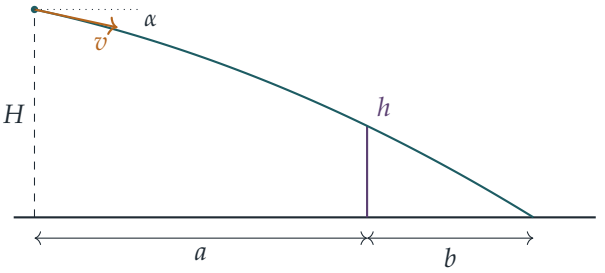
step.maths.org/questions/97-s2-q11

The key idea

Write down the trajectory as an equation for the drop below the server's height in terms of horizontal distance travelled. Then the two facts we are given (the ball is at height h when it has gone a, and at height 0 when it has gone $a+b$) are two equations in the two unknowns v and α. The bounds come from asking when the solutions of those equations actually make physical sense.

Solution

We model the ball as a particle moving under gravity alone (no air resistance or spin). Take the origin at the ground directly below the point where the ball is struck, with x horizontal towards the net and y vertically upwards.



The formula for v^2

Initially the velocity is $(v\cos\alpha, -v\sin\alpha)$. After time t,

$$x = vt\cos\alpha, \qquad y = H - vt\sin\alpha - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(v\cos\alpha)$ and using $\sec^2\alpha = 1+\tan^2\alpha$,

$$y = H - x\tan\alpha - \frac{gx^2(1+\tan^2\alpha)}{2v^2}. \qquad (*)$$

It is convenient to write $T = \tan\alpha$ and $k = \dfrac{g(1+T^2)}{2v^2}$, so that $y = H - xT - kx^2$.

The ball lands at $x = a + b$, so putting $y = 0$ there:

$$0 = H - (a+b)T - k(a+b)^2 \quad \Longrightarrow \quad k = \frac{H-(a+b)T}{(a+b)^2}. \tag{1}$$

Replacing k by its definition and rearranging for v^2 gives

$$v^2 = \frac{g(a+b)^2(1+\tan^2\alpha)}{2[H-(a+b)\tan\alpha]},$$

which is the first result.

The formula for $\tan\alpha$

The ball just clears the net, so $y = h$ at $x = a$. From $(*)$,

$$H - h = aT + ka^2.$$

Substituting k from (1) and multiplying through by $(a+b)^2$:

$$\begin{aligned}(H-h)(a+b)^2 &= aT(a+b)^2 + a^2\big[H-(a+b)T\big] \\ &= aT(a+b)\big[(a+b)-a\big] + a^2H \\ &= abT(a+b) + a^2H.\end{aligned}$$

Collecting the H terms, and using $(a+b)^2 - a^2 = b(2a+b)$,

$$ab(a+b)T = b(2a+b)H - (a+b)^2h.$$

Dividing by $ab(a+b)$, which is positive,

$$\tan\alpha = \frac{2a+b}{a(a+b)}\,H - \frac{a+b}{ab}\,h,$$

as required.

Bounds on H

Given a, b, h and H, the second formula fixes $\tan\alpha$ and then the first fixes v^2. Such a serve exists exactly when these values are admissible: $\tan\alpha > 0$ (the ball is hit *downwards*, so $0 < \alpha < \frac{\pi}{2}$) and $v^2 > 0$.

> **Ask yourself** The numerator of v^2 is clearly positive. So what does $v^2 > 0$ actually require?

It requires the denominator to be positive: $H - (a+b)\tan\alpha > 0$. Substituting the formula for $\tan\alpha$,

$$\begin{aligned}H - (a+b)\tan\alpha &= H - \frac{2a+b}{a}\,H + \frac{(a+b)^2}{ab}\,h \\ &= -\frac{a+b}{a}\,H + \frac{(a+b)^2}{ab}\,h = \frac{a+b}{a}\left[\frac{(a+b)h}{b} - H\right].\end{aligned}$$

Since $(a+b)/a > 0$, this is positive if and only if

$$H < \frac{(a+b)h}{b}.$$

For $\tan\alpha > 0$ we need

$$\frac{2a+b}{a(a+b)}H > \frac{a+b}{ab}h \quad\Longleftrightarrow\quad H > \frac{(a+b)^2 h}{b(2a+b)},$$

multiplying by the positive quantity $a(a+b)/(2a+b)$.

Both conditions together give

$$\boxed{\frac{(a+b)^2}{b(2a+b)}h \;<\; H \;<\; \frac{a+b}{b}h}$$

and conversely any H in this range gives a valid α and v. The interval is non-empty because $\dfrac{a+b}{2a+b} < 1$.

Both bounds have a clear meaning. The upper bound is the height at which the straight line from the landing point through the top of the net meets the server's vertical: gravity bends the path downwards, so the ball must start below that line (at $H = (a+b)h/b$ we would need infinite speed). The lower bound is the case $\alpha = 0$, a horizontal serve; serving from lower than this, the ball would have to be hit upwards to clear the net and still land at the given point.

Checks

- With $a = 12\,\text{m}$, $b = 6\,\text{m}$ and $h = 1\,\text{m}$ the bounds are $1.8\,\text{m} < H < 3\,\text{m}$. At $H = 1.8$ the formula gives $\tan\alpha = \frac{30}{216}(1.8) - \frac{18}{72} = 0$, as expected for the lower bound.
- Taking random a, b, h and H within the bounds, computing α and v from the two formulas and stepping the motion numerically, the ball passed through the top of the net and landed at $x = a+b$ every time (relative error below 10^{-14} for the exact path).
- Dimensions: each term of $\tan\alpha$ is a length divided by a length, and v^2 has dimensions of $g\times$ length, as it should.

Common slips

- Sign errors from the downward angle: the initial vertical velocity is $-v\sin\alpha$, and the $x\tan\alpha$ term in $(*)$ is subtracted.
- Stopping at "$v^2 > 0$" without noticing that it is a condition on the denominator, which only becomes a condition on H after substituting for $\tan\alpha$.

Beyond STEP

Here two boundary conditions (the net and the landing point) determine the launch conditions, rather than the other way round: a simple *boundary-value problem*, and the idea behind "shooting methods" for solving differential equations numerically. With air resistance and spin the equations no longer have closed-form solutions and this is exactly how one would proceed. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles with drag and much more.

WORKED SOLUTION

STEP II 1997, Question 12

Probability and Statistics • Binomial Parity Probability

STEP II 1997, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q12

The key idea

Track only whether the message is currently *right* or *wrong*. Each mishearing flips it, so the message that comes back to Albert is right exactly when an even number of the whispers went wrong. Picking out the even terms of a binomial expansion is what $(a+b)^n + (a-b)^n$ does, since the odd terms cancel. For the modified game, notice that after one failure Albert passes on a *wrong* message, so from then on each round needs an *odd* number of mishearings.

Solution

Three players

With n players there are n whispers: Albert to Bertha, Bertha to the next player, and so on round to Zebedee, then Zebedee back to Albert. Each whisper is heard correctly with probability p, independently. Whether Albert hears the truth depends only on how many whispers were misheard, and not on what the coin showed.

With three players there are three whispers, each correct with probability $\frac{2}{3}$. The message that reaches Albert is right if no whisper goes wrong, or if exactly two do (the second error undoes the first). So

$$\Pr(\text{won}) = \left(\tfrac{2}{3}\right)^3 + 3\left(\tfrac{1}{3}\right)^2\left(\tfrac{2}{3}\right) = \frac{8}{27} + \frac{6}{27} = \frac{14}{27}.$$

n players

Let $q = 1 - p$. The number of misheard whispers has the $\mathrm{B}(n, q)$ distribution, and the game is won when it is even:

$$P = \sum_{k \text{ even}} \binom{n}{k} q^k p^{n-k}.$$

Now

$$(p+q)^n + (p-q)^n = \sum_{k=0}^{n} \binom{n}{k} p^{n-k} q^k \left(1 + (-1)^k\right) = 2 \sum_{k \text{ even}} \binom{n}{k} p^{n-k} q^k,$$

because the odd-k terms cancel and the even-k terms appear twice. Since $p+q=1$ and $p-q = 2p-1$,

$$\boxed{P = \tfrac{1}{2}\left(1 + (2p-1)^n\right).}$$

For $n=3$, $p=\frac{2}{3}$ this gives $\frac{1}{2}\left(1+\frac{1}{27}\right) = \frac{14}{27}$, agreeing with the first part.

The modified game

Let N be the number of times Albert whispers to Bertha. Call each pass of the message all the way round the circle a *round*, so every round begins with Albert whispering.

> **Ask yourself** After a failed round, is the next round just a fresh copy of the first?

No. In the first round Albert sends the correct result. If he hears it wrongly from Zebedee, he passes on what he *heard*, which is wrong. Starting from a wrong message, the round ends with Albert hearing the truth exactly when an *odd* number of whispers go wrong. That has probability $1-P$. If the round fails again, Albert has once more heard (and passes on) a wrong message, so every later round is like this one. The rounds are independent, because the hearings are.

So the first round succeeds with probability P, and each later round succeeds with probability $1-P$ and fails with probability P. Hence $\Pr(N=1) = P$, and for $k \geq 2$

$$\Pr(N=k) = \underbrace{(1-P)}_{\text{round 1 fails}} \underbrace{P^{k-2}}_{\text{rounds 2,...,}k-1\text{ fail}} \underbrace{(1-P)}_{\text{round } k \text{ wins}} = (1-P)^2 P^{k-2}.$$

Assume $0 < p < 1$, so that $0 < P < 1$. Then, using the given sum with $x = P$,

$$\sum_{k=2}^{\infty} k P^{k-2} = \frac{1}{P}\left(\sum_{k=1}^{\infty} k P^{k-1} - 1\right) = \frac{1}{P}\left(\frac{1}{(1-P)^2} - 1\right),$$

and so

$$\begin{aligned} \mathrm{E}(N) &= P + (1-P)^2 \cdot \frac{1}{P}\left(\frac{1}{(1-P)^2} - 1\right) \\ &= P + \frac{1-(1-P)^2}{P} \\ &= P + (2-P) = 2. \end{aligned}$$

The expected number of times Albert whispers to Bertha is 2, whatever the number of players and whatever p is.

There is a quicker way to see this. Given that the first round fails, the number of further rounds is geometric with success probability $1-P$, so its mean is $1/(1-P)$. Hence $\mathrm{E}(N) = 1 + (1-P)\cdot\frac{1}{1-P} = 2$.

(At the edges: if $p = 1$, or if $p = 0$ with n even, then $P = 1$ and Albert whispers just once. If $p = 0$ with n odd, then $P = 0$ and the second round is certain to succeed, so $N = 2$.)

Checks

- Listing all 2^n patterns of right and wrong hearings with exact fractions reproduces $\frac{1}{2}(1 + (2p-1)^n)$ for $n \leq 8$ and several values of p, including $\Pr(\text{won}) = \frac{14}{27}$ for three players.
- $p = \frac{1}{2}$ gives $P = \frac{1}{2}$: the message is pure noise, as it should be.
- Simulating 2×10^5 modified games with $(n, p) = (3, \frac{2}{3})$, $(5, 0.9)$ and $(4, 0.3)$ gave average whisper counts 2.001, 1.996 and 1.994.

Common slips

- Counting $n - 1$ whispers instead of n: the last whisper, from Zebedee back to Albert, counts too.
- Treating every round of the modified game as winning with probability P. That gives $1/P$, which is wrong, because after a failure Albert passes on a wrong message.

Beyond STEP

The right/wrong state of the message is a two-state Markov chain, and $2p - 1$ is the second eigenvalue of its transition matrix. That is why $(2p-1)^n$ controls how fast the message forgets the coin. The same model, the binary symmetric channel, is the starting point of information theory and error-correcting codes. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and Markov chains at this level.

WORKED SOLUTION

STEP II 1997, Question 13

Probability and Statistics • Buffon's Needle

STEP II 1997, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

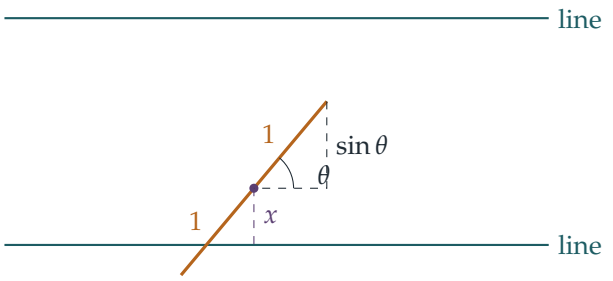
step.maths.org/questions/97-s2-q13

The key idea

This is Buffon's needle. Describe the needle's position by two independent random quantities: the distance of its centre from the nearest line, and the acute angle it makes with the lines. Once the centre is fixed, everything hinges on how far the needle reaches perpendicular to the lines, which is its half-length times the sine of that angle. Each part is then a question about a uniformly distributed angle.

Solution

We model "dropped at random" in the usual way: the distance X of the centre from the nearest line is uniform on $[0, 1]$ (the lines are 2 cm apart, so the nearest one is at most 1 cm away), and the acute angle θ between the needle and the lines is uniform on $[0, \frac{\pi}{2}]$, independently of X. (Taking the angle uniform on $[0, \pi)$ gives the same answers, since $\sin\theta$ has the same distribution either way.)



Each half of the needle has length 1, so the ends are at perpendicular distance $\sin\theta$ above and below the centre.

Part (i)

With the centre x from the nearest line, the needle crosses that line exactly when one end reaches past it, that is when

$$\sin\theta > x \iff \theta > \arcsin x,$$

since sin is increasing on $[0, \frac{\pi}{2}]$. As θ is uniform on $[0, \frac{\pi}{2}]$,

$$\mathrm{P}(\text{crosses nearest line} \mid X = x) = \frac{\frac{\pi}{2} - \arcsin x}{\frac{\pi}{2}} = 1 - \frac{2}{\pi}\arcsin x = \frac{2}{\pi}\arccos x.$$

Part (ii)

Suppose the needle crosses the nearest line, so $\theta > \arcsin x$. Measured along the needle, the crossing point is at distance $x/\sin\theta$ from the centre (the hypotenuse of a right-angled triangle with side x opposite θ). So the two pieces have lengths $1 + x/\sin\theta$ and $1 - x/\sin\theta$, and the shorter one is

$$S = 1 - \frac{x}{\sin\theta}.$$

As θ runs from $\arcsin x$ to $\frac{\pi}{2}$, S increases from 0 to $1 - x$, so S takes values in $[0, 1 - x]$. Conditioning on crossing, θ is uniform on $(\arcsin x, \frac{\pi}{2}]$, an interval of length $\frac{\pi}{2} - \arcsin x = \arccos x$. For $0 \le s \le 1 - x$,

$$S \le s \iff \frac{x}{\sin\theta} \ge 1 - s \iff \sin\theta \le \frac{x}{1-s} \iff \theta \le \arcsin\frac{x}{1-s},$$

where we used $1 - s > 0$ and $x/(1-s) \le 1$ (because $s \le 1 - x$), so the arcsine makes sense. Hence the cumulative distribution function of the shorter segment is

$$F(s) = \mathrm{P}(S \le s) = \begin{cases} 0, & s < 0, \\ \dfrac{\arcsin\left(\frac{x}{1-s}\right) - \arcsin x}{\arccos x}, & 0 \le s \le 1 - x, \\ 1, & s > 1 - x. \end{cases}$$

As a sanity check, the middle formula gives 0 at $s = 0$ and $(\frac{\pi}{2} - \arcsin x)/\arccos x = 1$ at $s = 1 - x$, so F is continuous.

Ask yourself Why is θ still uniform once we are told that the needle crosses the line?

Conditioning a uniform variable on lying in a sub-interval leaves it uniform on that sub-interval: for $\arcsin x \le a \le \frac{\pi}{2}$, $\mathrm{P}(\theta \le a \mid \theta > \arcsin x) = (a - \arcsin x)/(\frac{\pi}{2} - \arcsin x)$, which is linear in a. That is exactly what we used.

Part (iii)

The nearest line is at distance $X \le 1$, and every other line is at distance at least $2 - X \ge 1$. The needle reaches at most $\sin\theta \le 1$ from its centre, so it can cross a line other than the

nearest one only if $X = 1$ and $\theta = \frac{\pi}{2}$ exactly, which has probability zero. So the needle misses all the lines exactly when it misses the nearest line, and by part (i)

$$\mathrm{P}(\text{miss} \mid X = x) = \frac{2}{\pi} \arcsin x.$$

Since X is uniform on $[0, 1]$ (density 1), we average over x:

$$\mathrm{P}(\text{miss}) = \int_0^1 \frac{2}{\pi} \arcsin x \, \mathrm{d}x.$$

Integrating by parts with $u = \arcsin x$ and $\mathrm{d}v = \mathrm{d}x$,

$$\begin{aligned} \int_0^1 \arcsin x \, \mathrm{d}x &= \Big[x \arcsin x\Big]_0^1 - \int_0^1 \frac{x}{\sqrt{1 - x^2}} \, \mathrm{d}x \\ &= \frac{\pi}{2} - \Big[-\sqrt{1 - x^2}\Big]_0^1 = \frac{\pi}{2} - 1. \end{aligned}$$

Therefore

$$\mathrm{P}(\text{needle misses all the lines}) = \frac{2}{\pi}\left(\frac{\pi}{2} - 1\right) = 1 - \frac{2}{\pi} \approx 0.363.$$

Checks

- Part (i) behaves sensibly at the ends: at $x = 0$ the centre is on the line and the probability is 1; as $x \to 1$ only a perpendicular needle reaches, and the probability tends to 0.
- The probability of crossing is $2/\pi \approx 0.637$, Buffon's classical answer $2\ell/(\pi d)$ with $\ell = d = 2$.
- A simulation of 10^6 random drops gave a miss rate of 0.3626, against $1 - 2/\pi = 0.3634$; simulated values of $F(s)$ for $x = 0.2, 0.5, 0.7$ agreed with the formula to about 0.002.

Common slips

- Forgetting to condition in part (ii): dividing by $\frac{\pi}{2}$ instead of $\arccos x$ gives a "distribution function" that never reaches 1.
- Taking the shorter segment as $1 - x\sin\theta$ or $1 - x/\cos\theta$: check which side of the triangle is x and which angle is θ.
- In part (iii), multiplying by $\frac{1}{2}$ or using X uniform on $[0, 2]$ without adjusting the crossing condition; the nearest-line distance is uniform on $[0, 1]$.

Beyond STEP

Buffon's needle (1777) is one of the first problems in geometric probability, and the appearance of π means that dropping needles gives a (slow) Monte Carlo estimate of π.

The cleanest modern treatment uses the linearity of expectation: the expected number of crossings is proportional to the needle's length, even for a bent needle ("Buffon's noodle"), which leads into integral geometry and the Crofton formula. Grimmett and Welsh, *Probability: An Introduction*, discusses joint distributions and geometric probability problems of this kind.

WORKED SOLUTION

STEP II 1997, Question 14

Probability and Statistics • Poisson Processes

STEP II 1997, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q14

The key idea

Despite the mechanics setting, this is a Poisson process question. The kinematics just tells us which vehicles count: a faster vehicle catches me inside the tunnel only if it enters less than a certain time after me. Once we know that cut-off, everything becomes a question about Poisson counts. In the second half we use the fact that a Poisson stream whose vehicles are labelled fast or slow at random splits into two independent Poisson streams. We also need to notice that one slow vehicle shields me from every fast vehicle behind it.

Solution

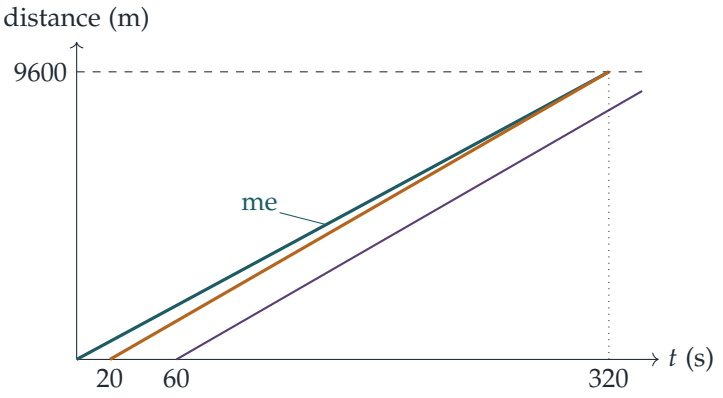
Throughout we model vehicles as particles (their lengths are ignored), so a vehicle that catches up simply travels at the speed of the one in front from then on. Measure time t in seconds from the moment I enter. Vehicles enter at a rate of 6 per minute, which is 0.1 per second, so the number entering in any interval of length τ seconds is $\mathrm{Po}(0.1\tau)$, and the counts in disjoint intervals are independent.

The first part: who can catch me?

I take $9600/30 = 320\,\mathrm{s}$ to get through the tunnel. Consider a vehicle at $32\,\mathrm{m\,s^{-1}}$ entering at time $t > 0$. Until it catches me, my distance in is $30s$ at time s and its distance is $32(s-t)$. These are equal when $2s = 32t$, i.e. $s = 16t$, at a distance of $480t$ metres into the tunnel. So it catches me before I leave exactly when

$$480t \leq 9600 \iff t \leq 20.$$

(Equivalently: it needs $9600/32 = 300\,\mathrm{s}$ to get through, so it reaches the exit no later than me exactly when $t + 300 \leq 320$.)



The distance–time graph shows me (teal), a fast vehicle entering at $t = 20$ (amber) and one entering at $t = 60$ (plum). The amber vehicle meets me just as we both reach the exit; any fast vehicle entering later, like the plum one, is still behind me when I leave.

Every other vehicle travels at $32\,\mathrm{m\,s^{-1}}$, so nothing can hold up a vehicle entering in $(0, 20]$ before it reaches me: any vehicle ahead of it, apart from me, is also travelling at $32\,\mathrm{m\,s^{-1}}$ until it too reaches me. Vehicles that entered before me are ahead of me and irrelevant. So the queue behind me when I leave consists of exactly the vehicles entering in $(0, 20]$, whose number is $\mathrm{Po}(0.1 \times 20) = \mathrm{Po}(2)$. The expected number of vehicles in the queue behind me is

$$\boxed{2}.$$

The second part: exactly two enter and catch me

Now each other vehicle is fast ($32\,\mathrm{m\,s^{-1}}$) or slow ($30\,\mathrm{m\,s^{-1}}$) with probability $\frac{1}{2}$, independently of everything else.

Ask yourself Does every fast vehicle entering within 20 seconds still catch me?

No. A slow vehicle entering at time $t' > 0$ keeps a constant gap of $30t'$ metres behind me, so it never catches me. A fast vehicle entering after it catches the slow vehicle (or is blocked by something behind it) and is stuck at $30\,\mathrm{m\,s^{-1}}$ from then on, so it never reaches me either. A vehicle entering in $(0, 20]$ therefore catches me if and only if it is fast *and* every vehicle entering between me and it is fast. A fast vehicle that enters after $t = 20$ cannot catch me even if nothing is in its way.

Let $K \sim \mathrm{Po}(2)$ be the number of vehicles entering in $(0, 20]$. For the event in this part, exactly two vehicles enter in that window and both catch me. That needs $K = 2$ with both vehicles fast:

$$\mathrm{P} = \mathrm{P}(K = 2) \times \left(\tfrac{1}{2}\right)^2 = \frac{2^2 \mathrm{e}^{-2}}{2!} \cdot \frac{1}{4} = \boxed{\tfrac{1}{2}\mathrm{e}^{-2}} \approx 0.0677.$$

The final part: exactly two queuing behind me

Now any number of vehicles may enter in the first 20 seconds. List the K vehicles entering in $(0, 20]$ in order of entry. By the argument above, the queue behind me at the exit is the unbroken run of fast vehicles at the start of this list. (No vehicle entering after $t = 20$ can join, since it can never catch me.) So there are exactly two vehicles in the queue if and only if

- $K = 2$ and both are fast: probability $\frac{1}{4}$ given $K = 2$; or
- $K \geq 3$ and the first two are fast and the third is slow: probability $\frac{1}{8}$ given $K = k \geq 3$.

Hence

$$\begin{aligned} \mathrm{P}(\text{queue of two}) &= \tfrac{1}{4}\,\mathrm{P}(K = 2) + \tfrac{1}{8}\,\mathrm{P}(K \geq 3) \\ &= \tfrac{1}{4} \cdot 2\mathrm{e}^{-2} + \tfrac{1}{8}\left(1 - \mathrm{e}^{-2} - 2\mathrm{e}^{-2} - 2\mathrm{e}^{-2}\right) \\ &= \tfrac{1}{2}\mathrm{e}^{-2} + \tfrac{1}{8} - \tfrac{5}{8}\mathrm{e}^{-2} \\ &= \boxed{\tfrac{1}{8}\left(1 - \mathrm{e}^{-2}\right)} \approx 0.108. \end{aligned}$$

This is larger than the previous answer, as it must be: that event is one of the ways of getting a queue of two.

> A neat alternative: split the stream into fast vehicles and slow vehicles, two independent Poisson processes each of rate $0.05\,\mathrm{s}^{-1}$. A queue of exactly two means that, in the first 20 seconds, the first two arrivals are fast and the next arrival (if there is one) is slow. Thinking in terms of these two streams is often the quickest route in Poisson questions with labelled events.

Checks

- Limiting case: if every vehicle were slow, nobody could catch me, and indeed the queue formula gives 0 when the probability of being fast is 0.
- With probability p of being fast in place of $\frac{1}{2}$, the same argument gives $p^2\mathrm{P}(K = 2) + p^2(1 - p)\mathrm{P}(K \geq 3)$. At $p = 1$ this is $\mathrm{P}(K = 2) = 2\mathrm{e}^{-2}$, matching the first part, where the queue length is exactly $\mathrm{Po}(2)$.
- We simulated 4×10^5 runs of the traffic with no overtaking. They gave a mean queue of 2.003 in the first part, and probabilities 0.0674 and 0.1076 in the second, against the exact 0.0677 and 0.1081.

Common slips

- Forgetting that a slow vehicle shields me from every fast vehicle behind it, and answering $\mathrm{P}(\text{exactly two fast in 20 s}) = \frac{1}{2}\mathrm{e}^{-1}$ for the final part.

- Using the rate 6 per minute with a time in seconds: in 20 seconds the mean is 2, not 120.
- Assuming the queue grows from vehicles entering at any time: only those entering within the first 20 seconds can reach me.

Beyond STEP

The second part uses *thinning*: if each event of a Poisson process of rate λ is kept independently with probability p, the kept and discarded events form independent Poisson processes of rates $p\lambda$ and $(1-p)\lambda$. Queues of vehicles that cannot overtake are a classic model in traffic flow theory, where slow vehicles produce "platoons" of the kind we found here. For Poisson processes and thinning, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 1997, Question 1

Pure • Exponential Power Series

STEP III 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q1

The key idea

Multiplying $e^x = \sum x^n/n!$ by a polynomial shifts the series along. The coefficient of x^n is then a combination of $1/n!$, $1/(n-1)!$, $1/(n-2)!, \ldots$, and over the common denominator $n!$ the numerator is a polynomial in n. Putting $x = 1$ turns the identity of power series into an identity for a numerical series. So the whole question is about choosing the polynomial that produces the numerator we want. The *falling factorials* n, $n(n-1)$, $n(n-1)(n-2)$ are the natural building blocks, because $n(n-1)\cdots(n-k+1)/n! = 1/(n-k)!$.

Solution

Throughout, $e^x = \sum_{n\geq 0} x^n/n!$ converges absolutely for every real x. Multiplying it by a polynomial gives a finite sum of such series, so we may add the series term by term and then set $x = 1$. We use the convention $1/m! = 0$ for negative integers m. This lets every sum below start at $n = 0$.

Part (i)

We have

$$(x^2+5x+4)\,e^x = \sum_{n\geq 0}\frac{x^{n+2}}{n!} + 5\sum_{n\geq 0}\frac{x^{n+1}}{n!} + 4\sum_{n\geq 0}\frac{x^n}{n!}.$$

To collect the coefficient of x^n, reindex the first two sums. The coefficient is

$$\begin{aligned}\frac{1}{(n-2)!} + \frac{5}{(n-1)!} + \frac{4}{n!} &= \frac{n(n-1)+5n+4}{n!}\\ &= \frac{n^2+4n+4}{n!} = \frac{(n+2)^2}{n!}.\end{aligned}$$

This holds for every $n \geq 0$, including $n = 0$ and $n = 1$, where the convention $1/m! = 0$ removes the missing terms. (For $n = 0$ the coefficient is $4 = 2^2/0!$; for $n = 1$ it is

$5 + 4 = 9 = 3^2/1!$.) Hence

$$(x^2 + 5x + 4)\,\mathrm{e}^x = \sum_{n\geq 0} \frac{(n+2)^2}{n!}\,x^n,$$

and putting $x = 1$ gives

$$10\,\mathrm{e} = \frac{2^2}{0!} + \frac{3^2}{1!} + \frac{4^2}{2!} + \frac{5^2}{3!} + \cdots = 4 + \frac{3^2}{1!} + \frac{4^2}{2!} + \cdots,$$

as required.

Part (ii)

Now the general term is $(n+1)^2/n!$, so we want a quadratic $x^2 + bx + c$ whose series has the coefficient

$$\frac{n(n-1) + bn + c}{n!} = \frac{(n+1)^2}{n!}.$$

Ask yourself How do we find b and c without guessing?

Write the target numerator in falling factorials: $n^2 + 2n + 1 = n(n-1) + 3n + 1$. Matching gives $b = 3$ and $c = 1$. So, by exactly the argument of part (i),

$$(x^2 + 3x + 1)\,\mathrm{e}^x = \sum_{n\geq 0} \frac{(n+1)^2}{n!}\,x^n,$$

and putting $x = 1$ gives

$$(1 + 3 + 1)\,\mathrm{e} = 5\,\mathrm{e} = 1 + \frac{2^2}{1!} + \frac{3^2}{2!} + \frac{4^2}{3!} + \cdots.$$

Part (iii)

The general term is $(n+1)^3/n!$, so we now need a cubic. The coefficient of x^n in $(ax^3 + bx^2 + cx + d)\,\mathrm{e}^x$ is

$$\frac{a}{(n-3)!} + \frac{b}{(n-2)!} + \frac{c}{(n-1)!} + \frac{d}{n!} = \frac{a\,n(n-1)(n-2) + b\,n(n-1) + c\,n + d}{n!}.$$

We expand the target in falling factorials. Since $n^3 = n(n-1)(n-2) + 3n(n-1) + n$ and $n^2 = n(n-1) + n$,

$$\begin{aligned}(n+1)^3 &= n^3 + 3n^2 + 3n + 1\\ &= \big[n(n-1)(n-2) + 3n(n-1) + n\big] + 3\big[n(n-1) + n\big] + 3n + 1\\ &= n(n-1)(n-2) + 6n(n-1) + 7n + 1.\end{aligned}$$

(A quick check at $n = 1$: both sides equal 8. At $n = 2$: $27 = 0 + 12 + 14 + 1$.) So $a = 1$, $b = 6$, $c = 7$, $d = 1$ and

$$(x^3 + 6x^2 + 7x + 1)\,\mathrm{e}^x = \sum_{n\geq 0} \frac{(n+1)^3}{n!}\,x^n.$$

Putting $x = 1$,

$$1 + \frac{2^3}{1!} + \frac{3^3}{2!} + \frac{4^3}{3!} + \cdots = (1 + 6 + 7 + 1)\,\mathrm{e} = 15\,\mathrm{e}.$$

Ask yourself Why can't we just square or cube the series term by term?

Because $\sum (n+1)^3/n!$ is not built from $\sum (n+1)/n!$ in any simple way. The sum is linear in the numerator, so what we can do is split the numerator into pieces $n(n-1)\cdots(n-k+1)$, each of which cancels against $n!$ to give a shifted copy of the series for e. Each such piece contributes exactly e at $x = 1$. The answer is therefore e times the sum of the falling-factorial coefficients, $1 + 6 + 7 + 1 = 15$.

Checks

- Partial sums up to $n = 60$ give $27.1828\ldots$, $13.5914\ldots$ and $40.7742\ldots$, agreeing with 10e, 5e and 15e to machine precision.
- The coefficients in the three numerator identities were checked exactly for $n = 0, \ldots, 39$.
- Consistency: part (ii)'s series is $\sum (n+1)^2/n!$ and part (i)'s is $\sum (n+2)^2/n!$. The difference $\sum (2n+3)/n! = 2\sum n/n! + 3\sum 1/n! = 2\mathrm{e} + 3\mathrm{e} = 5\mathrm{e}$ is indeed $10\mathrm{e} - 5\mathrm{e}$.

Common slips

- Forgetting that the first two terms ($n = 0, 1$) need separate attention when sums are reindexed. The convention $1/m! = 0$ for $m < 0$ handles them, but you should check them.
- Matching $(n+1)^3$ against powers n^3, n^2, n instead of falling factorials, and then picking the wrong cubic.

Beyond STEP

The sums $\sum_{n \geq 0} n^k/n!$ are $B_k\,\mathrm{e}$, where B_k is the kth Bell number (the number of ways to partition a set of k objects): $B_0, \ldots, B_4 = 1, 1, 2, 5, 15$. This is Dobiński's formula. Indeed, writing $m = n + 1$, part (iii) is $\sum_{m \geq 1} m^4/m! = B_4\,\mathrm{e} = 15\,\mathrm{e}$, and the coefficients $1, 6, 7, 1$ are the Stirling numbers of the second kind $1, 7, 6, 1$ in reverse order. Rearranging and combining power series term by term is justified by absolute convergence, a theme of first-year analysis. See Abbott, *Understanding Analysis*, for the theory, and Siklos, *Advanced Problems in Mathematics*, for more questions of this kind.

WORKED SOLUTION

STEP III 1997, Question 2

Pure • Curve Sketching

STEP III 1997, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q2

The key idea

Taking logarithms turns $x^y = y^x$ into $y \ln x = x \ln y$, and dividing by xy gives $\mathrm{f}(x) = \mathrm{f}(y)$. So the second half is really a question about the first: once we know how many inputs give each output of f, we know how many partners y each x has. The equation always has the obvious solution $y = x$; the interest is in the other one.

Solution

The graph of f

Differentiating with the quotient rule,

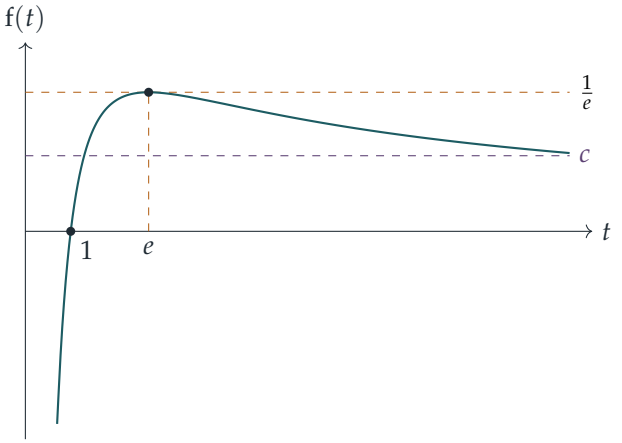
$$\mathrm{f}'(t) = \frac{t \cdot \frac{1}{t} - \ln t}{t^2} = \frac{1 - \ln t}{t^2}, \qquad \mathrm{f}''(t) = \frac{2\ln t - 3}{t^3}.$$

So $\mathrm{f}'(t) > 0$ for $0 < t < e$ and $\mathrm{f}'(t) < 0$ for $t > e$: f is strictly increasing on $(0, e]$ and strictly decreasing on $[e, \infty)$. The maximum value is

$$\mathrm{f}(e) = \frac{1}{e}.$$

Other features for the sketch:

- $\mathrm{f}(t) < 0$ for $0 < t < 1$, $\mathrm{f}(1) = 0$, and $\mathrm{f}(t) > 0$ for $t > 1$.
- As $t \to 0^+$, $\ln t \to -\infty$ and $1/t \to +\infty$, so $\mathrm{f}(t) \to -\infty$: the y-axis is an asymptote.
- As $t \to \infty$, $\mathrm{f}(t) \to 0^+$, since $\ln t$ grows more slowly than t. So the t-axis is an asymptote.
- There is a point of inflection at $t = e^{3/2}$, where $\mathrm{f} = \frac{3}{2}e^{-3/2} \approx 0.33$. Nobody needs it for the counting, but it makes the sketch honest.



How many t give a value c?

Because f is continuous and strictly monotonic on each of $(0, e]$ and $[e, \infty)$, each piece takes any value it reaches exactly once. On $(0, e]$, f increases from $-\infty$ to $\frac{1}{e}$; on $[e, \infty)$ it decreases from $\frac{1}{e}$ towards 0, never reaching it. Hence, for a given value c:

- $c > \frac{1}{e}$: **no** values of t;
- $c = \frac{1}{e}$: **one** value, $t = e$;
- $0 < c < \frac{1}{e}$: **two** values, one in $(1, e)$ and one in (e, ∞) (the horizontal line c in the sketch);
- $c \leq 0$: **one** value, in $(0, 1]$, because the decreasing branch stays strictly positive.

Counting solutions of $x^y = y^x$

For $x, y > 0$ both sides are positive, so we may take logarithms; ln is one-to-one, so nothing is gained or lost:

$$x^y = y^x \iff y \ln x = x \ln y \iff \frac{\ln x}{x} = \frac{\ln y}{y} \iff \mathrm{f}(y) = \mathrm{f}(x),$$

where we divided by $xy > 0$. So, for fixed x, the number of positive y is the number of t with $\mathrm{f}(t) = c$ where $c = \mathrm{f}(x)$. One of them is always $y = x$.

? **Ask yourself** Which of the four cases above can $c = \mathrm{f}(x)$ actually fall into?

Since c is a value of f, the case $c > \frac{1}{e}$ cannot happen. Translating the other cases back into conditions on x:

- $0 < x \leq 1$: then $c = \mathrm{f}(x) \leq 0$, so there is exactly **one** y, namely $y = x$.
- $x = e$: then $c = \frac{1}{e}$, so there is exactly **one** y, namely $y = e$.

- $1 < x < e$ or $x > e$: then $0 < c < \frac{1}{e}$, so there are exactly **two** values of y: $y = x$ and one other. If $1 < x < e$ the other lies in (e, ∞); if $x > e$ it lies in $(1, e)$.

For example, $x = 2$ has partner $y = 4$, since $2^4 = 4^2 = 16$.

Sketching the solution set

The set consists of the line $y = x$ (for all $x > 0$) together with a second curve made of the "other" solutions, which lives entirely in $x > 1$, $y > 1$. Its shape:

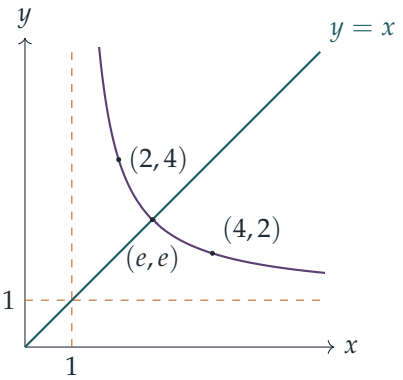
- *Symmetry.* The equation is unchanged when x and y are swapped, so the curve is symmetric in the line $y = x$.
- *Decreasing.* As x increases through $(1, e)$, $\mathrm{f}(x)$ increases, so the partner $y > e$ must move to where f is larger, i.e. y decreases. Similarly for $x > e$. So y is a decreasing function of x on this branch.
- *Asymptotes.* As $x \to 1^+$, $\mathrm{f}(x) \to 0^+$, so the partner $y \to \infty$: the line $x = 1$ is an asymptote. By symmetry, as $x \to \infty$, $y \to 1^+$, and $y = 1$ is an asymptote.
- *Crossing the line $y = x$.* As $x \to e$, the partner also tends to e, so the curve passes through (e, e) and crosses the line there. Near $t = e$, Taylor's theorem with $\mathrm{f}'(e) = 0$ and $\mathrm{f}''(e) = -e^{-3}$ gives $\mathrm{f}(t) \approx \frac{1}{e} - \frac{(t-e)^2}{2e^3}$. So $\mathrm{f}(x) = \mathrm{f}(y)$ forces $(y - e)^2 \approx (x - e)^2$, i.e. $y - e \approx \pm(x - e)$. The plus sign is the line $y = x$. The minus sign is our curve, which therefore crosses the line at right angles, with gradient -1.

? Ask yourself Can we write the curve down explicitly?

Yes, if we parametrise by the ratio. Put $y = tx$ with $t > 0$, $t \neq 1$. Then $x^{tx} = (tx)^x$; taking x-th roots gives $x^t = tx$, so $x^{t-1} = t$, and

$$x = t^{1/(t-1)}, \qquad y = t^{t/(t-1)}.$$

At $t = 2$ this gives $(2, 4)$, and at $t = \frac{1}{2}$ it gives $(4, 2)$. As $t \to 1$, both coordinates tend to e, because $t^{1/(t-1)} \to e$. This is how the curve below was drawn.



Checks

- $2^4 = 4^2$ is the famous integer example, and it sits on the curve, in $1 < x < e$ and $x > e$ as predicted.
- Numerically, a fine grid search puts the maximum of f at $t \approx 2.718$ with value $0.367879 \approx 1/e$. Bisection gives the partner of $x = 1.01$ as $y \approx 659$, and of $x = 100$ as $y \approx 1.050$. These are consistent with the asymptotes $x = 1$ and $y = 1$.
- A central difference at $x = e$ gives gradient -1.0000 for the second branch.

Common slips

- Saying every $x \neq e$ has two partners. For $0 < x \leq 1$ the value $\mathrm{f}(x)$ is not positive, and only $y = x$ works.
- Forgetting the line $y = x$ in the final sketch, or drawing the other branch so that it touches the line at (e, e) instead of crossing it.
- Dividing by $\ln x$ or $\ln y$ (which can be zero) instead of by xy.

Beyond STEP

The second curve is a level set of $\mathrm{f}(x) - \mathrm{f}(y)$, and (e, e) is a point where the gradient of that function vanishes. That is why two branches cross there instead of one smooth curve passing through. The implicit function theorem, met in first-year analysis, explains when a level set is locally a graph and what happens when it isn't. Abbott's *Understanding Analysis* is a friendly place to build the foundations. The equation $x^y = y^x$ itself has a long history; Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

WORKED SOLUTION

STEP III 1997, Question 3

Pure • Roots of Unity

STEP III 1997, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q3

The key idea

The nth roots of unity $1, \omega, \omega^2, \ldots, \omega^{n-1}$ are exactly the roots of $z^n - 1$, so that polynomial factorises completely; dividing out the factor $z - 1$ gives the first identity. Put the polygon in the complex plane with O at the origin and the vertices become $re^{i\theta}$ times the roots of unity. Then the vector sum is a geometric series, and the squared chord lengths come from expanding $|\mathbf{a}_k - \mathbf{a}_1|^2$ with a dot product.

Solution

The product identity

For $k = 0, 1, \ldots, n-1$ we have $(\omega^k)^n = e^{2\pi i k} = 1$, so each ω^k is a root of $z^n - 1 = 0$. These n numbers are distinct, because their arguments $2\pi k/n$ are distinct and all lie in $[0, 2\pi)$. A polynomial of degree n with leading coefficient 1 and n distinct roots is the product of the corresponding linear factors, so

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}).$$

On the other hand, summing a geometric series (or just multiplying out),

$$z^n - 1 = (z-1)(1 + z + z^2 + \cdots + z^{n-1}).$$

For every $z \neq 1$ we may divide both expressions by $z - 1$, which gives

$$(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}) = 1 + z + z^2 + \cdots + z^{n-1} \qquad (z \neq 1).$$

Ask yourself The question says "any complex number z". What about $z = 1$, where we can't divide?

Both sides are polynomials of degree $n - 1$, and they agree at the infinitely many points $z \neq 1$. Their difference is then a polynomial with infinitely many roots, so it is the zero polynomial. (Alternatively, both sides are continuous in z, so let $z \to 1$.) Hence the identity holds for every complex z, including $z = 1$, as required.

The vector sum

Take the plane of the circle as the complex plane with O at the origin, and write a_k for the complex number representing A_k. If $a_1 = re^{i\theta}$, then labelling the vertices in order anticlockwise (consecutive vertices subtend angle $2\pi/n$ at O),

$$a_k = re^{i\theta}\,\omega^{k-1}, \qquad k = 1, 2, \ldots, n.$$

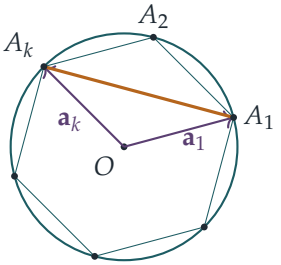
(Labelling clockwise instead just replaces ω by $\bar{\omega} = \omega^{n-1}$, which runs through the same set of roots; and the sum doesn't depend on the order of the vertices anyway.)

The vector $\overrightarrow{OA_k}$ corresponds to the complex number a_k, and adding vectors corresponds to adding complex numbers. So we need

$$\sum_{k=1}^{n} a_k = re^{i\theta}\left(1 + \omega + \omega^2 + \cdots + \omega^{n-1}\right).$$

The bracket is the right-hand side of the identity at $z = \omega$. Since ω appears among the factors on the left (for $n \geq 2$), the left-hand side has the factor $\omega - \omega = 0$, so the bracket is 0. (Equally, it is a geometric series with ratio $\omega \neq 1$, so it equals $(\omega^n - 1)/(\omega - 1) = 0$.) Therefore

$$\overrightarrow{OA_1} + \overrightarrow{OA_2} + \cdots + \overrightarrow{OA_n} = \mathbf{0}.$$



The sum of squared distances

Now write $\mathbf{a}_k = \overrightarrow{OA_k}$ as a vector, so $|\mathbf{a}_k| = r$ for every k and $\overrightarrow{A_1A_k} = \mathbf{a}_k - \mathbf{a}_1$. Expanding with the dot product,

$$\begin{aligned} |A_1A_k|^2 &= (\mathbf{a}_k - \mathbf{a}_1)\cdot(\mathbf{a}_k - \mathbf{a}_1) \\ &= |\mathbf{a}_k|^2 + |\mathbf{a}_1|^2 - 2\,\mathbf{a}_1\cdot\mathbf{a}_k \\ &= 2r^2 - 2\,\mathbf{a}_1\cdot\mathbf{a}_k. \end{aligned}$$

This also holds for $k = 1$, where both sides are 0. Summing over $k = 1, \ldots, n$ and using the fact that the dot product distributes over addition,

$$\begin{aligned} \sum_{k=1}^{n} |A_1A_k|^2 &= 2r^2n - 2\,\mathbf{a}_1\cdot\left(\sum_{k=1}^{n}\mathbf{a}_k\right) \\ &= 2r^2n - 2\,\mathbf{a}_1\cdot\mathbf{0} = 2r^2n, \end{aligned}$$

by the previous part. This is the required result.

Checks

- $n = 2$: the identity reads $z - (-1) = 1 + z$. The two points are ends of a diameter, so the sum is $0 + (2r)^2 = 4r^2 = 2r^2 \cdot 2$.
- $n = 4$, $r = 1$: the squared distances from one corner of the square are $0, 2, 4, 2$, totalling $8 = 2 \cdot 1^2 \cdot 4$.
- Numerically, for 2000 random $n \leq 20$, radii and rotations, the vector sum vanished and the sum of squared chords matched $2r^2 n$ to within 10^{-14}; the product identity was checked at random z (and at $z = 1$) to the same accuracy.

Common slips

- Dividing by $z - 1$ and forgetting that the identity is then only proved for $z \neq 1$.
- Writing $|\mathbf{a}_k - \mathbf{a}_1|^2 = |\mathbf{a}_k|^2 - |\mathbf{a}_1|^2$: the cross term $-2\,\mathbf{a}_1 \cdot \mathbf{a}_k$ is the whole point.

Beyond STEP

Putting $z = 1$ in the product identity and taking moduli gives $|1 - \omega|\,|1 - \omega^2| \cdots |1 - \omega^{n-1}| = n$: the product of the chord lengths from one vertex of a regular n-gon inscribed in a unit circle is exactly n, a neat companion to the sum of squares here. Factorising $z^n - 1$ over the roots of unity is the starting point for cyclotomic polynomials and for the discrete Fourier transform, both met early in a university course. Needham's *Visual Complex Analysis* is a beautiful place to see this kind of geometry done with complex numbers.

WORKED SOLUTION

STEP III 1997, Question 4

Pure • AM-GM Inequality

STEP III 1997, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q4

The key idea

The question as printed has a misprint: a polynomial with four roots must be a quartic, and the pattern of coefficients $1, 4, 6, 4, 1$ tells us the intended polynomial is

$$\mathrm{f}(x) = x^4 - 4ax^3 + 6b^2x^2 - 4c^3x + d^4.$$

The coefficients are chosen so that each of a, b^2, c^3, d^4 is an *average*: an average of the roots, of the products of pairs, of the products of triples, and of the product of all four. Each inequality then comes from the AM–GM inequality applied to the roots of f, f′ or f″, with Rolle's theorem making sure those roots are still distinct and positive.

Solution

Write $\mathrm{f}(x) = (x-p)(x-q)(x-r)(x-s)$. Comparing coefficients,

$$p+q+r+s = 4a, \qquad pq+pr+ps+qr+qs+rs = 6b^2,$$
$$pqr+qrs+rsp+spq = 4c^3, \qquad pqrs = d^4.$$

Part (i)

Let $P = pqrs = d^4$. Since none of the roots is zero,

$$pqr = \frac{P}{s}, \quad qrs = \frac{P}{p}, \quad rsp = \frac{P}{q}, \quad spq = \frac{P}{r}.$$

The map $t \mapsto P/t$ is one-to-one on the positive reals, and p, q, r, s are distinct, so the four products are distinct.

Part (ii)

The four triple products are distinct positive numbers, so by the AM–GM inequality

$$c^3 = \frac{pqr+qrs+rsp+spq}{4} > \left(pqr \cdot qrs \cdot rsp \cdot spq\right)^{1/4} = \left((pqrs)^3\right)^{1/4} = (d^4)^{3/4} = d^3.$$

Each root appears in exactly three of the four products, which is why the product is $(pqrs)^3$. Since $c, d > 0$ and $t \mapsto t^3$ is increasing, $c^3 > d^3$ gives $c > d$.

Part (iii)

Label the roots so that $0 < p < q < r < s$. The polynomial f is differentiable and vanishes at p and q, so by Rolle's theorem f′ has a root α with $p < \alpha < q$. In the same way there are roots $\beta \in (q, r)$ and $\gamma \in (r, s)$. These open intervals don't overlap, so $\alpha < \beta < \gamma$ are distinct. As f′ is a cubic it has no other roots. Note also that all three are positive, because $\alpha > p > 0$.

Part (iv)

Differentiating,

$$\mathrm{f}'(x) = 4x^3 - 12ax^2 + 12b^2x - 4c^3 = 4\big(x^3 - 3ax^2 + 3b^2x - c^3\big),$$

so α, β, γ are the roots of $x^3 - 3ax^2 + 3b^2x - c^3$, and

$$\alpha + \beta + \gamma = 3a, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = 3b^2, \qquad \alpha\beta\gamma = c^3.$$

This is the same pattern one degree down, so we copy parts (i) and (ii). The pair products are $\alpha\beta = c^3/\gamma$, $\beta\gamma = c^3/\alpha$, $\gamma\alpha = c^3/\beta$, which are distinct because α, β, γ are distinct and positive. By the AM–GM inequality,

$$b^2 = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{3} > \big((\alpha\beta\gamma)^2\big)^{1/3} = c^2,$$

and since $b, c > 0$, we get $b > c$.

Ask yourself Why do we need part (iii) before we can use AM–GM on f′?

The AM–GM inequality only applies to positive reals, and we need the strict version. A cubic needn't have three real roots at all. Rolle's theorem is what guarantees that α, β, γ are real, positive and distinct, and so the pair products are too.

Part (v)

Apply the same argument once more. Since f′ has distinct roots $\alpha < \beta < \gamma$, Rolle's theorem gives roots $\mu \in (\alpha, \beta)$ and $\nu \in (\beta, \gamma)$ of

$$\mathrm{f}''(x) = 12x^2 - 24ax + 12b^2 = 12\big(x^2 - 2ax + b^2\big),$$

and these are distinct and positive. So $\mu + \nu = 2a$ and $\mu\nu = b^2$, and by the AM–GM inequality for two numbers,

$$a = \frac{\mu + \nu}{2} > \sqrt{\mu\nu} = b.$$

(Equivalently, $x^2 - 2ax + b^2$ has two distinct real roots, so its discriminant $4a^2 - 4b^2$ is positive, and $a > b$ because both are positive.) Altogether we have shown

$$a > b > c > d.$$

Checks

- Roots $1, 2, 3, 4$: then $4a = 10$, $6b^2 = 35$, $4c^3 = 50$, $d^4 = 24$, so $a = 2.5$, $b \approx 2.415$, $c \approx 2.321$, $d \approx 2.213$, in the right order.
- If all four roots were equal to k, then $\mathrm{f}(x) = (x - k)^4$ and $a = b = c = d = k$. That is why distinctness is needed for the strict inequalities.
- We tested 20 000 random sets of four distinct positive roots, including nearly coincident ones, and $a > b > c > d$ held every time.

Common slips

- Applying AM–GM to the roots of f directly. That gives $a > d$, but it can't separate the middle terms.
- Forgetting to check that the numbers fed into AM–GM are positive and distinct, which is the whole point of parts (i) and (iii).
- Going from $c^3 > d^3$ or $b^2 > c^2$ to $c > d$ or $b > c$ without noting that all the letters are positive.

Beyond STEP

The quantities a, b, c, d are the *symmetric means* of the roots, and the chain $a > b > c > d$ is a special case of *Maclaurin's inequality*: for positive reals, the k-th root of the average of the k-fold products decreases as k increases. The trick of differentiating to lower the degree while keeping the means unchanged is also the standard route to Newton's inequalities. For much more, see Steele, *The Cauchy–Schwarz Master Class*, or the classic by Hardy, Littlewood and Pólya, *Inequalities*.

WORKED SOLUTION

STEP III 1997, Question 5

Pure • Cycloid Arc Length

STEP III 1997, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q5

The key idea

A point on the rim of a rolling wheel traces a *cycloid*. Once we write down where that point is in terms of the angle the wheel has turned through, the question is just an arc-length integral. The rolling condition (no slipping) is what links the centre's motion to the angle, and a half-angle identity makes the square root in the arc-length formula collapse.

Solution

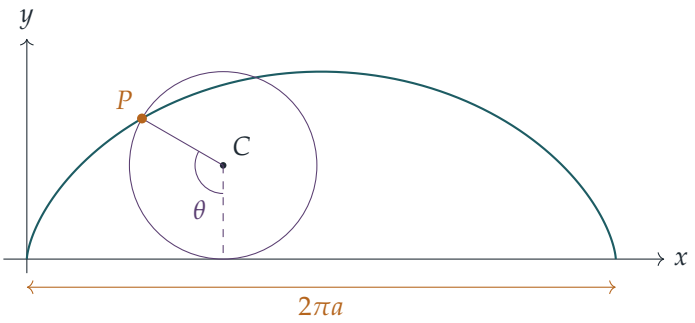
Setting up the path

Let the wheel have radius a and roll without slipping along the x-axis. Start with the marked point P touching the ground at the origin, and let θ be the angle (in radians) through which the wheel has turned.

Because there is no slipping, the length of ground covered equals the length of rim that has touched it, so the centre C has moved a distance $a\theta$ and sits at $(a\theta,\, a)$. Relative to C, the point P started directly below it and has turned through θ (clockwise, as the wheel rolls to the right), so P is at $C + (-a\sin\theta,\, -a\cos\theta)$. Hence

$$x = a(\theta - \sin\theta), \qquad y = a(1 - \cos\theta), \qquad 0 \le \theta \le 2\pi,$$

for one revolution. In that time the wheel moves forward a distance $2\pi a$.



The arc length

Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = a(1-\cos\theta), \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = a\sin\theta,$$

so

$$\begin{aligned}\left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2 &= a^2\left(1 - 2\cos\theta + \cos^2\theta + \sin^2\theta\right)\\ &= 2a^2(1-\cos\theta)\\ &= 4a^2\sin^2\tfrac{\theta}{2},\end{aligned}$$

using $1-\cos\theta = 2\sin^2\frac{\theta}{2}$.

> **Ask yourself** When we take the square root, is it $2a\sin\frac{\theta}{2}$ or $2a\left|\sin\frac{\theta}{2}\right|$, and does it matter here?

The square root is always $2a\left|\sin\frac{\theta}{2}\right|$. For $0 \le \theta \le 2\pi$ we have $0 \le \frac{\theta}{2} \le \pi$, where $\sin\frac{\theta}{2} \ge 0$, so the modulus can be dropped over exactly one revolution. (Over a longer range we would have to split the integral.)

The distance travelled by P is therefore

$$s = \int_0^{2\pi} 2a\sin\tfrac{\theta}{2}\,\mathrm{d}\theta = \Big[-4a\cos\tfrac{\theta}{2}\Big]_0^{2\pi} = -4a(-1) + 4a = 8a.$$

The ratio

The wheel moves $2\pi a$ while the point on its rim travels $8a$, so the required ratio is

$$\frac{2\pi a}{8a} = \frac{\pi}{4} \approx 0.785.$$

It doesn't depend on the size of the wheel, as it shouldn't: scaling the wheel scales both distances by the same factor.

Checks

- The point travels further than the wheel ($8a > 2\pi a \approx 6.28a$), which makes sense: it moves forward *and* goes up and down by $2a$ each way. In fact $8a$ is exactly four diameters.
- A crude bound: the path goes from $(0,0)$ up to $(\pi a, 2a)$ and down to $(2\pi a, 0)$, so it is at least $2\sqrt{\pi^2 a^2 + 4a^2} \approx 7.45a$, and at most $2\pi a + 4a \approx 10.28a$. The value $8a$ sits comfortably between.
- Numerically, following the rim point at 200 000 positions as the wheel rolls and adding up the straight-line steps gives 7.99999999992 for $a = 1$, so the ratio is $0.78539816\ldots = \pi/4$.

Common slips

- Getting the signs in the parametrisation wrong (for example $x = a(\theta + \sin\theta)$), which gives a path that moves backwards relative to the wheel. Check it at $\theta = \pi$: P should be at the top, $(\pi a, 2a)$.
- Leaving the integrand as $a\sqrt{2 - 2\cos\theta}$ and trying to integrate it directly; the half-angle identity is the step that makes it easy.
- Writing the ratio upside down: the question asks for the wheel's distance over the point's distance, which is $\pi/4$, not $4/\pi$.

Beyond STEP

The cycloid is one of the most celebrated curves in mathematics. It is the *brachistochrone*, the curve of fastest descent under gravity between two points, and the *tautochrone*, along which a bead reaches the bottom in the same time wherever it starts. Both facts are proved with the calculus of variations, which you meet in a first- or second-year course on mechanics or methods; this question's arc-length calculation is the first step in each. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1997, Question 6

Pure • Chebyshev Differential Equation

STEP III 1997, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q6

The key idea

The factor $1 - x^2$ next to $x = \cos\theta$ suggests one thing: $1 - \cos^2\theta = \sin^2\theta$. Once we change variable carefully with the chain rule, the awkward equation in x becomes the equation for simple harmonic motion in θ. We solve that and use the two side conditions to fix the constants. The recurrence is then a compound-angle identity in disguise. Throughout, n is a non-negative integer. The condition $y_n(x) = (-1)^n y_n(-x)$ only makes sense in that case.

Solution

Changing the variable

Let $x = \cos\theta$ with $0 < \theta < \pi$, so $\sin\theta > 0$ and $\mathrm{d}x/\mathrm{d}\theta = -\sin\theta \neq 0$. We write y for y_n to keep the notation light. By the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta} = -\frac{1}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}.$$

To differentiate again with respect to x, we differentiate with respect to θ and multiply by $\mathrm{d}\theta/\mathrm{d}x = -1/\sin\theta$:

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= -\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(-\frac{1}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}\right) = \frac{1}{\sin\theta}\left(\frac{1}{\sin\theta}\frac{\mathrm{d}^2y}{\mathrm{d}\theta^2} - \frac{\cos\theta}{\sin^2\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)\\ &= \frac{1}{\sin^2\theta}\frac{\mathrm{d}^2y}{\mathrm{d}\theta^2} - \frac{\cos\theta}{\sin^3\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta}.\end{aligned}$$

Now substitute, using $1 - x^2 = \sin^2\theta$ and $x = \cos\theta$:

$$\begin{aligned}(1-x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - x\frac{\mathrm{d}y}{\mathrm{d}x} + n^2y &= \frac{\mathrm{d}^2y}{\mathrm{d}\theta^2} - \frac{\cos\theta}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta} + \frac{\cos\theta}{\sin\theta}\frac{\mathrm{d}y}{\mathrm{d}\theta} + n^2y\\ &= \frac{\mathrm{d}^2y}{\mathrm{d}\theta^2} + n^2y.\end{aligned}$$

The two first-derivative terms cancel exactly. So the given equation is equivalent to $\dfrac{\mathrm{d}^2y_n}{\mathrm{d}\theta^2} + n^2y_n = 0$, as required.

Ask yourself Why isn't $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ simply $\dfrac{1}{\sin^2\theta}\dfrac{\mathrm{d}^2y}{\mathrm{d}\theta^2}$?

Because $\mathrm{d}y/\mathrm{d}x$ is a *product*: $-1/\sin\theta$ times $\mathrm{d}y/\mathrm{d}\theta$. Both factors depend on θ, so the product rule gives two terms. The extra term is exactly what cancels the $-x\,\mathrm{d}y/\mathrm{d}x$ term, which is why the question has that term in it.

Solving for y_n

For $n \geq 1$ the general solution of $y'' + n^2y = 0$ is

$$y_n = A\cos n\theta + B\sin n\theta.$$

For $n = 0$ the equation is $y'' = 0$, so $y_0 = A + B\theta$.

The condition at $x = 1$. $x = 1$ corresponds to $\theta = 0$, and $y_n(1) = 1$ gives $A = 1$ in both cases.

The symmetry condition. Replacing x by $-x$ means replacing θ by $\pi - \theta$, since $\cos(\pi - \theta) = -\cos\theta$, and $\pi - \theta$ is again in $[0, \pi]$. For an integer $n \geq 1$,

$$\cos n(\pi - \theta) = (-1)^n\cos n\theta, \qquad \sin n(\pi - \theta) = -(-1)^n\sin n\theta.$$

So $(-1)^n y_n(-x) = (-1)^n\left[A(-1)^n\cos n\theta - B(-1)^n\sin n\theta\right] = A\cos n\theta - B\sin n\theta$. Setting this equal to $y_n(x) = A\cos n\theta + B\sin n\theta$ for all θ gives $2B\sin n\theta = 0$, so $B = 0$. For $n = 0$ the condition reads $1 + B\theta = 1 + B(\pi - \theta)$ for all θ, which again forces $B = 0$.

Hence, for every non-negative integer n,

$$\boxed{y_n = \cos n\theta}, \qquad \text{that is,} \quad y_n(x) = \cos(n\arccos x) \quad (|x| \leq 1).$$

The relation $\theta = \arccos x$ takes $[-1, 1]$ onto $[0, \pi]$, which is why the result holds for $|x| \leq 1$. The endpoints $x = \pm 1$ ($\theta = 0, \pi$) are covered by continuity.

The deduction

With $n = 0$ and $n = 1$: $y_0 = \cos 0 = 1$ and $y_1 = \cos\theta = x$.

For the recurrence, add the compound-angle formulae

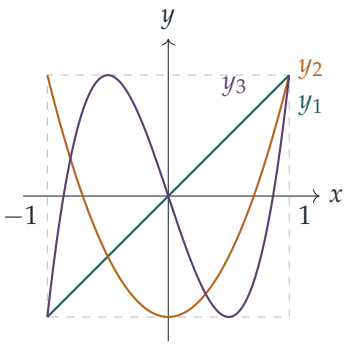
$$\cos(n+1)\theta = \cos n\theta\cos\theta - \sin n\theta\sin\theta, \qquad \cos(n-1)\theta = \cos n\theta\cos\theta + \sin n\theta\sin\theta.$$

The sine terms cancel, leaving $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$. In terms of x,

$$y_{n+1} - 2xy_n + y_{n-1} = 0 \qquad (n \geq 1,\ |x| \leq 1),$$

which is the required result.

Starting from $y_0 = 1$ and $y_1 = x$, the recurrence produces polynomials: $y_2 = 2x^2 - 1$, $y_3 = 4x^3 - 3x$, $y_4 = 8x^4 - 8x^2 + 1$, and so on.



Each y_n stays between -1 and 1 on $[-1, 1]$, as it must, being a cosine.

Checks

- $y_2 = 2x^2 - 1$: then $(1 - x^2) \cdot 4 - x \cdot 4x + 4(2x^2 - 1) = 0$. It satisfies the equation with $n = 2$, and $y_2(1) = 1$, and it is even.
- The parity of y_n matches $(-1)^n$. For example, $y_3 = 4x^3 - 3x$ is odd.
- Numerically, for $x = 0.3$: $\cos(2 \arccos 0.3) = -0.82 = 2(0.3)^2 - 1$. A finite-difference check of the original equation for $y_n = \cos(n \arccos x)$, $0 \leq n \leq 8$, gave residuals below 10^{-5}.

Common slips

- Forgetting the product rule in the second derivative, so the first-derivative terms don't cancel.
- Keeping the $\sin n\theta$ solution. It satisfies the differential equation and $y_n(1) = 1$ (since $\sin 0 = 0$), but it fails the symmetry condition. That condition is there precisely to remove it.
- Forgetting the separate case $n = 0$, where the general solution is $A + B\theta$, not a pair of cosines and sines.

Beyond STEP

The polynomials y_n are the *Chebyshev polynomials* $T_n(x)$. The given equation is Chebyshev's equation. The second solution we discarded, $\sin(n \arccos x)$, is $\sqrt{1 - x^2}$ times a polynomial. Chebyshev polynomials are central to approximation theory. Of all monic polynomials of degree n, $2^{1-n} T_n$ has the smallest maximum size on $[-1, 1]$, which is why the zeros of T_n are the best interpolation points for many numerical methods. For series solutions of equations like this one, and the special functions that come out of them, see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 1997, Question 7

Pure • Irrationality of e

STEP III 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q7

The key idea

The series a_n is exactly what is left of $n!\,\mathrm{e}$ once you strip off its whole-number part: multiply the series $\mathrm{e} = \sum 1/k!$ by $n!$ and every term up to $k = n$ becomes an integer, while the tail becomes a_n. The geometric series b_n is there to act as an easy upper bound for a_n, term by term. Once we know the tail is strictly between 0 and 1, a rational e would make some $n!\,\mathrm{e}$ an integer with a non-zero fractional part, which is absurd.

Solution

Part (i)

The series b_n is geometric, with first term $\frac{1}{n+1}$ and common ratio $r = \frac{1}{n+1}$. Since $n \geq 1$ we have $0 < r \leq \frac{1}{2} < 1$, so it converges, and

$$b_n = \frac{\frac{1}{n+1}}{1 - \frac{1}{n+1}} = \frac{\frac{1}{n+1}}{\frac{n}{n+1}} = \frac{1}{n}.$$

Part (ii)

Write the kth terms of the two series as

$$u_k = \frac{1}{(n+1)(n+2)\cdots(n+k)}, \qquad v_k = \frac{1}{(n+1)^k}.$$

Each of the k factors $n+1, n+2, \ldots, n+k$ in the denominator of u_k is at least $n+1$, so $(n+1)(n+2)\cdots(n+k) \geq (n+1)^k$, and therefore

$$0 < u_k \leq v_k \quad \text{for every } k \geq 1.$$

Equality holds only for $k = 1$; for $k \geq 2$ the factor $n+2$ is strictly bigger than $n+1$, so $u_k < v_k$.

All the terms are positive and bounded by those of the convergent series b_n, so a_n converges (by comparison), and $a_n \geq u_1 > 0$. For the upper bound,

$$b_n - a_n = \sum_{k\geq 1}(v_k - u_k) \;\geq\; v_2 - u_2 = \frac{1}{(n+1)^2} - \frac{1}{(n+1)(n+2)} > 0,$$

because every term $v_k - u_k$ is non-negative. Hence $a_n < b_n = \frac{1}{n}$, and together

$$0 < a_n < \frac{1}{n},$$

as required.

Ask yourself Why isn't "each term of a_n is at most the corresponding term of b_n" enough to give a *strict* inequality?

Termwise $\leq$ only gives $a_n \leq b_n$ in the limit; strict inequalities can turn into equalities when you add infinitely many terms. That is why we pick out one term, $k = 2$, where the gap is a definite positive amount, and note that all the other gaps are ≥ 0.

Part (iii)

Using the series $\mathrm{e} = \sum_{k=0}^{\infty} \frac{1}{k!}$, multiply by $n!$ and split at $k = n$:

$$n!\,\mathrm{e} = \underbrace{\sum_{k=0}^{n} \frac{n!}{k!}}_{N} + \sum_{k=n+1}^{\infty} \frac{n!}{k!}.$$

For $k \leq n$, $\frac{n!}{k!} = (k+1)(k+2)\cdots n$ is an integer, so N is a positive integer. For $k = n + j$ with $j \geq 1$,

$$\frac{n!}{(n+j)!} = \frac{1}{(n+1)(n+2)\cdots(n+j)} = u_j,$$

so the tail is exactly a_n. Thus

$$n!\,\mathrm{e} = N + a_n, \qquad N \in \mathbb{Z}, \quad 0 < a_n < \tfrac{1}{n} \leq 1,$$

using part (ii) and $n \geq 1$. A number of the form (integer) + (something in $[0, 1)$) has that integer as its integer part, so $[n!\,\mathrm{e}] = N$, and therefore

$$a_n = n!\,\mathrm{e} - [n!\,\mathrm{e}].$$

Part (iv)

Suppose, for a contradiction, that $\mathrm{e} = \frac{p}{q}$ with p, q positive integers. Take $n = q$. Then

$$n!\,\mathrm{e} = q! \cdot \frac{p}{q} = (q-1)!\,p$$

is an integer, so it equals its own integer part, and part (iii) gives $a_q = 0$. But part (ii) says $a_q > 0$. This contradiction shows that e is irrational.

(Any $n \geq q$ would do equally well, since q divides $n!$; the point is only that some $n!\,\mathrm{e}$ would have to be a whole number.)

Checks

For $n = 1$: $1!\,\mathrm{e} = 2.71828\ldots$, so the fractional part is $0.71828\ldots$, and summing $\frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots$ gives $0.71828\ldots$ too, comfortably below $b_1 = 1$. For $n = 5$, $a_5 \approx 0.19382$, just under $\frac{1}{5}$: the bound gets relatively tighter as n grows, because the first terms of a_n and b_n agree. A computer check with exact fractions confirms $0 < a_n < \frac{1}{n}$ and $n!\,\mathrm{e} = N + a_n$ with $[n!\,\mathrm{e}] = N$ for every n from 1 to 29.

Common slips

- Claiming $a_n < b_n$ simply because $u_k \leq v_k$ for all k. You need one strictly smaller term (here $k = 2$) to keep the inequality strict.
- Forgetting why $\sum_{k \leq n} n!/k!$ is an integer: it is because $k!$ divides $n!$ when $k \leq n$.
- In part (iv), only using $0 < a_n$ or only $a_n < 1$. Both halves matter: $a_n < 1$ (from part (ii)) is what identifies a_n as the fractional part, and $a_n > 0$ is what gives the contradiction.

Beyond STEP

This is Fourier's proof that e is irrational, and its pattern (a quantity that would have to be an integer is shown to lie strictly between 0 and 1) is the template for many irrationality proofs, including those for π and e^r for rational $r \neq 0$. In a first analysis course you will justify the comparison test and the series for e properly, and see that the speed at which $\sum 1/k!$ converges is what makes the argument work: e is approximated by fractions far better than any rational number could be. Abbott's *Understanding Analysis* covers the series side, and Davenport's *The Higher Arithmetic* is a lovely introduction to rational approximation.

WORKED SOLUTION

STEP III 1997, Question 8

Pure • Diagonalising Quadratic Forms

STEP III 1997, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q8

The key idea

A symmetric matrix describes a quadratic form $ax^2 + 2bxy + cy^2$, and the conjugated matrix $R_{-\alpha}AR_\alpha$ describes the same form in axes rotated through α. Making it diagonal removes the xy term, so in the new axes the ellipse is in standard position and its semi-axes can be read off. Since rotations preserve distance from the origin, those semi-axes are the extreme distances we want.

Solution

Throughout, write

$$R_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}, \qquad R_{-\alpha} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} = R_\alpha^{\mathsf{T}}.$$

Part (i)

Let $C = \cos\alpha$ and $S = \sin\alpha$. First

$$AR_\alpha = \begin{pmatrix} aC + bS & -aS + bC \\ bC + cS & -bS + cC \end{pmatrix},$$

and multiplying on the left by $R_{-\alpha}$, the top-right entry is

$$\begin{aligned} C(-aS + bC) + S(-bS + cC) &= b(C^2 - S^2) - (a - c)SC \\ &= b\cos 2\alpha - \tfrac{1}{2}(a - c)\sin 2\alpha. \end{aligned}$$

Because A is symmetric, so is $R_{-\alpha}AR_\alpha = R_\alpha^{\mathsf{T}}AR_\alpha$ (its transpose is itself), so the bottom-left entry is the same and one condition does both. The matrix is diagonal exactly when

$$\tan 2\alpha = \frac{2b}{a - c} \quad (a \neq c), \qquad \text{so we may take} \qquad \alpha = \tfrac{1}{2}\arctan\frac{2b}{a - c}.$$

If $a = c$ the condition is $b\cos 2\alpha = 0$, and $\alpha = \frac{1}{4}\pi$ works (any α works if also $b = 0$, when A is already diagonal).

Part (ii)

Expanding the equation of the ellipse,

$$x^2 + y^2 + 4xy\cot 2\theta + 4x^2\cot^2 2\theta = 1,$$

while $\begin{pmatrix} x & y \end{pmatrix} A \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2bxy + cy^2$. Matching coefficients,

$$a = 1 + 4\cot^2 2\theta, \qquad b = 2\cot 2\theta, \qquad c = 1.$$

Then $a - c = 4\cot^2 2\theta \neq 0$ (as $0 < 2\theta < \frac{1}{2}\pi$), and

$$\frac{2b}{a-c} = \frac{4\cot 2\theta}{4\cot^2 2\theta} = \tan 2\theta,$$

so $\alpha = \theta$ satisfies the condition from part (i): $R_{-\theta}AR_\theta$ is diagonal.

For the diagonal entries, the same multiplication as in part (i) gives

$$\text{top-left} = aC^2 + 2bSC + cS^2, \qquad \text{bottom-right} = aS^2 - 2bSC + cC^2,$$

now with $C = \cos\theta$, $S = \sin\theta$. Put $k = \cot 2\theta$ and $\mu = \cos 2\theta$, and use $2SC = \sin 2\theta$, $k\sin 2\theta = \mu$, $C^2 = \frac{1}{2}(1+\mu)$, $S^2 = \frac{1}{2}(1-\mu)$, and $k^2 = \mu^2/(1-\mu^2)$. For the bottom-right entry,

$$\begin{aligned}(1+4k^2)S^2 - 2k\sin 2\theta + C^2 &= 1 + 4k^2S^2 - 2\mu \\ &= 1 - 2\mu + \frac{2\mu^2(1-\mu)}{(1-\mu)(1+\mu)} = \frac{(1-2\mu)(1+\mu)+2\mu^2}{1+\mu} \\ &= \frac{1-\mu}{1+\mu} = \frac{2\sin^2\theta}{2\cos^2\theta} = \tan^2\theta.\end{aligned}$$

In exactly the same way the top-left entry is $1 + 2\mu + \dfrac{2\mu^2}{1-\mu} = \dfrac{1+\mu}{1-\mu} = \cot^2\theta$. So

$$R_{-\theta}AR_\theta = \begin{pmatrix} \cot^2\theta & 0 \\ 0 & \tan^2\theta \end{pmatrix}.$$

Ask yourself Is there a quick way to confirm these two entries without redoing the algebra?

Yes. Conjugating doesn't change the trace or the determinant. Here $\det A = ac - b^2 = 1 + 4k^2 - 4k^2 = 1$, and indeed $\cot^2\theta\tan^2\theta = 1$. Also $\operatorname{tr} A = 2 + 4\cot^2 2\theta$, and since $\cot\theta - \tan\theta = \dfrac{\cos^2\theta - \sin^2\theta}{\sin\theta\cos\theta} = 2\cot 2\theta$,

$$\cot^2\theta + \tan^2\theta = (\cot\theta - \tan\theta)^2 + 2 = 4\cot^2 2\theta + 2.$$

Part (iii)

Introduce new coordinates by $\begin{pmatrix} x \\ y \end{pmatrix} = R_\theta \begin{pmatrix} X \\ Y \end{pmatrix}$, so that the X- and Y-axes are the old axes rotated through θ. Since $\begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} X & Y \end{pmatrix} R_\theta^{\mathsf{T}} = \begin{pmatrix} X & Y \end{pmatrix} R_{-\theta}$, the equation of the ellipse becomes

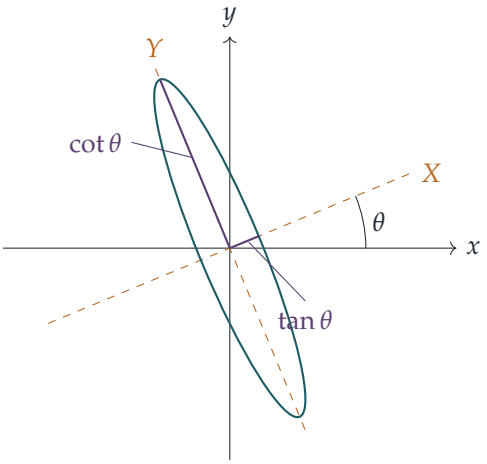
$$\begin{pmatrix} X & Y \end{pmatrix} R_{-\theta} A R_\theta \begin{pmatrix} X \\ Y \end{pmatrix} = 1, \qquad \text{that is} \qquad X^2 \cot^2\theta + Y^2 \tan^2\theta = 1.$$

This is symmetric under $(X, Y) \mapsto (-X, -Y)$, so the centre is the origin. A rotation preserves lengths, so the distance from the centre to a point is $r = \sqrt{x^2 + y^2} = \sqrt{X^2 + Y^2}$.

Because $0 < \theta < \frac{1}{4}\pi$, we have $0 < \tan\theta < 1 < \cot\theta$, so $\tan^2\theta < \cot^2\theta$. For any point on the ellipse,

$$\tan^2\theta\,(X^2 + Y^2) \;\le\; X^2\cot^2\theta + Y^2\tan^2\theta = 1 \;\le\; \cot^2\theta\,(X^2 + Y^2),$$

so $\tan^2\theta \le r^2 \le \cot^2\theta$, that is, $\tan\theta \le r \le \cot\theta$. The left inequality is an equality when $X = 0$, at $(X, Y) = (0, \pm\cot\theta)$, and the right one when $Y = 0$, at $(X, Y) = (\pm\tan\theta, 0)$. Both bounds are attained, so the minimum and maximum distances from the centre to the ellipse are $\tan\theta$ and $\cot\theta$, as required.



The figure shows the case $\theta = \frac{1}{8}\pi$, with the rotated axes dashed.

Checks

- $(0, \pm 1)$ lies on the ellipse, and $\tan\theta < 1 < \cot\theta$, consistent with the bounds.
- For $\theta = \frac{1}{8}\pi$ the major-axis end in the original coordinates is $\cot\theta\,(-\sin\theta, \cos\theta) = (-\cos\theta, \cos^2\theta/\sin\theta) \approx (-0.924,\ 2.230)$. Here $\cot 2\theta = 1$, and $0.924^2 + (2.230 - 1.848)^2 \approx 0.854 + 0.146 = 1$, as it should be.

- Numerically, scanning 200 000 directions round the ellipse for several values of θ gave, for $\theta = 0.3$, a minimum distance of 0.309336 and a maximum of 3.232728, matching $\tan 0.3$ and $\cot 0.3$.

Common slips

- Matching the xy coefficient to b rather than $2b$: the form is $ax^2 + 2bxy + cy^2$.
- Mixing up which rotation goes where: with $\mathbf{x} = R_\theta \mathbf{X}$ the row vector is $\mathbf{X}^\mathsf{T} R_{-\theta}$, which is why $R_{-\theta} A R_\theta$ appears.
- Thinking the axis with the larger coefficient is the longer one: $\cot^2\theta X^2$ has the larger coefficient, so the ellipse is *short* in the X direction.

Beyond STEP

This is the two-dimensional case of the spectral theorem: every real symmetric matrix can be diagonalised by a rotation, and its eigenvalues are the diagonal entries. The eigenvalues of a quadratic form classify conics and quadric surfaces, and the same idea gives principal axes of inertia in mechanics and principal components in statistics. Strang's *Introduction to Linear Algebra* treats symmetric matrices and quadratic forms very readably.

WORKED SOLUTION

STEP III 1997, Question 9

Mechanics • Statics with Elastic Springs

STEP III 1997, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q9

The key idea

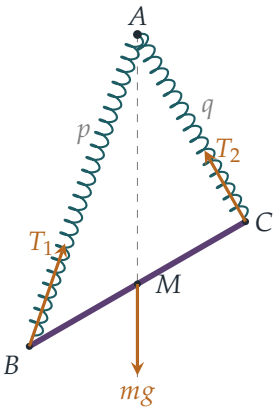
Only three forces act on the rod: the two spring tensions and its weight. When a body is in equilibrium under three non-parallel forces, their lines of action must pass through a single point. Both tensions point along lines through A, so the weight must pass through A too, which puts the midpoint of the rod directly below A. After that, horizontal resolution almost does the rest.

Solution

We model the rod as uniform and rigid, with mass m and midpoint M. The springs are light, obey Hooke's law with the same constant κ, and are both stretched. Let the springs AB and AC have natural lengths a and b and stretched lengths p and q, so the tensions are

$$T_1 = \kappa(p - a), \qquad T_2 = \kappa(q - b),$$

where both are positive, since the springs are stretched.



Step 1: M is directly below A

Take moments about A for the rod. The tension T_1 acts along BA and T_2 along CA, so both lines pass through A and neither has a moment about A. Hence the weight, acting vertically through M, must also have zero moment about A, so the vertical line through M passes through A. Thus M lies directly below A; let $h > 0$ be its depth below A.

Step 2: resolve horizontally

Set up axes with origin A, x horizontal (in the vertical plane of the rod) and y vertically upwards. Then $M = (0, -h)$, and since M is the midpoint of BC we can write

$$B = (-d,\ -h - e), \qquad C = (d,\ -h + e)$$

for some d, e. Here d is the horizontal half-width of the rod, and $d \neq 0$ exactly because the rod is not vertical.

The tension in AB acts on the rod at B in the direction from B to A, that is, along the vector $\overrightarrow{BA} = (d,\ h + e)$, which has length p. So this force is $\dfrac{T_1}{p}(d,\ h + e)$. Similarly, the tension at C is $\dfrac{T_2}{q}(-d,\ h - e)$.

Resolving horizontally,

$$\frac{T_1}{p}\, d - \frac{T_2}{q}\, d = 0,$$

and since $d \neq 0$,

$$\frac{T_1}{p} = \frac{T_2}{q}.$$

? **Ask yourself** Where exactly did we use the fact that the rod is not vertical?

Only in cancelling d. If the rod hung vertically, both springs would lie along the same vertical line, horizontal resolution would read $0 = 0$, and the tensions could be in any ratio, subject only to the vertical balance. Such vertical equilibria really do occur, and in them the result is generally false.

Step 3: bring in Hooke's law

Substituting the tensions,

$$\frac{\kappa(p - a)}{p} = \frac{\kappa(q - b)}{q} \quad \Longrightarrow \quad 1 - \frac{a}{p} = 1 - \frac{b}{q} \quad \Longrightarrow \quad \frac{a}{p} = \frac{b}{q},$$

so

$$\frac{p}{q} = \frac{a}{b}.$$

The ratio of the stretched lengths equals the ratio of the natural lengths, as required. ■

For completeness, resolving vertically gives $\frac{T_1}{p}(h+e)+\frac{T_2}{q}(h-e)=mg$, that is, $2h\,\frac{T_1}{p}=mg$. We don't need this, but it fixes the common value of T/length.

Checks

- *Interpretation.* Since $p/q=a/b$, each spring is stretched by the same factor: $p=\lambda a$ and $q=\lambda b$ for some $\lambda>1$. So the two springs behave like one uniform spring scaled up in proportion, which is a pleasing way to remember the answer.
- *Numerical.* We found equilibria by minimising the potential energy $mgy_M+\frac{1}{2}\kappa(p-a)^2+\frac{1}{2}\kappa(q-b)^2$ numerically for random m,κ,a,b and rod lengths. For example, with $a/b=0.709972$, a tilted equilibrium gave $p/q=0.709972$, with M directly below A to within 10^{-7}. The same search also turned up vertical equilibria with $p/q\neq a/b$ (for example, $a/b=1.526$ but $p/q=1.192$), which confirms that the hypothesis is needed.

Common slips

- Assuming the tensions are equal, or that the springs make equal angles with the vertical. Neither is true in general; what matches is tension divided by length.
- Taking moments about M and getting tangled in angles. Moments about A are much cleaner, because both tensions pass through A.
- Never using the condition that the rod is not vertical. A complete answer says where it is needed.

Beyond STEP

The vector $\frac{T}{p}\overrightarrow{BA}$ we used is the natural way to write a spring force. A spring with zero natural length exerts a force exactly proportional to the displacement vector, and such "ideal" springs turn many equilibrium problems into linear algebra. More generally, an equilibrium is a stationary point of the potential energy, which is the idea behind our numerical check and the starting point of Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both statics and the energy viewpoint carefully.

WORKED SOLUTION

STEP III 1997, Question 10

Mechanics • Rolling with Slipping

STEP III 1997, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q10

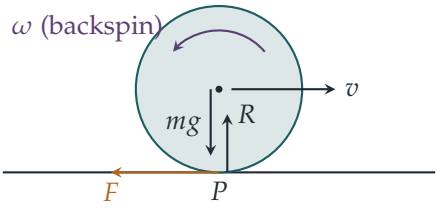
This question has a full guided treatment in *Beyond STEP*, Section 11.6, *Backspin*: https://beyondstep.keithmcnulty.org/ch11/backspin.html

The key idea

What matters is whether the bottom of the marble is sliding over the table. While it slides, the friction has a fixed size and a fixed direction, so both the linear and the angular motion are uniformly decelerated and we can write them down exactly. Sliding ends when the contact point comes to rest relative to the table, and after that the marble rolls with no friction at all. So the whole question is about which happens first: $v = 0$, $\omega = 0$, or the slip velocity reaching zero.

Solution

Model. The marble is a uniform solid sphere of mass m and radius a, so its moment of inertia about a diameter is $I = \frac{2}{5}ma^2$. Take v as the velocity of the centre (positive in the direction of v_0) and ω as the angular velocity in the *backward* sense (positive for backspin), so $v(0) = v_0 > 0$ and $\omega(0) = \omega_0 > 0$. The finger only starts the motion; afterwards the forces are the weight mg, the normal reaction $R = mg$ and the friction F at the contact point P.



With backspin, the bottom of the marble moves forward relative to the centre at speed

$a\omega$, so the velocity of the contact point is

$$s = v + a\omega.$$

Initially $s = v_0 + a\omega_0 > 0$: the marble skids forward over the table, so friction acts *backwards*, with constant magnitude F. Write $F = mk$.

Equations of motion while sliding. Resolving horizontally, $m\dot{v} = -F$. Taking moments about the centre, the backward friction at the bottom turns the marble in the forward-rolling sense, so it reduces the backspin:

$$\tfrac{2}{5}ma^2\,\dot{\omega} = -Fa \quad\Longrightarrow\quad \dot{\omega} = -\frac{5k}{2a}.$$

Both are constant, so as long as the marble keeps sliding forward,

$$v = v_0 - kt, \qquad a\omega = a\omega_0 - \tfrac{5}{2}kt, \qquad s = v_0 + a\omega_0 - \tfrac{7}{2}kt.$$

The slip velocity therefore falls to zero at

$$t_1 = \frac{2(v_0 + a\omega_0)}{7k},$$

and until then friction keeps the same direction, so these formulae hold on the whole interval $0 \le t \le t_1$, even if v or ω changes sign along the way.

Ask yourself What happens at $t = t_1$, and why doesn't the marble then start to slide the other way?

At t_1 the contact point is at rest, so $v = -a\omega$: the marble is rolling. If it carries on rolling at constant speed, nothing needs to push it, so friction (which only has to be as large as necessary once there is no slip) is zero, and rolling at constant velocity is consistent with all the equations. So from t_1 onwards the marble rolls uniformly with velocity

$$V = v(t_1) = v_0 - \tfrac{2}{7}(v_0 + a\omega_0) = \frac{5v_0 - 2a\omega_0}{7}.$$

Part (i)

To end up completely still, the final rolling velocity must be zero: $V = 0$, so $5v_0 = 2a\omega_0$, that is

$$\frac{v_0}{a\omega_0} = \frac{2}{5}.$$

Check that the picture makes sense: with this ratio, $v = 0$ at $t = v_0/k$ and $\omega = 0$ at $t = 2a\omega_0/(5k) = v_0/k$, the same moment, and this is also t_1. So linear and angular motion die away together while the marble is still sliding, and then it stays at rest (no slip, no friction).

Part (ii)

Now the marble must end up rolling backwards with speed $v_0/7$, so $V = -v_0/7$:

$$\frac{5v_0 - 2a\omega_0}{7} = -\frac{v_0}{7} \quad \Longrightarrow \quad 6v_0 = 2a\omega_0 \quad \Longrightarrow \quad \frac{v_0}{a\omega_0} = \frac{1}{3}.$$

With this ratio, v reaches zero at $t = v_0/k$, which is before $t_1 = 2(v_0 + 3v_0)/(7k) = 8v_0/(7k)$. At that instant the marble is momentarily stationary but still has backspin $\omega = \omega_0 - \frac{5v_0}{2a} = \frac{v_0}{2a} > 0$, so the bottom is still skidding forward and friction keeps pulling it back. At t_1 we get $v = -v_0/7$ and $a\omega = v_0/7$, so it rolls back towards its starting point at speed $v_0/7$, as required.

Ask yourself Is there a quicker way to get V without solving for t_1?

Yes. Take angular momentum about a fixed point O on the table's surface on the line of motion. The friction acts along the table surface, so it has no moment about O, and the moments of R and mg cancel. In the forward-rolling sense, the angular momentum about O is mva (the centre is at height a) minus the spin part $\frac{2}{5}ma^2\omega$, so this is conserved:

$$mv_0a - \tfrac{2}{5}ma^2\omega_0 = maV + \tfrac{2}{5}ma^2 \cdot \frac{V}{a} = \tfrac{7}{5}maV,$$

giving $V = (5v_0 - 2a\omega_0)/7$ again.

Checks

- With no spin ($\omega_0 = 0$) the formula gives $V = \frac{5}{7}v_0$, the standard result for a ball launched skidding without spin.
- With $v_0 = 0$ and backspin alone, $V = -\frac{2}{7}a\omega_0 < 0$: the marble moves backwards, as anyone who has flicked a backspun marble knows.
- Stepping the equations numerically (with the friction switching off once the slip velocity reaches zero) gives a final velocity of $-0.142857\ldots = -\frac{1}{7}$ for $v_0 = 1$, $a\omega_0 = 3$, and a dead stop for $v_0 = 2$, $a\omega_0 = 5$.

Common slips

- Stopping the clock when $v = 0$. The marble only stops sliding when $v + a\omega = 0$; at $v = 0$ it may still have spin, as in part (ii).
- Getting the sign of the torque wrong: backward friction at the bottom *reduces* backspin.
- Keeping the friction on after rolling starts. Once there is no slip, constant velocity needs no friction, so the friction drops to zero.

Beyond STEP

The friction here is Coulomb's model of sliding friction: a constant size while surfaces slip, and whatever is needed (up to a limit) when they don't. The angular-momentum shortcut is an instance of a general principle in rigid-body dynamics: choose the reference point so the awkward unknown force has no moment. Snooker and billiards players exploit exactly this physics with screw-back shots. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and rolling with slipping in this spirit.

WORKED SOLUTION

STEP III 1997, Question 11

Mechanics • Rigid Body Rotation

STEP III 1997, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q11

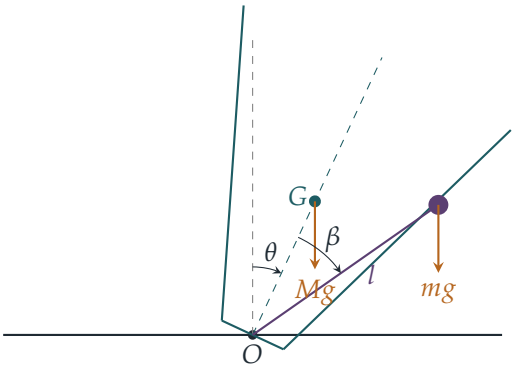
The key idea

While they touch, the bell and clapper turn about the same point O with the same angular velocity, so the only thing that can pull them apart is a difference in angular *acceleration*. Ask how fast each would accelerate on its own: the clapper and bell separate at the instant the bell, swinging freely, would accelerate just as fast as the free clapper. The contact force is then exactly zero, and the angular velocity never enters.

Solution

The model. The bell is a rigid body free to turn about the horizontal axis through O, with moment of inertia Mk^2 about it. The clapper is a particle of mass m on a light rigid rod of length l, also pivoted at O. The pivot is smooth and there is no air resistance. Measure every angle from the *upward* vertical, positive in the direction the clapper lies from the axis. Let θ be the angle of the bell's axis OG. While the two are in contact, the clapper rod stays at angle β to the axis, so it is at angle $\theta + \beta$.

Inside the bell, the clapper ball leans against the inner wall on the side towards which it would fall. So the wall pushes the ball back (towards smaller angles) and, by Newton's third law, the ball pushes the bell forwards. Let $\Gamma \geq 0$ be the moment about O of this contact force (including any friction). It acts as $-\Gamma$ on the clapper and $+\Gamma$ on the bell.



Equations of motion in contact. Taking moments about O for each body separately (the reaction at the pivot and the tension in the rod have no moment about O):

$$\text{clapper:}\quad ml^2\ddot{\theta} = mgl\sin(\theta+\beta) - \Gamma,$$
$$\text{bell:}\quad Mk^2\ddot{\theta} = Mgh\sin\theta + \Gamma.$$

The weight mg acts at distance l from O along a line at angle $\theta+\beta$ to the vertical, so its moment is $mgl\sin(\theta+\beta)$; similarly for Mg at G. Both bodies share the same $\ddot{\theta}$ because they move together.

Adding the equations eliminates Γ:

$$\ddot{\theta} = \frac{g(ml\sin(\theta+\beta) + Mh\sin\theta)}{ml^2 + Mk^2}.$$

Substituting back into the clapper equation and simplifying,

$$\Gamma = \frac{mMgl}{ml^2+Mk^2}\Big(k^2\sin(\theta+\beta) - hl\sin\theta\Big).$$

At release, $\theta = 0$ and $\Gamma = \dfrac{mMglk^2\sin\beta}{ml^2+Mk^2} > 0$, so the clapper does press on the bell at first, as it should.

> **Ask yourself** Why does separation happen exactly when Γ reaches zero, whatever the angular velocity?

The contact can only push, so it needs $\Gamma \geq 0$. While $\Gamma > 0$ the free clapper would accelerate faster than the bell, so it is held against the wall and the two move together. Because both turn about the same point O with the same angular velocity, the gap between the ball and the wall can only open through a difference in angular acceleration. Once Γ would turn negative, the bell's own acceleration $gh\sin\theta/k^2$ exceeds the clapper's $g\sin(\theta+\beta)/l$, and the wall swings away from the ball. So separation happens at the first θ with $\Gamma = 0$. (Such a θ always exists before $\theta = \pi$, since the bracket is $k^2\sin\beta > 0$ at $\theta = 0$ and $-k^2\sin\beta < 0$ at $\theta = \pi$.)

The separation angle. Put $\alpha = \theta + \beta$, the angle of the clapper rod with the upward vertical at separation. Then $\Gamma = 0$ gives

$$k^2\sin\alpha = hl\sin(\alpha-\beta) = hl(\sin\alpha\cos\beta - \cos\alpha\sin\beta).$$

Here $0 < \alpha < \pi$, so $\sin\alpha \neq 0$, and $\sin\beta \neq 0$. Dividing by $hl\sin\alpha\sin\beta$:

$$\frac{k^2}{hl\sin\beta} = \cot\beta - \cot\alpha,$$

that is,

$$\cot\alpha = \cot\beta - \frac{k^2}{hl}\operatorname{cosec}\beta,$$

as required. Notice that neither mass appears.

Checks

- *A short clapper.* As $l \to 0$ the correction term blows up, so $\cot\alpha \to -\infty$ and $\alpha \to \pi$. Sensible: a short pendulum has a large free angular acceleration $g\sin\alpha/l$, so a clapper on a very short rod keeps pushing the bell until $\sin\alpha$ is nearly zero, that is, until the pair is nearly upside down.
- *Equivalent form.* The condition is just $\dfrac{g\sin\alpha}{l} = \dfrac{gh\sin(\alpha-\beta)}{k^2}$: the free clapper and the free bell have equal angular accelerations.
- *Numerically.* With $k = 1$, $h = 0.6$, $l = 0.8$ and $\beta = 30°$, the formula gives $\cot\alpha = \sqrt{3} - 25/6 \approx -2.43$, so $\alpha \approx 157.7°$. Integrating the combined motion step by step from release, and stopping when Γ first hits zero, gives the same angle to many decimal places, for $M/m = 5$ and for $M/m = 500$ alike.

Common slips

- Using energy conservation to find $\dot\theta$ and then trying a "normal reaction" from circular motion. The angular velocity is common to both bodies and plays no part in whether they separate.
- Measuring α from the bell's axis rather than from the upward vertical. The rod is at β to the axis throughout the contact, so $\alpha = \theta + \beta$.
- Forgetting that Mk^2 is the moment of inertia about O, not about G. No parallel-axis correction is needed here.

Beyond STEP

This is a neat example of the general principle for finding when a constraint breaks: solve the constrained motion, compute the constraint force, and find where it changes sign. Lagrangian mechanics makes this systematic using Lagrange multipliers, which turn out to be the constraint forces. The same idea gives the classic particle leaving the surface of a sphere. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation about a fixed axis and is excellent preparation.

WORKED SOLUTION

STEP III 1997, Question 12

Probability and Statistics • Coupon Collector Problem

STEP III 1997, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q12

The key idea

Part (i) is the geometric distribution, and its probability generating function hands us the mean and variance by differentiation: $\mathrm{E}(K) = \mathrm{G}'(1)$ and $\mathrm{E}\big(K(K-1)\big) = \mathrm{G}''(1)$. In part (ii), the trick is to split the long wait for a full set into 52 short waits, one for each *new* card. Each short wait is a "toss until the first head" experiment, so part (i) does all the work.

Solution

Part (i)

The first head comes on toss k exactly when the first $k-1$ tosses are tails and toss k is a head. The tosses are independent, so

$$\mathrm{P}(K = k) = q^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

Hence

$$\mathrm{G}(s) = \sum_{k=1}^{\infty} pq^{k-1}s^k = ps\sum_{j=0}^{\infty}(qs)^j = \frac{ps}{1-qs},$$

summing the geometric series, which converges for $|qs| < 1$ (in particular for s near 1 when $p > 0$). That is the required result.

Differentiating,

$$\mathrm{G}'(s) = \frac{p(1-qs) + psq}{(1-qs)^2} = \frac{p}{(1-qs)^2},$$
$$\mathrm{G}''(s) = \frac{2pq}{(1-qs)^3}.$$

Since $\mathrm{G}'(s) = \sum k\,\mathrm{P}(K=k)s^{k-1}$ and $\mathrm{G}''(s) = \sum k(k-1)\mathrm{P}(K=k)s^{k-2}$, putting $s = 1$ and using $1 - q = p$ gives

$$\mathrm{E}(K) = \mathrm{G}'(1) = \frac{p}{p^2} = \frac{1}{p}, \qquad \mathrm{E}\big(K(K-1)\big) = \mathrm{G}''(1) = \frac{2pq}{p^3} = \frac{2q}{p^2}.$$

Therefore

$$\begin{aligned}\mathrm{Var}(K) &= \mathrm{E}\big(K(K-1)\big) + \mathrm{E}(K) - \big(\mathrm{E}(K)\big)^2 \\ &= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q+p-1}{p^2} = \frac{q}{p^2}.\end{aligned}$$

So $\mathrm{E}(K) = \dfrac{1}{p}$ and $\mathrm{Var}(K) = \dfrac{q}{p^2}$.

Part (ii)

Let N_i be the number of draws made after we have seen exactly $i-1$ different cards, up to and including the draw that shows the ith different card. So $N_1 = 1$ (the first card is always new), and

$$N = N_1 + N_2 + \cdots + N_{52}.$$

Ask yourself Why is each N_i a "first head" variable like K?

Once we have seen $i-1$ different cards, $53-i$ of the 52 cards are still unseen. Because we draw with replacement, every draw is independent of the past and shows an unseen card with the same probability

$$p_i = \frac{53-i}{52}.$$

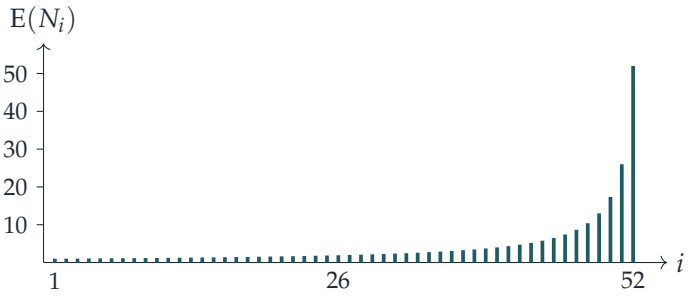
Call such a draw a "head". Then N_i is the number of draws up to and including the first head, which is exactly the variable K of part (i) with $p = p_i$. Hence

$$\mathrm{E}(N_i) = \frac{1}{p_i} = \frac{52}{53-i}.$$

Expectation is linear (we don't need independence of the N_i for this), so

$$\begin{aligned}\mathrm{E}(N) &= \sum_{i=1}^{52} \frac{52}{53-i} = 52\left(\frac{1}{52} + \frac{1}{51} + \cdots + \frac{1}{2} + \frac{1}{1}\right) \\ &= 52\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{52}\right).\end{aligned}$$

The early waits are short and the last few dominate: the very last card alone takes 52 draws on average.



To estimate, use $1 + \frac{1}{2} + \cdots + \frac{1}{n} \approx 0.5 + \ln n$ with $n = 52$. For $\ln 52$: $2.7^2 = 7.29$ and $7.29^2 \approx 53.1$, so $e^4 \approx 53 \approx 52$, giving $\ln 52 \approx 4$. Therefore

$$\mathrm{E}(N) \approx 52\,(0.5 + 4) = 52 \times 4.5 = 234.$$

So on average we need roughly 234 draws, about four and a half times the size of the pack, to see every card.

Checks

- In (i), $\mathrm{G}(1) = p/(1-q) = 1$, as any probability generating function must. With a fair coin ($p = \frac{1}{2}$) we get $\mathrm{E}(K) = 2$ and $\mathrm{Var}(K) = 2$; and as $p \to 1$ the variance $q/p^2 \to 0$, as it should when the first toss is almost certainly a head.
- The exact value is $52H_{52} \approx 235.98$, and a simulation of 10^5 full collections gave a mean of 235.7. Our estimate of 234 is close; the small shortfall comes from the approximation $0.5 + \ln n$ (a sharper one is $\ln n + 0.577$) and from rounding $\ln 52 \approx 3.95$ up to 4.

Common slips

- Getting the success probability of N_i upside down: the chance of a *new* card is $(53 - i)/52$, which shrinks as the collection grows.
- Taking $\mathrm{Var}(K) = \mathrm{G}''(1) - \mathrm{G}'(1)^2$, forgetting the $+\mathrm{G}'(1)$ term; $\mathrm{G}''(1)$ is $\mathrm{E}(K(K-1))$, not $\mathrm{E}(K^2)$.
- Thinking the N_i must be independent before adding expectations. Linearity of expectation holds regardless (though here they are in fact independent).

Beyond STEP

This is the *coupon collector's problem*: with n equally likely types, the expected time to collect all of them is $nH_n \approx n \ln n$. Undergraduate courses go further and show that the time is sharply concentrated around $n \ln n$, with fluctuations of order n, and probability generating functions become a standard tool for sums of independent variables like these. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and the geometric distribution carefully; Blitzstein and Hwang, *Introduction to Probability*, discusses the coupon collector problem with many variations.

WORKED SOLUTION

STEP III 1997, Question 13

Probability and Statistics • Normal Moment Generating Functions

STEP III 1997, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q13

The key idea

Everything rests on one integral: the moment generating function of a standard normal, found by completing the square in the exponent. After that, independence turns the mgf of a sum into a product, and the uniqueness of mgfs lets us read off distributions. The final part is the same calculation in disguise: a power of $Z = \mathrm{e}^{\mu+\sigma X}$ is just an exponential of a multiple of X.

Solution

Part (i)

Write $M(\theta) = \mathrm{E}(\mathrm{e}^{\theta X})$. Then

$$M(\theta) = \int_{-\infty}^{\infty} \mathrm{e}^{\theta x} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(x^2-2\theta x)}\,\mathrm{d}x.$$

Completing the square, $x^2 - 2\theta x = (x-\theta)^2 - \theta^2$, so

$$M(\theta) = \mathrm{e}^{\frac{1}{2}\theta^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(x-\theta)^2}\,\mathrm{d}x = \mathrm{e}^{\frac{1}{2}\theta^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}u^2}\,\mathrm{d}u,$$

substituting $u = x - \theta$. The last integral is the total integral of the standard normal density, which is 1. Hence, for every real θ,

$$\boxed{\mathrm{E}(\mathrm{e}^{\theta X}) = \mathrm{e}^{\frac{1}{2}\theta^2}.}$$

Part (ii)

Since X and Y are independent, so are $\mathrm{e}^{\theta a X}$ and $\mathrm{e}^{\theta b Y}$, and the expectation of a product of independent random variables is the product of their expectations. Using part (i) with θ replaced by $a\theta$ and by $b\theta$ (and the fact that Y has the same distribution as X),

$$\mathrm{E}\left(\mathrm{e}^{\theta(aX+bY)}\right) = \mathrm{E}(\mathrm{e}^{a\theta X})\,\mathrm{E}(\mathrm{e}^{b\theta Y}) = \mathrm{e}^{\frac{1}{2}a^2\theta^2}\,\mathrm{e}^{\frac{1}{2}b^2\theta^2} = \mathrm{e}^{\frac{1}{2}(a^2+b^2)\theta^2}.$$

Ask yourself Why does matching the mgfs settle the question of distribution?

A moment generating function that exists for all θ near 0 determines the distribution uniquely. So $aX + bY$ has the same distribution as X exactly when its mgf equals $\mathrm{e}^{\frac{1}{2}\theta^2}$ for all θ, that is, when

$$\mathrm{e}^{\frac{1}{2}(a^2+b^2)\theta^2} = \mathrm{e}^{\frac{1}{2}\theta^2} \quad \text{for all } \theta.$$

Taking logarithms and choosing any $\theta \neq 0$, this holds if and only if

$$\boxed{a^2 + b^2 = 1.}$$

(The "only if" direction is also clear without uniqueness: $\mathrm{Var}(aX + bY) = a^2 + b^2$ must equal $\mathrm{Var}(X) = 1$.) In general $aX + bY$ is normal with mean 0 and variance $a^2 + b^2$; for instance $(X + Y)/\sqrt{2}$ is again standard normal.

Part (iii)

For any real θ,

$$Z^\theta = \left(\mathrm{e}^{\mu+\sigma X}\right)^\theta = \mathrm{e}^{\mu\theta}\,\mathrm{e}^{(\sigma\theta)X},$$

and $\mathrm{e}^{\mu\theta}$ is a constant, so by part (i) with θ replaced by $\sigma\theta$,

$$\mathrm{E}(Z^\theta) = \mathrm{e}^{\mu\theta}\,\mathrm{E}\left(\mathrm{e}^{(\sigma\theta)X}\right) = \mathrm{e}^{\mu\theta}\,\mathrm{e}^{\frac{1}{2}\sigma^2\theta^2} = \mathrm{e}^{\mu\theta+\frac{1}{2}\sigma^2\theta^2},$$

as required.

Taking $\theta = 1$ gives the mean:

$$\boxed{\mathrm{E}(Z) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2}.}$$

Taking $\theta = 2$ gives $\mathrm{E}(Z^2) = \mathrm{e}^{2\mu+2\sigma^2}$, so

$$\begin{aligned}\mathrm{Var}(Z) &= \mathrm{E}(Z^2) - \left(\mathrm{E}(Z)\right)^2 = \mathrm{e}^{2\mu+2\sigma^2} - \mathrm{e}^{2\mu+\sigma^2} \\ &= \mathrm{e}^{2\mu+\sigma^2}\left(\mathrm{e}^{\sigma^2} - 1\right).\end{aligned}$$

Checks

- Differentiating $M(\theta) = \mathrm{e}^{\frac{1}{2}\theta^2}$ gives $M'(0) = 0$ and $M''(0) = 1$: mean 0, variance 1, as they should be.
- If $\sigma = 0$ then $Z = \mathrm{e}^{\mu}$ is constant; our formulas give mean e^{μ} and variance 0.
- Numerically, with $\mu = 0.3$ and $\sigma = 0.5$, integrating against the normal density gives $\mathrm{E}(Z) \approx 1.5296$ and $\mathrm{Var}(Z) \approx 0.6645$, matching $\mathrm{e}^{0.425}$ and $\mathrm{e}^{0.85}(\mathrm{e}^{0.25} - 1)$. A simulation of $0.6X + 0.8Y$ gave sample variance 1.000.

Common slips

- Writing $\mathrm{E}(Z) = \mathrm{e}^{\mu}$ by "plugging in the mean of X". In general $\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$, and here the convexity of the exponential pushes the mean up by the factor $\mathrm{e}^{\frac{1}{2}\sigma^2}$.
- Giving the condition as $a + b = 1$: it is the variances, not the coefficients, that add.

Beyond STEP

The random variable Z has the *log-normal* distribution, the standard model for quantities that grow by multiplying many small random factors, such as share prices in the Black–Scholes model. Part (ii) is a glimpse of a striking fact: the normal is essentially the only distribution with finite variance whose suitably scaled sums stay in the same family, which is tied up with the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, covers moment generating functions, the normal distribution and the central limit theorem at exactly this level.

WORKED SOLUTION

STEP III 1997, Question 14

Probability and Statistics • Products of Random Variables

STEP III 1997, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q14

The key idea

The perimeter is a linear function of the length and breadth, so its mean and variance come straight from the rules for sums of independent variables. The area is a product, and for independent variables the expectation of a product is the product of the expectations; applying this to LB and to L^2B^2 gives the mean and variance. To show dependence, the cleanest route is to find a non-zero covariance, because independent variables always have zero covariance.

Solution

Let L and B be the length and breadth of a randomly chosen plate, so that L and B are independent with

$$\mathrm{E}(L) = \mu_1, \quad \mathrm{Var}(L) = \sigma_1^2, \qquad \mathrm{E}(B) = \mu_2, \quad \mathrm{Var}(B) = \sigma_2^2.$$

Let $P = 2(L + B)$ be the perimeter and $A = LB$ the area. We shall use $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$ repeatedly, so that $\mathrm{E}(L^2) = \mu_1^2 + \sigma_1^2$ and $\mathrm{E}(B^2) = \mu_2^2 + \sigma_2^2$.

The perimeter

By linearity of expectation,

$$\mathrm{E}(P) = 2(\mu_1 + \mu_2).$$

Since L and B are independent, the variance of their sum is the sum of their variances, so

$$\mathrm{Var}(P) = 2^2\big(\mathrm{Var}(L) + \mathrm{Var}(B)\big) = 4(\sigma_1^2 + \sigma_2^2), \qquad \text{s.d.}(P) = 2\sqrt{\sigma_1^2 + \sigma_2^2}.$$

The area

Because L and B are independent, $\mathrm{E}(LB) = \mathrm{E}(L)\,\mathrm{E}(B)$, so

$$\mathrm{E}(A) = \mu_1\mu_2.$$

For the variance we need $\mathrm{E}(A^2) = \mathrm{E}(L^2B^2)$.

Ask yourself Can we still split $\mathrm{E}(L^2B^2)$ into a product?

Yes: functions of independent variables are independent, so L^2 and B^2 are independent and $\mathrm{E}(L^2B^2) = \mathrm{E}(L^2)\,\mathrm{E}(B^2)$. Therefore

$$\begin{aligned}\mathrm{Var}(A) &= \mathrm{E}(L^2)\,\mathrm{E}(B^2) - \mu_1^2\mu_2^2 \\ &= (\mu_1^2+\sigma_1^2)(\mu_2^2+\sigma_2^2) - \mu_1^2\mu_2^2 \\ &= \sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2,\end{aligned}$$

and

$$\mathrm{s.d.}(A) = \sqrt{\sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2}.$$

Notice that the variance of the product is *not* just $\sigma_1^2\sigma_2^2$: the spread in each dimension is magnified by the typical size of the other.

Perimeter and area are not independent

If P and A were independent we would have $\mathrm{E}(PA) = \mathrm{E}(P)\,\mathrm{E}(A)$, that is, $\mathrm{Cov}(P, A) = 0$. We compute the covariance. First,

$$\mathrm{E}(PA) = 2\,\mathrm{E}(L^2B) + 2\,\mathrm{E}(LB^2).$$

Using independence again (now of L^2 and B, and of L and B^2),

$$\mathrm{E}(L^2B) = (\mu_1^2+\sigma_1^2)\mu_2, \qquad \mathrm{E}(LB^2) = \mu_1(\mu_2^2+\sigma_2^2).$$

Hence

$$\begin{aligned}\mathrm{Cov}(P, A) &= \mathrm{E}(PA) - \mathrm{E}(P)\,\mathrm{E}(A) \\ &= 2\big[(\mu_1^2+\sigma_1^2)\mu_2 + \mu_1(\mu_2^2+\sigma_2^2)\big] - 2(\mu_1+\mu_2)\mu_1\mu_2 \\ &= 2\big(\mu_2\sigma_1^2 + \mu_1\sigma_2^2\big).\end{aligned}$$

Lengths and breadths are positive, so $\mu_1 > 0$ and $\mu_2 > 0$; the standard deviations are given to be non-zero, so $\sigma_1^2 > 0$ and $\sigma_2^2 > 0$. Thus $\mathrm{Cov}(P, A) > 0$.

Since independent variables have zero covariance, P and A cannot be independent. This makes sense: a plate that is larger than average in either direction tends to have both a larger perimeter and a larger area.

Ask yourself If the covariance had come out as zero, would that have shown P and A were independent?

No. The implication only runs one way: independence forces zero covariance, but zero covariance does not force independence. So a *non-zero* covariance proves dependence, which is exactly what we have; a zero covariance would have told us nothing.

Checks

- Take L and B independent, each equal to 1 or 3 with probability $\frac{1}{2}$. Then $\mu_i = 2$, $\sigma_i^2 = 1$ and the formula gives $\mathrm{Cov}(P, A) = 2(2 + 2) = 8$, which matches direct enumeration of the four equally likely plates. Dependence is also visible directly: $\mathrm{P}(P = 4) = \mathrm{P}(A = 1) = \frac{1}{4}$, but $\mathrm{P}(P = 4 \text{ and } A = 1) = \frac{1}{4} \neq \frac{1}{16}$, since both events say $L = B = 1$.
- A simulation with $\mu_1 = 5$, $\mu_2 = 3$, $\sigma_1 = 0.7$, $\sigma_2 = 0.4$ gave s.d.$(A) \approx 2.919$ against the formula's 2.913, and $\mathrm{Cov}(P, A) \approx 4.56$ against 4.54.
- If $\sigma_2 \to 0$ (fixed breadth μ_2), s.d.$(A) \to \mu_2\sigma_1$, as it should for $A = \mu_2 L$.

Common slips

- Writing $\mathrm{Var}(LB) = \sigma_1^2\sigma_2^2$, or $\mathrm{Var}(LB) = \mathrm{Var}(L)\mathrm{Var}(B) + \ldots$ with terms missing; always go through $\mathrm{E}(L^2B^2)$.
- Forgetting to square the factor 2 in $\mathrm{Var}(2(L + B))$.
- Arguing that P and A are dependent "because both depend on L and B". That is intuition, not proof; the covariance calculation (or an explicit example of unequal probabilities) is what settles it.

Beyond STEP

The distinction between uncorrelated and independent random variables is central in first-year probability: independence means the joint distribution factorises, which is much stronger than a single moment condition. The calculation for $\mathrm{Var}(LB)$ is also the starting point for the "propagation of errors" formulas used throughout experimental science. Grimmett and Welsh, *Probability: An Introduction*, treats independence, covariance and products of random variables carefully.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

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